Estimation with Recurrent Event Data Under an Informative Monitoring Period

by

Akim Adekpedjou

Bachelor of Science
University of Niamey, Niger, 1998

Master of Science
University of Michigan, Ann-Arbor, 2002

Submitted in Partial Fulfillment of the

Requirements for the Degree of Doctor of Philosophy in the

Department of Statistics

University of South Carolina

2007

Major Professor	Chairman, Examining Committee
Committee Member	Committee Member
Committee Member	Dean of The Graduate School

Acknowledgments

First and foremost, I would like to express my sincere gratitude to my advisor, Dr. Edsel Peña, for his guidance, his insightful comments and suggestions, and his availability any time I needed him throughout my research. I am fortunate to have had an advisor who is so motivated and committed to research, yet also takes time to advise students with the same tenacity. I would also like to thank him for his financial support through the NIH Grant GM056182. This financial support made this dissertation possible. For all this, I will be grateful forever.

I am grateful to Dr. Donald Edwards for serving on my committee and for the financial support of the department through the graduate teaching assistantship; without which, this dissertation would not have been possible. I am also grateful to Dr. Tebbs for serving on my committee. I would like to thank Dr. James Lynch for not only serving on my committee, but also for providing partial financial support through the NSF Grant DMS 0243594.

My sincere thanks go to Dr. Elizabeth Slate for serving on my committee and for driving back and forth from Charleston to Columbia to attend my proposal defense and my dissertation defense. Thanks also for your inputs on my work.

Thanks to my research colleagues, Jonathan Quiton, Laura Taylor and Alex McLain for helpful discussions during our research meetings. I thank Jonathan Quiton for suggestions and input on the simulation studies portion of my research. And finally, I thank my wife Leila and my parents for their encouragement and support during these years.

ABSTRACT

We consider a study which monitors the occurrence of a recurrent event for n subjects or units. Of interest is the problem of parametric and semiparametric estimation of the inter-event or gap-time distribution when the data is subject to informative censoring. We first address the case where the gap-times are assumed to be independent and identically distributed with some parametric survival distribution function $\bar{F}(t, \boldsymbol{\theta})$; where $\boldsymbol{\theta}$ is some p-dimensional parameter. It is further assumed that the recurrent event is observed until some random time, τ, whose distribution G depends on that of F through the relationship $\bar{G} = \bar{F}^{\beta}$, for some $\beta > 0$; the so-called Koziol-Green model. We present finite and asymptotic properties of the maximum likelihood estimates of β and $\boldsymbol{\theta}$ as well as the estimator $\bar{F}(t)$. Next, we derive Nelson-Aalen and Kaplan-Meier type estimators by embedding the problem in an appropriate counting process. Finite and asymptotic properties of the semiparametric estimators are also established. The proposed estimators in both cases are compared to those derived ignoring informative censoring to ascertain efficiency gained that results by taking into account informative censoring. It will be shown that the proposed estimators are more efficient than those derived ignoring informative censoring. Moreover, in the special case of the homogeneous Poisson process, the asymptotic relative efficiency is shown to be bounded by approximately 0.65.

Table of Contents

Acknowledgments **ii**

Abstract **iv**

1 Introduction **1**

1.1 Review of Literature 3

1.2 The Koziol-Green Model 6

1.3 Specific Aims 8

2 Mathematical Preliminaries **10**

3 Maximum Likelihood Estimation **23**

3.1 Stochastic Process Formulation and MLE 23

3.1.1 Introduction 23

3.1.2 Development of the Likelihood Process and MLE 24

3.2 Asymptotic Properties of the MLE 29

3.2.1 Preliminaries 29

3.2.2 Reformulated Process 30

3.2.3 Regularity Conditions and Asymptotic Results 33

3.2.4 Expression of the Asymptotic Covariance Matrix 35

3.2.5 Weak Convergence Result: Parametric Case 39

3.2.6 Estimation Ignoring Informative Censoring 42

3.3 Efficiency Comparison 44

4 Estimation in the Homogeneous Poisson Process (HPP) Case 47

4.1 Maximum Likelihood Estimator of θ and β 48

4.2 Asymptotic Properties in the HPP case 49

4.3 Efficiency Comparison 52

5 Simulation Study in the Parametric Case 58

5.1 Simulation Results in the General Case 59

5.1.1 Results by exploiting the KG model in the general case 59

5.1.2 Results by ignoring the KG Model in the general case 63

5.1.3 Efficiency comparison in the general case 67

5.2 Simulation Results in HPP Case 70

6 Semiparametric Estimation 72

6.1 Introduction 72

6.2 Doubly-Indexed Process 73

6.3 Method of Moment Estimators 76

6.4 Estimation of β 80

6.5 Asymptotic Properties 84

6.5.1 Consistency of the Proposed Estimators 84

6.5.2 Weak Convergence Result: Semiparametric case 93

6.6 Efficiency Comparison 101

6.6.1 Comparison Semiparametric/Nonparametric Estimator 101

6.6.2 Efficiency Comparison in the HPP Case 103

7 Finite Sample Properties in the Semiparametric Case 106

7.1 Simulation design 106

7.2 Discussion of Simulation Results 108

7.2.1 Estimating β 108

7.2.2 Small Sample Properties 109

8 Application 123

8.1 Migrating Motor Complex Data (MMC data) 123

8.1.1 Description of the data 123

8.1.2 Parametric Estimation 124

8.1.3 Semiparametric Estimation 125

8.2 Air Conditioning Data 125

8.2.1 Parametric Estimation 126

9 Summary and Future Work 130

9.1 Summary 130

9.2 Future Work 132

9.2.1 Technical Issues 132

9.2.2 Estimation with Frailties 133

Bibliography **135**

List of Figures

1.1 The recurrent event setting with sum-quota accrual scheme. 2

4.1 Efficiency versus p in the HPP case. 57

4.2 Efficiency versus time when $\beta \to 0$ in the HPP case. 57

5.1 Biases in the estimator of $\bar{F}(t)$ for different values of n and β under the KG model. True distribution is Weibull (1, 2). 61

5.2 Variances in the estimator of $\bar{F}(t)$ for different values of n and β under the KG model. True distribution is Weibull (1, 2). 62

5.3 MSE in the estimator of $\bar{F}(t)$ for different values of n and β under the KG model. True distribution is Weibull (1, 2) 63

5.4 Biases in the estimator of $\bar{F}(t)$ for different values of n and β when ignoring the KG model. True distribution is Weibull (1, 2) 65

5.5 Variances in the estimator of $\bar{F}(t)$ for different values of n and β when ignoring the KG model. True distribution is Weibull (1, 2). 66

5.6 MSE in the estimator of $\bar{F}(t)$ for different values of n and β when ignoring the KG model. True distribution is Weibull (1, 2). 67

5.7 Efficiency of the estimator obtained ignoring informative censoring versus that obtained when it is taken into account for different values of n and β when the true distribution is assumed to be Weibull(1, .5). . 68

5.8 Efficiency of the estimator obtained ignoring informative censoring versus that obtained when it is taken into account for different values of n and β when the true distribution is assumed to be Weibull(1, 2). . 69

5.9 Efficiency of the estimator obtained ignoring informative censoring versus that obtained when it is taken into account for different values of n and β in the homogeneous Poisson process case. 71

7.1 Graph of bias for different values of n and β under Weibull(1, .5) in the semiparametric case. 111

7.2 Biases for different values of n and β under Weibull(1, 1) in the semiparametric case. 112

7.3 Biases for different values of n and β under Weibull(1, 2) in the semiparametric case. 113

7.4 Variances for different values of n and β under Weibull(1, .5) in the semiparametric case. 114

7.5 Variances for different values of n and β under Weibull(1, 1) in the semiparametric case. 115

7.6 Variances for different values of n and β under Weibull(1, 2) in the semiparametric case. 116

7.7 MSE for different values of n and β under Weibull(1, .5) in the semiparametric case. 117

7.8 MSE for different values of n and β under Weibull(1, 1) in the semiparametric case. 118

7.9 MSE for different values of n and β under Weibull(1, 2) in the semiparametric case. 119

7.10 MSE for different values of n and β under Weibull(1, 2) in the semiparametric case. 120

7.11 Efficiency for different values of n and β under Weibull(1, .5) in the semiparametric case. 121

7.12 Efficiency for different values of n and β under Weibull(1, 1) in the semiparametric case. 122

8.1 Migrating Motor Complex Data (MMC). 124

8.2 Survivor curves estimates for the MMC data. The thick curve is obtained when the KG model is ignored and the curve with break is obtained when the KG model is assumed. 126

8.3 Survivor curves estimates for the MMC data. The thick curve is obtained when the KG model is ignored and the curve with break is obtained when the KG model is assumed. 127

8.4 Survivor curves estimates for the Air conditioning data in the parametric case. 128

8.5 Survivor curves estimates for the Air conditioning data in the semi-parametric case. 129

List of Tables

6.1 Efficiency of the semiparametric estimator to the nonparametric estimator in the HPP case for $\theta = 1$ and $\beta = .5$ 104

6.2 Efficiency of the semiparametric estimator to the nonparametric estimator in the HPP case for $\theta = \beta = 1$ 105

6.3 Efficiency of the semiparametric estimator to the nonparametric estimator in the HPP case for $\theta = 1$ and $\beta = 2$ 105

6.4 Efficiency in of the semiparametric estimator to the nonparametric estimator the HPP case for $\theta = 1$ and $\beta = 5$ 105

7.1 Simulated biases and standard errors for $\hat{\beta}$ for $\mu = .5$. 109

7.2 Simulated biases and standard errors for $\hat{\beta}$ for $\mu = 2$. 109

7.3 Simulated biases and standard errors for $\hat{\beta}$ for $\mu = 1$. 110

Chapter 1

Introduction

Recurrent events are frequently encountered in studies with longitudinal designs. In fact, this type of data can be encountered in a wide variety of settings: biomedical, such as keeping track of hospitalization of patients with a chronic disease [cf. Byar [10], Gail and Brown [17], Klein et al. [26], Aalen and Husbeye [23]]; mechanical, such as keeping track of failure of a car; engineering, such as keeping track of a failure of air conditioning machines [cf. Proschan [33] in which the dataset reports repairs of Boeing 720 jet air-conditioners]; actuarial science, such as keeping track of a claim from a given insurer, etc.... A pictorial representation of this type of data for one subject is given in Figure 1.1.

In this figure, the T_js represent the inter-event times, while τ is the random time at which monitoring of the subject ceases. The S_js are the calendar times of event occurrences, and K is the total number of event occurrences for the subject. There is also a possibility to include covariates, the X_is, but in this dissertation, we will not

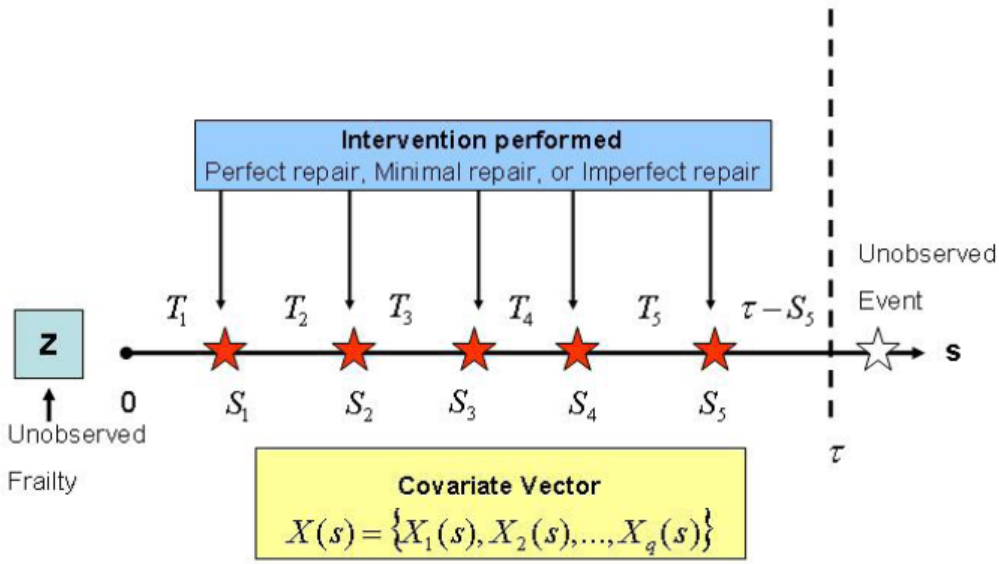

Figure 1.1: The recurrent event setting with sum-quota accrual scheme.

include covariates. In addition, no intervention [that is, the performance of a minimal, perfect or imperfect repair at each event occurrence] is performed since we are going to assume that the inter-event times are independent and identically distributed, the so-called i.i.d. renewal assumption. Another possibility is the inclusion of a frailty, Z, that are unobserved random effect components used frequently in survival analysis to model association among the inter-event times. Frailty will be included in the derivation of estimators in future research. If K is the total number of event occurrences, S_K is the calendar time of the last event occurrence. An important property inherent in the recurrent events setting is the sum-quota $[\sum_{j=1}^{K} T_j \leq \tau]$ accrual scheme wherein, given a finite observation period, the number of event occurrences and censoring time depend on the successive inter-event times, which implies both K and $\tau - S_K$ are informative concerning the distribution of the inter-event times. Consequently, censoring is informative and dependent in this recurrent events setting

[cf. Peña et al. [30]]. The Kaplan-Meier (KM) estimator [cf. Kaplan and Meier [25]] has been shown to be an asymptotically efficient estimator of the survivor function of the event time under the assumption of independent censoring and a noninformative monitoring period. However, this efficiency is lost if informative censoring is present; that is, if the survivor function of the end of observation time carries additional information about the distribution of the inter-events or gap-times. It is then of interest to develop new estimators in this recurrent events setting that account for the informative censoring and to compare them with those derived ignoring informative censoring, thereby ascertaining any efficiency gain that is achieved by exploiting a model that accounts for informative censoring.

1.1 Review of Literature

Many authors have worked on both parametric and nonparametric estimation of the event time distribution in the random censorship [single event setting] and recurrent event setting. However, with recurrent events, little attention has been given to the sum-quota accrual nature of the data which leads to an informative censoring, and which, as a consequence, could lead to loss in statistical efficiency. The problem of event time distribution estimation has been considered by Gill [[19], [20]], Vardi [[42], [41]], Cheng and Lin [12], Wang and Chang [44], Peña et al. [30]. Gill [20] dealt with the problem of nonparametric inference for renewal processes in a life testing setting. Vardi [41] presented an algorithm for obtaining the maximum likelihood (ML) estimator of the survivor function when the underlying interoccurence times

are arithmetic. Cheng and Lin [12] showed that their proposed ML type estimator is more efficient than the classical KM estimator under the proportional hazards model in the single event setting. In addition, they proved that the heavier the censoring, the greater the asymptotic relative efficiency (ARE) of their estimator. Sellke [36], in the case of a single unit, considered the problem of establishing weak convergence of a Nelson-Aalen (NA) type estimator when the length of the monitoring period increases indefinitely. Peña et al.[30], using counting process approach, rigorously developed NA and KM type estimators in the recurrent events setting, and derived their asymptotic properties when the number of units becomes unbounded. It is worth mentioning here that Aalen [1] was the first to study event time modeling and its statistical inference through the counting process framework.

The nature of the data to be considered here, however, contains important issues that differentiate the single event from the recurrent events setting. Let us suppose we have n independent subjects in the study [subjects could be people, air conditioning machines, car engines, etc.]. Assume also that each subject is observed until some random time. As mentioned earlier, the frequency of event occurrences is a random variable (r.v.) whose distribution depends on the interocurrence times distribution. Moreover, the number of events during the observation period is also informative concerning the distribution of the event time. Furthermore, the last observation for each subject is always right-censored and depends on the observation window and previous interoccurrence times for the given subject [cf. Peña et al. [30]]. As a consequence, censoring is dependent and informative in this recurrent event setting. As mentioned earlier, the KM estimator, in a single event setting, has been shown to be

an asymptotically efficient estimator of the distribution of the gap-time if no additional assumptions are made on the data [cf. Gill and van der Vaart [40]]. However, this efficiency is lost if informative censoring is present. For instance, in the single event setting, Cheng and Lin [12] have shown that their proposed ML type estimator is more efficient than the classical KM estimator under a proportional hazards model and gave a bound for the ARE of their estimator with respect to the KM estimator. With recurrent event data, not much has been done in terms of efficiency comparison and, we believe, it is important to account for this informative censoring so that possible loss of statistical efficiency can be limited. The question now is how do we account for informative censoring? Many models of informative censoring have been proposed in the literature. Link [47] proposed a model where the event of censoring is related to the frailty of the individual, and showed, in the case of exponential frailty, that the KM estimator can produce substantial error in the estimation of survival probability. Wang et al. [45] proposed various models where the occurrence of recurrent events are modeled by a subject-specific, non-stationary Poisson process via a latent variable. Siannis [38] considered a parametric model where the parameter represents the level of dependence between the failure and the censoring process. Another model for informative censoring well-known in survival analysis literature is the Koziol-Green model [cf. Koziol Green [27]]. In their 1976 paper, Koziol and Green assume existence of some parameter, $\beta > 0$, such that the survivor function of the censoring r.v. [which, in this research, will be the end of observation time] is some power of the survivor function of event time when right censored data [to be defined later] are available. β was interpreted as the censoring parameter, and $\frac{\beta}{\beta+1}$ is the

proportion of subjects that have been censored. Among all the models of informative censoring proposed in the literature, the Koziol-Green model is the most convenient to show, theoretically, that informative censoring could lead to loss in statistical efficiency in cases where any pattern of censoring exists and is not accounted for in estimation process. Moreover, as pointed out in Chen et al. [11], a key advantage of the model is its tractability. For instance, in Theorem 2.1 of their paper, an important independence property pertaining to the characterization of the Koziol-Green model greatly simplifies the computations of the mean and variance of the estimator of the survivor function, although this independence property does not arise in the recurrent event setting. However, it has been pointed out by many authors [cf. Csörgő and Faraway [15]; Gather and Pawlitschko [18]] that the assumption of the Koziol-Green model is also restrictive and that very few real data sets exist which fit this model. Consequently, Gather and Pawlitschko [18] proposed a generalization of the Koziol-Green model where only a part of the censored observations are supposed to be informatively censored. The Koziol-Green model, however, will be exploited as a model of informative censoring in this research and is described in details in the next section.

1.2 The Koziol-Green Model

Let $X_1, ..., X_n$ be an i.i.d. sequence of nonnegative r.v's having common continuous distribution function $F(t) = P(X_1 \leq t)$. The X_is are censored in the right by i.i.d. positive random variables $Y_1, ..., Y_n$, the so-called censoring times, which have

a common distribution function G, and are independent of the X_is. In the right-censorship model, we only observe

$$Z_i = X_i \wedge Y_i \quad \text{and} \quad \delta_i = I\{X_i \leq Y_i\}, \quad i = 1, 2, ..., n;$$

where $a \wedge b$ denotes the minimum of a and b, and $I\{\cdot\}$ the indicator function. The above model is referred to as the general random censorship model. In the random censorship model, the most commonly used nonparametric estimator of the survivor function $\bar{F}(t) = 1 - F(t)$ is the KM estimator [cf. Kaplan and Meier [25]]. If no additional assumptions are made, the KM estimator is asymptotically efficient for $\bar{F}$ [cf. Gill and van der Vaart, [40]]. However, this efficiency is lost if informative censoring is present, that is, if the survival function $\bar{G} = 1 - G$ of the censoring times carries additional information about $\bar{F}$. For a more rigorous definition of informative censoring, we refer the reader to Andersen et al. [3]. Koziol and Green [27] introduced an appealing and convenient model of informative censoring. In their 1976 paper, they assume existence of some $\beta \geq 0$ such that F and G are related via

$$\bar{G}(\cdot) = \bar{F}(\cdot)^{\beta}. \tag{1.1}$$

In Koziol and Green [27], β was interpreted as the censoring parameter and we have

$$P(X_i \leq Y_i) = \int_{\mathbb{R}} (1 - G(t))dF(t) = \int_{\mathbb{R}} (1 - F(t))^{\beta} dF(t) = \frac{1}{\beta + 1}. \tag{1.2}$$

As a consequence, from (1.2), $P(\text{a subject is censored}) = \frac{\beta}{\beta+1}$. It can be seen that as β increases, the proportion of censored observations also increases. Hereafter, we will simply call it the KG model. Csörgö [14] presents a comprehensive review of the work pertaining to the KG model. Instances of the use of this structure can be found in Chen et al. [11], Cheng and Lin [12] and Hollander and Peña [22], among others. For instance, Hollander and Peña [22] developed confidence bands for the survivor function under the KG model. Chen et al. [11] studied the exact bias and variance of the KM estimator under the KG model and compared its exact variance to the asymptotic variance. Cheng and Lin [12] proposed a ML-type estimator of the survivor function under the KG model and compared it to the classical KM estimator. They showed that, under the KG model, their proposed estimator is more efficient. We now discuss the aims of this dissertation.

1.3 Specific Aims

The major aims of this research are to derive parametric and nonparametric estimators of the gap-time distribution in the recurrent events setting that account for informative censoring provided by the KG model. The specific aims are:

1. To derive parametric and semiparametric estimators of the survivor function under the KG model and obtain their asymptotic properties. We will first assume that the survivor function depends on some p dimensional parameter $\boldsymbol{\theta}$, and will derive maximum likelihood estimators of $\boldsymbol{\theta}$ and β, and consequently an estimator of the gap-time distribution will be proposed. Next, we will as-

sume that the gap-time survival function does not have any specific form. A semiparametric estimator of the survivor function and that of the cumulative hazard will be obtained under the model in (1.1).

2. To examine the impact of an informative monitoring period. We will compare our proposed estimators to those derived ignoring informative censoring in order to ascertain any efficiency gained.

3. To investigate the special case where the inter-event times are i.i.d. exponentially distributed with constant failure rate θ; that is, the homogeneous Poisson process (HPP) case and obtain closed-form expressions for the estimators of θ and β as well as a bound for the ARE. This case will also be considered for the semiparametric estimation.

4. To conduct simulation studies in order to study small to moderate sample size properties of the proposed estimators of the gap-time distribution as well as the ARE.

Chapter 2

Mathematical Preliminaries

The theory of counting processes via the intensity process has drawn the attention of researchers for years. Bremaud [9] used the theory of square integrable martingales and stochastic integrals to study counting processes. In his 1975 dissertation from The University of California at Berkeley entitled 'Statistical Inference for a Family of Counting Processes', Aalen [1] showed that the same approach could be used to analyze data in reliability and survival analysis settings. This mathematical section is included in this dissertation to facilitate its reading. A detailed discussion of this framework can be found in Andersen et al. [3], Fleming and Harrington [16], Chung and Williams [13].

Let $\big(\Omega, \mathcal{F}, P\big)$ be a complete probability space and $T = [0, \tau] \subset \mathbb{R}$ be an interval of time. The following definitions, theorems, discussion are from Andersen et al. [3].

Definition 2.1 *A filtration* $\boldsymbol{F} = \{\mathcal{F}_t,\ t \in T\}$ *on* $\big(\Omega, \mathcal{F}, P\big)$ *is an increasing family of*

σ-algebras, that is, $\forall t \leq s, \quad \mathcal{F}_t \subseteq \mathcal{F}_s \subseteq \mathcal{F}$.

Note here that in the case of a stochastic process, $\mathcal{F}_t$ could be taken to be all information generated by the process up to time t, called the natural history of the process. Obviously, any process is adapted to its history. From now on, we will denote by $\mathbf{F}$ the natural filtration associated with the probability space $\big(\Omega, \mathcal{F}, P\big)$.

Definition 2.2 *A stochastic process $\boldsymbol{X}$ is called cadlag (in French, continue à droite, limité à gauche) if its sample paths are right continuous with left hand limits for almost all w.*

Definition 2.3 *A stochastic process $\boldsymbol{X} = \{X_t, t \geq 0\}$ is:*

1. *Integrable if* $\sup_{t \in T} E(|X(t)|) < \infty$;
2. *Square integrable if* $\sup_{t \in T} E(X(t)^2) < \infty$;
3. *Bounded if there exists a finite constant Γ such that* $P\left\{\sup_{t \in T} |(X(t)| < \Gamma\right\} = 1$.

From now on, we will only consider cadlag stochastic processes.

Definition 2.4 *A collection $\boldsymbol{M} = \{M_t, t \geq 0\}$ is an $\boldsymbol{F}$- martingale if $\boldsymbol{M}$ is $\boldsymbol{F}$-adapted and*

1. $E(|M_t|) < \infty$ *for all* $t \in T$;
2. $E(M_t|\mathcal{F}_s) = M_s$ *a.s* $\quad \forall s < t$.

In the previous definition, if we change the relation in 2) from $=$ to $\geq$, we get a submartingale. On the other hand, if we change it to $\leq$, one gets a supermartingale.

In this research, many of the processes have properties that hold for local martingales which we define as follows.

Definition 2.5 *(Local Martingale)*

1. *A stochastic process* $\boldsymbol{M} = \{M_t, t \geq 0\}$ *is a local martingale with respect to a filtration* $\boldsymbol{F}$ *if there exists a sequence* $\{\tau_n\}$, $n \in \mathbb{N}$, *of stopping times such that, for each* n, $M_n = \{\boldsymbol{M}(t \wedge \tau_n) : 0 \leq t < \infty\}$ *is an* $\boldsymbol{F}$*-martingale.*

2. *If* M_n *above is a martingale and is a square integrable process,* M_n *is called a square integrable martingale and* $\boldsymbol{M}$ *is called a local square integrable martingale.*

We now discuss the notion of a predictable process. Predictability will arise when dealing with integrals with respect to a martingale process. Informally, a predictable process $\mathbf{X}$, is one whose behavior at time t is determined by the information on $[0, t)$. We now formally define a predictable process.

Definition 2.6 *The* σ*-algebra generated by all the sets of the form:*

1. $[0] \times A, \quad A \in \mathcal{F}_0$, *and*

2. $(a, b] \times A$, $0 \leq a < b < \infty$, $A \in \mathcal{F}_a$

is called the predictable σ*-algebra for* $\boldsymbol{F}$*, where* $\mathcal{F}_0$ *is the information at time* 0.

A predictable process is a process that is measurable with respect to the predictable σ-algebra. The next lemma characterizes a predictable process.

Lemma 2.1 *Let* $\boldsymbol{F}$ *be a filtration, and* $\boldsymbol{X}$ *a left-continuous real-valued process adapted to* $\boldsymbol{F}$*. Then* $\boldsymbol{X}$ *is predictable.*

Proposition 2.1 *Let $\boldsymbol{X}$ be an $\mathcal{F}_t$-predictable process. Then, for any $t > 0$, $X(t)$ is $\mathcal{F}_{t-}$-measurable.*

We will continue our discussion of martingales by the notion of martingale transform and quadratic variation. We first provide an important theorem for decomposition of a sub-martingale.

Theorem 2.1 ***(Doob-Meyer Decomposition, Fleming and Harrington [16])*** *Let $\boldsymbol{M} = \{M_t : t \geq 0\}$ be a right-continuous, nonnegative submartingale with respect to the filtration $\boldsymbol{F}$. Then there exists a right-continuous martingale $\widetilde{M(t)}$ and an increasing right-continuous predictable process $A(t)$ such that $M(t) = \widetilde{M(t)} + A(t)$ a.s.*

Note that if M is a martingale with $E(M^2(t)) < \infty$ for $t > 0$, Jensen's inequality implies that $M^2(t)$ is a submartingale, and the corresponding right-continuous predictable process $A(t)$ of $M^2(t)$ is called the *compensator* of $M^2(t)$. The compensator has a special notation given in the next corollary.

Corollary 2.1 *Let $\boldsymbol{M}$ be a cadlag martingale with respect to $\boldsymbol{F}$. Then, there exists a unique increasing right-continuous predictable process denoted by $\langle \boldsymbol{M}, \boldsymbol{M} \rangle(t)$ called the predictable quadratic variation process of $\boldsymbol{M}$, such that $\langle \boldsymbol{M}, \boldsymbol{M} \rangle(0) = 0$ a.s, $E\langle \boldsymbol{M}, \boldsymbol{M} \rangle(t) < \infty$ for all t and $\{\boldsymbol{M}^2(t) - \langle \boldsymbol{M}, \boldsymbol{M} \rangle(t) : t \geq 0\}$ is a right-continuous martingale.*

Stochastic integration is the subject of the next topic. A complete discussion of stochastic integration can be found in Chung and Williams [13].

Let M be a martingale with bounded variation and H a bounded predictable process. Define $t \mapsto M'(t) = \int_0^t H(w)dM(w)$.

M' exists as a Lebesgue-Stieljes integral for all paths of H and M and it can be shown that M' is a martingale. In the counting process framework, we are interested in the stochastic integral of a predictable process with respect to a local square integrable martingale. The next theorem is on the predictable variation of $M'(t)$.

Theorem 2.2 *Suppose M is a finite variation local square integrable martingale, H is a predictable process and $\int_0^t H^2 d\langle M\rangle$ is locally integrable. Then, $\int_0^t HdM$ is a local square integrable martingale and its quadratic variation process is given by*

$$\left\langle \int HdM \right\rangle (t) = \int_0^t H^2 d\langle M\rangle.$$

The above theorem can be further generalized to a vector of martingales **M** and **M'** and matrices **H** and **K** of predictable processes. In that case, the predictable covariation process is given by

$$\left\langle \int \mathbf{H}d\mathbf{M}, \int \mathbf{K}d\mathbf{M}' \right\rangle = \int_0^t \mathbf{H}d\langle \mathbf{M}, \mathbf{M}'\rangle \mathbf{K}^t; \tag{2.1}$$

where A^t denote the transpose of a matrix A. We will continue our discussion with the theory of counting processes. A multivariate counting process is a stochastic process which can be thought of as counting the occurrences [as time proceeds] of k different types of events.

Definition 2.7 *Suppose a filtration $\boldsymbol{F}$ on $(\Omega, \mathcal{F}, P)$ is given. A multivariate counting*

process $\boldsymbol{N} = (N_1, \ldots, N_k)$ *is a vector of* k $\boldsymbol{F}$*-adapted cadlag processes for which:*

1. $N_i(0) = 0 \quad \forall i = 1, 2, \ldots, k;$

2. *Their jumps are of size one and no two components can jump at the same time;*

3. *Their paths are nondecreasing and piecewise constant.*

Note that because the components of a counting process $\mathbf{N}$ are adapted, cadlag, locally bounded and nondecreasing, they are local submartingales. So, by the Doob-Meier decomposition, there exists a compensator of N_i, say Λ_i. Λ_i is referred to as the cumulative intensity process of the counting process N_i. We will be interested in dealing with the case where Λ_i is absolutely continuous; that is, there exists an intensity process λ_i such that $\Lambda_i(t) = \int_0^t \lambda_i(s)ds$, for all $t \in T$. Hence, the Λ_is are nondecreasing predictable processes such that $\Lambda_i(0) = 0$ for all t and $M_i \equiv N_i - \Lambda_i$ is a local martingale for all i.

The next proposition, taken from Andersen et al. [3], connects stochastic integration, martingales and counting processes.

Proposition 2.2 *Let* $\boldsymbol{N}$ *be a multivariate counting process and let* $\boldsymbol{\Lambda} = \int \lambda$ *be its associated vector of compensator processes such that each component of* $\boldsymbol{\Lambda}$ *is absolutely continuous. Let* $\boldsymbol{M} = \boldsymbol{N} - \boldsymbol{\Lambda}$ *be the resulting vector of local martingales. If* $\boldsymbol{H}$ *is a vector of locally bounded and predictable processes, then* $\int \boldsymbol{H} d\boldsymbol{M}$ *are vectors of local square integrable martingales with a quadratic variation process given by*

$$\left\langle \int \boldsymbol{H} d\boldsymbol{M} \right\rangle = \int \boldsymbol{H} diag\{\lambda\} \boldsymbol{H}^t ds;$$

where diag($\boldsymbol{\lambda}$) is the diagonal matrix of associated intensity processes.

We will now turn our attention to the construction of likelihood based on realization of counting processes. When such data are available, the likelihood function can be written in a product integral form, which is just a continuous version of the simple product, $\prod$. It is also used in the representation of the survivor function in terms of its cumulative hazard. The idea of constructing likelihood with intensity process data is due to Jacod [24]. Following Jacod's formula, let $\triangle N_i(t) = N_i(t) - N_i(t-)$ be the jump process, and let our intensity process depend on some parameter θ which could be a p-dimensional parameter. When dealing with a parametric model, the hazard rate, at time t, is a function of θ, say $\lambda_i(t, \theta)$. The likelihood in $[0, t]$ can be written as

$$L(\theta, t) = \prod_{i=1}^{n} \prod_{w \in [0,t]} \Big\{ \lambda_i(w, \theta)^{\triangle N_i(w)} \times (1 - \lambda_i(w, \theta))^{1 - \triangle N_i(w)} \Big\}; \tag{2.2}$$

where $N_i(t)$ is the counting process for each subject i in the study. The likelihood in (2.2) could be further simplified by using Taylor expansion. Observe that $1 - \lambda_i(w, \theta)dw \simeq \exp(-\lambda_i(w, \theta))dw$. This yields the likelihood function:

$$L(\theta, t) \propto \prod_{i=1}^{n} \left[\prod_{w \in [0,t]} \Big\{ \lambda_i(w, \theta)^{\triangle N_i(w)} \Big\} \times \exp\Big\{ - \int_0^t \lambda_i(w, \theta) dw \Big\} \right]. \tag{2.3}$$

From (2.3), we obtain the log-likelihood process by

$$l(\theta, t) = \sum_{i=1}^{n} \left\{ \int_0^t \log\{\lambda_i(w, \theta)\} dN_i(w) - \int_0^t \lambda_i(w, \theta) dw \right\}. \tag{2.4}$$

Thus, the score process $U_\theta(\theta, t)$ is obtained by taking the derivative of $l(\theta, t)$ with

respect to θ

$$\begin{aligned} U_\theta(\theta,t) &= \sum_{i=1}^{n}\left\{\int_0^t \frac{\partial}{\partial\theta}\log\{\lambda(w,\theta)\}dN_i(w) - \int_0^t \frac{\partial}{\partial\theta}\lambda_i(w,\theta)dw\right\} \\ &= \sum_{i=1}^{n}\left\{\int_0^t \frac{\partial}{\partial\theta}\log\{\lambda_i(w,\theta)\}dM_i(t)\right\}. \end{aligned} \tag{2.5}$$

We will now give a result that will play a major role in obtaining asymptotic properties of estimators. The Rebolledo's [34] martingale central limit theorem is utilized in obtaining weak convergence results of properly standardized estimators. Let $D[0,t]^k$ be the space of $\mathbb{R}^k$ -valued cadlag functions on T endowed with the Skorohod topology [for a detailed discussion of Skorohod topology, cf. Billingsley [7], Shorack and Wellner [37]]. The theorem is given below and can be found in Andersen et al. [3].

Theorem 2.3 ***(Rebolledo's Martingale Central Limit Theorem)***
For each $n = 1, 2, ...,$ let $\boldsymbol{M}^{(n)} = (M_1^n, M_2^n, ..., M_k^n)$ be vectors of local square-integrable martingales where each may be defined on different samples spaces with respect to different filtration. For $\epsilon > 0$, let $\boldsymbol{M}_\epsilon^{(n)}$ be a vector of local square integrable martingales such that $|M_h^n - M_{\epsilon h}^{(n)}|$ is a local square integrable martingale and $|\Delta M_h^{(n)} - \Delta M_{\epsilon h}^{(n)}| \leq \epsilon$. Let $\langle \boldsymbol{M}^{(n)}\rangle$, $n = 1, 2, ...$ be the $k \times k$ matrix of processes with elements $\langle M_h^{(n)}, M_{h'}^{(n)}\rangle$. Assume the following conditions for $T_0 \subseteq T$:

1. *There exists a matrix of deterministic functions $\boldsymbol{V}(t)$ such that $\langle \boldsymbol{M}^{(n)}\rangle(t) \xrightarrow{p} \boldsymbol{V}(t), \ \forall t \in T_0, \quad$ as $n \to \infty$;*

2. *$\langle M_{\epsilon h}^{(n)}(t)\rangle \xrightarrow{p} 0, \quad \forall t \in T_0, \quad h$ and $\epsilon > 0 \quad$ as $n \to \infty$.*

Then,

$$\left(M^{(n)}(t_1), \ldots, M^{(n)}(t_k)\right) \xrightarrow{d} \left(M^{(\infty)}(t_1), \ldots, M^{(\infty)}(t_k)\right) \quad \forall t_1, \ldots, t_k \in T_0.$$

Moreover, if T_0 is dense in T and contains τ if $\tau \in T$, then the same conditions imply that

$$\boldsymbol{M}^{(n)} \xrightarrow{d} \boldsymbol{M}^{(\infty)} \quad in \quad D(T)^k \quad as\ n \to \infty,$$

where $\boldsymbol{M}^{(\infty)}$ is a vector of continuous Gaussian martingales.

We will now discuss the Glivenko-Cantelli theorem. The Glivenko-Cantelli theorem says that the empirical distribution function is a uniform approximation of the true distribution function [cf. Shorack and Wellner [37]].

Let $\{X_n, n \geq 1\}$ be i.i.d random variables with common distribution function F. Assume F is unknown, and, on the basis of a sample $X_1, ..., X_n$, we seek to estimate F. The nonparametric estimator is the empirical distribution function and is defined by

$$\hat{F}_n(x, w) = \frac{1}{n} \sum_{i=1}^{n} I_{[X_i(w) \leq x]}.$$

By the Strong Law of Large Numbers, we get for each fixed x, $\hat{F}_n(x) \xrightarrow{a.s.} F(x)$.

Theorem 2.4 *(Glivenko-Cantelli) Define*

$$D_n := \sup_x |\hat{F}_n(x) - F(x)|.$$

Then, $D_n \xrightarrow{a.s.} 0$ as $n \to \infty$; which gives a stronger result.

Another concept to be discussed in this chapter is the notion of weak-convergence in metric spaces along with the functional delta -method. Weak-convergence [i.e convergence of probability measures] of stochastic processes generalizes convergence in distribution of real-valued random variables. The following discussions are from Billinglsey [7]. We start with a definition of weak-convergence.

Definition 2.8 *Let F_n, $n = 1, 2, ...,$ and F be distribution functions on the real line. We say that F_n* converges weakly *to F as $n \to \infty$ (written $F_n \Rightarrow F$) if $F_n(x) \to F(x)$ at all points of continuity of F.*

The next definition pertains to the weak-convergence of a sequence of probability measures.

Definition 2.9 *Suppose $\mathcal{B}$ is the Borel σ-algebra of $\mathbb{R}$ and let P_n and P be probability measures on $(\mathbb{R}, \mathcal{B})$. Let ∂B be the boundary of any set $B \in \mathcal{B}$. We say that P_n* converges weakly *to P as $n \to \infty$ if $P_n(B) \to P(B)$ for each $B \in \mathcal{B}$ for which $P(\partial B) = 0$. We write $P_n \Rightarrow P$.*

The next lemma characterizes weak-convergence of distribution functions associated with probability measures.

Lemma 2.2 *Let P and $\{P_n\}$ be probability measures on $(\mathbb{R}, \mathcal{B})$ corresponding to distribution functions F and F_n. Then $F_n \Rightarrow F$ if $P_n \Rightarrow P$.*

We now give a more precise statement of weak-convergence in the space of continuous functions. Let Γ be a metric space equipped with the uniform metric, defining

the distance between two points x and y [that is functions] of Γ as

$$\rho(x, y) = \|x - y\| = \sup_t |x(t) - y(t)|.$$

The above distance can be used to define the open sets and hence the σ-algebra generated by the open sets. Let $\mathcal{S}$ be the smallest σ-algebra generated by the open sets, called the Borel σ-algebra. Before we give the statement of weak-convergence on metric spaces, we need the notion of relative compactness, which is given next.

Definition 2.10 *Let* Π *be a family of probability measures on* $(\Gamma, \mathcal{S})$. Π *is* relatively compact *if every sequence of elements of* Π *contains a weakly convergent subsequence.*

And now, the weak-convergence on the metric space $(C[0, t], \mathcal{S})$, where $C[0, t]$ is the space of continuous functions on $[0, t]$ equipped with the uniform topology.

Theorem 2.5 *Suppose* P_n *and* P *are probability measures on* $(C[0, t], \mathcal{S})$. *If the finite-dimensional distributions of* P_n *converge weakly to those of* P *and if* $\{P_n\}$ *is relatively compact, then* $P_n \Rightarrow P$.

A very important result in weak-convergence theory is *The Continuous Mapping Theorem* which extends weak-convergence of a sequence of measures from one metric space $(\Gamma, \mathcal{S})$ into another metric space $(\Gamma', \mathcal{S}')$.

Theorem 2.6 ***The Continuous Mapping Theorem:***
Suppose f *is a continuous mapping from one metric space* $(\Gamma, \mathcal{S})$ *into another* $(\Gamma', \mathcal{S}')$. *If* $P_n \Rightarrow P$ *on* Γ, *then* $P_n f^{-1} \Rightarrow P f^{-1}$ *on* Γ'.

We now give the functional version of the delta-method. To do so, we will need the concept of differentiability on a metric space equipped with the topology described above. Details of the discussion of this notion can be found in Andersen et al. [3].

Definition 2.11 *Let Γ and Γ' be two metric spaces. A function $\phi : \Gamma \to \Gamma'$ is compactly or Hadamard differentiable at a point $\theta \in \Gamma$ if and only if a continuous, linear map $d\phi(\theta) : \Gamma \to \Gamma'$ exists (called the derivative of ϕ at point θ) such that for all real sequences $a_n \to \infty$ and all convergent sequences $h_n \to h \in \Gamma$,*

$$a_n\{\phi(\theta + a_n^{-1}h_n) - \phi(\theta)\} \to d\phi(\theta).h \quad as \quad n \to \infty.$$

This allows us to give the functional delta-method.

Theorem 2.7 *(Functional Delta-Method, Andersen et al. [3])*
Let $\{T_n\}$ be a sequence of random elements of Γ, $a_n \to \infty$ a real sequence, such that $a_n(T_n - \theta) \xrightarrow{d} Z$, for some fixed point θ in Γ and a random element Z of Γ. Suppose $\phi : \Gamma \to \Gamma'$ is compactly differentiable at θ. Then

$$a_n(\phi(T_n) - \phi(\theta)) \xrightarrow{d} d\phi(\theta).Z,$$

and, moreover,

$$a_n\{\phi(T_n) - \phi(\theta)\} \quad and \quad d\phi(\theta).a_n(T_n - \theta)$$

are asymptotically equivalent in the sense that they have asymptotically the same distribution.

The last concept to be discussed in the Mathematical Preliminaries chapter is the

notion of concave functions and convergence that follows from it. The next theorem pertains to uniform convergence of random concave functions to a concave function on a compact subset of an open convex subset of $\mathbb{R}^p$.[cf. Andersen and Gill [2]]

Theorem 2.8 *Let E be an open convex subset of $\mathbb{R}^p$ and let F_1, F_2, ..., be a sequence of random concave functions on E such that for all $x \in E$, $F_n(x) \xrightarrow{p} f(x)$ as $n \to \infty$, where f is some real function on E. Then, f is concave and for all compact $A \subset E$,*

$$\sup_{x \in A} |F_n(x) - f(x)| \xrightarrow{p} 0 \quad as \quad n \to \infty.$$

The next corollary deals with the convergence of maximums of a sequence of random concave functions to the maximum of the limiting concave function.

Corollary 2.2 *Under the same conditions as in Theorem 2.8, suppose f has a unique maximum at $\hat{x} \in E$. Let $\hat{X}_n$ maximize F_n. Then $\hat{X}_n \xrightarrow{p} \hat{x}$ as $n \to \infty$.*

Chapter 3

Maximum Likelihood Estimation

3.1 Stochastic Process Formulation and MLE

3.1.1 Introduction

Denote by $(\Omega, \mathcal{F}, P)$ the common probability space on which all the random variables are defined. Suppose there are n independent subjects in the study. For the ith subject, let $\{S_{ij}, j = 1, 2, ..., \}$ be the calendar time of event occurrences and $\{T_{ij},\ j = 1, 2, \ldots\}$ be the successive interoccurence times of the recurrent event of interest. We have $S_{i0} = 0$, $S_{ij} = \Sigma_{k=1}^{j} T_{ik}$ and $T_{ij} = S_{ij} - S_{i,j-1}$. These $T_{ij}s$ are assumed to be i.i.d nonnegative r.v.'s with a common absolutely continuous distribution function $F(t) = P(T_{ij} \leq t)$. The associated renewal function is $\rho(t) = \Sigma_{n=0}^{\infty} F^{\star(n)}(t)$ for $t \geq 0$, where $F^{\star(n)}$ is the nth convolution of F, the distribution of S_{in}. Suppose $F(t) = F(t, \boldsymbol{\theta})$ where $\boldsymbol{\theta} \in \Theta \subset \mathbb{R}^p$, and the hazard rate is $\lambda_F(t, \boldsymbol{\theta}) = \frac{f(t,\boldsymbol{\theta})}{1-F(t,\boldsymbol{\theta})} = \frac{f(t,\boldsymbol{\theta})}{\bar{F}(t,\boldsymbol{\theta})}$. Suppose that for subject i, $i = 1, 2, \ldots, n$, the recurrent event is observed over a random period $[0, \tau_i]$,

where the $\tau_i s$ are right-censoring r.v's., i.i.d according to some distribution function G. The τ_is may, for example, represent an administrative constraint. Furthermore, we assume that τ_i and T_{ij} are mutually independent and that the probability of observing a finite number of events in $[0, \tau_i]$ is one; that is $P(K_i < \infty) = 1$, where for each i, $K_i = \max\{k \in \mathbb{N} : S_{ik} \leq \tau_i\}$, the total number of event occurrences for each subject. Hence, the observed data for the ith subject are

$$(K_i, \tau_i, T_{i1}, \ldots, T_{iK_i}, \tau_i - S_{iK_i}). \tag{3.1}$$

Note further that $\tau_i - S_{iK_i}$ is the right-censoring variable for T_{i,K_i+1}. To model informative censoring, we assume the KG model; that is, there exists $\beta \geq 0$ such that $\bar{G}(t, \boldsymbol{\theta}) = \bar{F}(t, \boldsymbol{\theta})^{\beta}$. Note also that this relationship is equivalent to $\Lambda_G(\cdot) = \beta\Lambda_F(\cdot)$, where $\Lambda_F(\cdot) = \int_0^{\cdot} \lambda_F(w)dw$ is the cumulative hazard function of F. In survival analysis, this model is also referred to as the proportional hazards model. Assuming the data takes the form in (3.1), we will derive the ML estimators of $\boldsymbol{\theta}$ and β, and consequently, the estimator of the gap-time distribution F.

3.1.2 Development of the Likelihood Process and MLE

In this section, we will derive the likelihood, score process, and obtain the ML estimators of the parameters of interest. To this end, for $i = 1, 2, \ldots, n$, and $j = 1, 2, \ldots$, define, for $s \in \mathbb{R}$,

$$N_i^{\dagger}(s) = \sum_{j=1}^{\infty} I\{S_{ij} \leq s \wedge \tau_i\}; \quad Y_i^{\dagger}(s) = I\{\tau_i \geq s\} \quad \text{and} \quad N_i^{\tau}(s) = I\{\tau_i \leq s\}. \tag{3.2}$$

Note that the processes in (3.2) count the number of failures for subject i; indicate whether subject i is still under observation at calendar time s; and if censoring time has already occurred, respectively. We will now further augment our probability space by the natural filtration $\mathbf{F} = \{\mathcal{F}_s : s \geq 0\}$, where $\mathcal{F}_s$ is given by

$$\mathcal{F}_s = \mathcal{F}_0 \vee \left\{ \bigvee_{i=1}^{n} \sigma\{N_i^\dagger(w), Y_i^\dagger(w+), 0 \leq w \leq s\} \right\};$$

where $\mathcal{F}_0$ represents the σ-field containing all information available at time 0 and $\mathcal{F}_s$, the σ-field containing all information generated on all subjects up to time s.

We will now introduce a process that will give the connection between gap-time and calendar time. For this, we define the process

$$R_i(w) = w - S_{iN_i^\dagger(w-)} \quad \text{for} \quad i = 1, 2, ..., n.$$

The above process indicates the length of time that has elapsed since the last event occurrence for each i, also called the backward recurrence time. Note further that this process is $\mathbf{F}$-adapted and left-continuous, and hence $\mathbf{F}$-predictable. Let us introduce some more notation. At calendar time s, let

$$A_i^\dagger(s; \boldsymbol{\theta}) = \int_0^s Y_i^\dagger(w) \lambda_F(R_i(w), \boldsymbol{\theta}) dw \quad \text{and} \quad A_i^\tau(s; \boldsymbol{\theta}, \beta) = \int_0^s Y_i^\dagger(w) \beta \lambda_F(w, \boldsymbol{\theta}) dw. \tag{3.3}$$

Following the Mathematical Preliminaries discussion, $A_i^\dagger(s; \boldsymbol{\theta})$ and $A_i^\tau(s; \boldsymbol{\theta}, \beta)$ are,

respectively, the compensators of $N_i^\dagger(s)$ and $N_i^\tau(s)$, and as a consequence

$$M_i^\dagger(s;\boldsymbol{\theta}) = N_i^\dagger(s) - A_i^\dagger(s;\boldsymbol{\theta}) \quad \text{and} \quad M_i^\tau(s;\boldsymbol{\theta},\beta) = N_i^\tau(s) - A_i^\tau(s;\boldsymbol{\theta},\beta) \tag{3.4}$$

are, for each i, square integrable martingales with respect to the filtration $\mathbf{F}$.

With all the processes and notations needed for estimation in place, we are ready to construct the likelihood function on which estimation of the parameters of interest; that is, $\boldsymbol{\theta}$ and β, will be based. Noting that we have two simultaneous processes, the process counting the event occurrences and the other indicating whether subject i has been censored; from (2.2), the full likelihood process is

$$\begin{aligned}
L(s;\boldsymbol{\theta},\beta) &= \prod_{i=1}^{n}\Big\{\{\prod_{w=0}^{s}[dA_i^\tau(w;\boldsymbol{\theta},\beta)]^{\triangle N_i^\tau(w)} \times [1-dA_i^\tau(w;\boldsymbol{\theta},\beta)]^{1-\triangle N_i^\tau(w)}\} \\
&\quad \times [dA_i^\dagger(w,\boldsymbol{\theta})]^{\triangle N_i^\dagger(w)} \times \{[1-dA_i^\dagger(w,\boldsymbol{\theta})]^{1-\triangle N_i^\dagger(w)}\}\Big\} \\
&= \prod_{i=1}^{n}\Big\{\prod_{w=0}^{s}[Y_i^\dagger(w)\beta\lambda_F(w,\boldsymbol{\theta})dw]^{\triangle N_i^\tau(w)}[1-Y_i^\dagger(w)\beta\lambda_F(w,\boldsymbol{\theta})dw]^{1-\triangle N_i^\tau(w)} \\
&\quad \times [Y_i^\dagger(w)\lambda_F(R_i(w),\boldsymbol{\theta})dw]^{\triangle N_i^\dagger(w)}[1-Y_i^\dagger(w)\lambda_F(R_i(w),\boldsymbol{\theta})dw]^{1-\triangle N_i^\dagger(w)}\Big\} \\
&= \prod_{i=1}^{n}\prod_{w=0}^{s}\Big\{Y_i^\dagger(w)^{\triangle N_i^\dagger(w)+\triangle N_i^\tau(w)}\lambda_F(R_i(w),\boldsymbol{\theta})^{\triangle N_i^\dagger(w)}\{\beta\lambda_F(w,\boldsymbol{\theta})\}^{\triangle N_i^\tau(w)}\Big\} \\
&\quad \times \exp\left\{-\sum_{i=1}^{n}\int_0^s Y_i^\dagger(w)[\lambda_F(R_i(w),\boldsymbol{\theta})+\beta\lambda_F(w,\boldsymbol{\theta})]dw\right\}.
\end{aligned} \tag{3.5}$$

By taking the logarithm of the likelihood process, we obtain the log-likelihood process

$$\begin{aligned}
l(s;\boldsymbol{\theta},\beta) &= \sum_{i=1}^{n}\left\{\int_0^s \log(Y_i^\dagger(w)\beta\lambda_F(w,\boldsymbol{\theta}))dN_i^\tau(w) + \int_0^s \log(Y_i^\dagger(w)\lambda_F(R_i(w),\boldsymbol{\theta}))dN_i^\dagger(w)\right\} \\
&\quad -\sum_{i=1}^{n}\left\{\int_0^s Y_i^\dagger(w)\beta\lambda_F(w,\boldsymbol{\theta})dw + \int_0^s Y_i^\dagger(w)\lambda_F(R_i(w),\boldsymbol{\theta})dw\right\}.
\end{aligned} \tag{3.6}$$

For convenience, for a vector $\mathbf{a}$, let $\mathbf{a}^0 = 1$, $\mathbf{a}^{\otimes 1} = a$, $\mathbf{a}^{\otimes 2} = \mathbf{a}\mathbf{a}^t$ and $\nabla_{\theta_j} = \frac{\partial}{\partial \theta_j}$, $\quad j = 1, 2, ..., p$. Hence,

$$(\nabla_{\theta_1} \log\{\lambda_F(w, \boldsymbol{\theta})\}, ..., \nabla_{\theta_p} \log\{\lambda_F(w, \boldsymbol{\theta})\})^t = \boldsymbol{\nabla}_{\boldsymbol{\theta}} \log \lambda_F(w, \boldsymbol{\theta}) \equiv \varphi(w, \boldsymbol{\theta}). \tag{3.7}$$

Now, using the above notation, and assuming that differentiation with respect to $\boldsymbol{\theta}$ and β and the integration operation may be interchanged, the following score processes could be obtained, at calendar time s for $\boldsymbol{\theta}$ and β, by taking the derivative of (3.6) with respect to $\boldsymbol{\theta}$ and β, respectively:

$$\begin{aligned}
U_{\boldsymbol{\theta}}(s; \boldsymbol{\theta}, \beta) &= \sum_{i=1}^{n} \Bigg\{ \int_0^s \varphi(w, \boldsymbol{\theta}) dN_i^{\tau}(w) + \int_0^s \varphi(R_i(w), \boldsymbol{\theta}) dN_i^{\dagger}(w) \\
&\quad - \int_0^s Y_i^{\dagger}(w) \{\varphi(w, \boldsymbol{\theta}) \beta \lambda_F(w, \boldsymbol{\theta}) + \varphi(R_i(w), \boldsymbol{\theta}) \lambda_F(R_i(w), \boldsymbol{\theta})\} dw \Bigg\} \\
&= \sum_{i=1}^{n} \Bigg\{ \int_0^s \varphi(w, \boldsymbol{\theta}) \{dN_i^{\tau}(w) - Y_i^{\dagger}(w) \lambda_G(w, \boldsymbol{\theta})\} dw \\
&\quad + \int_0^s \varphi(R_i(w), \boldsymbol{\theta}) \{dN_i^{\dagger}(w) - Y_i^{\dagger}(w) \lambda_F(R_i(w), \boldsymbol{\theta})\} dw \Bigg\} \\
&= \sum_{i=1}^{n} \Bigg\{ \int_0^s \varphi(w, \boldsymbol{\theta}) M_i^{\tau}(dw; \boldsymbol{\theta}, \beta) + \int_0^s \varphi(R_i(w), \boldsymbol{\theta}) M_i^{\dagger}(dw, \boldsymbol{\theta}) \Bigg\}; \quad (3.8)
\end{aligned}$$

$$\begin{aligned}
U_{\beta}(s; \boldsymbol{\theta}, \beta) &= \sum_{i=1}^{n} \int_0^s \frac{1}{\beta} dN_i^{\tau}(w) - \sum_{i=1}^{n} \int_0^s Y_i^{\dagger}(w) \lambda_F(w, \boldsymbol{\theta}) dw \\
&= \sum_{i=1}^{n} \int_0^s \frac{1}{\beta} \{dN_i^{\tau}(w) - Y_i^{\dagger}(w) \beta \lambda_F(w, \boldsymbol{\theta})\} dw \\
&= \sum_{i=1}^{n} \int_0^s \frac{1}{\beta} M_i^{\tau}(dw; \boldsymbol{\theta}, \beta). \qquad (3.9)
\end{aligned}$$

Note that in the above expressions of $U_{\boldsymbol{\theta}}(s;\boldsymbol{\theta},\beta)$ and $U_{\beta}(s;\boldsymbol{\theta},\beta)$, we have used the facts that $dM_i^{\tau}(w;\boldsymbol{\theta},\beta) = dN_i^{\tau}(w) - Y_i^{\dagger}(w)\lambda_G(w,\boldsymbol{\theta})dw$ and $dM_i^{\dagger}(w;\boldsymbol{\theta},\beta) = dN_i^{\dagger}(w) - Y_i^{\dagger}(w)\lambda_F(w,\boldsymbol{\theta})dw$. For brevity of notation, let

$$\mathbf{H}_i(w;\boldsymbol{\theta},\beta) = \begin{pmatrix} \varphi(w,\boldsymbol{\theta}) & \varphi(R_i(w),\boldsymbol{\theta}) \\ \frac{1}{\beta} & 0 \end{pmatrix};$$

$$\mathbf{M}_i(w;\boldsymbol{\theta},\beta) = \Big(M_i^{\tau}(w;\boldsymbol{\theta},\beta), M_i^{\dagger}(w;\boldsymbol{\theta},\beta)\Big)^t; \tag{3.10}$$

and

$$\mathbf{U}(w;\boldsymbol{\theta},\beta) = \Big(U_{\boldsymbol{\theta}}(w;\boldsymbol{\theta},\beta)^t, U_{\beta}(w;\boldsymbol{\theta},\beta)^t\Big)^t. \tag{3.11}$$

Therefore, (3.8), (3.9), (3.10) and (3.11) all together yield

$$\mathbf{U}(s;\boldsymbol{\theta},\beta) = \sum_{i=1}^{n} \int_0^s \mathbf{H}_i(w;\boldsymbol{\theta},\beta) d\mathbf{M}_i(w;\boldsymbol{\theta},\beta). \tag{3.12}$$

Next, let $s^{\star} \in \mathbb{R}$ with $s^{\star} \geq \max_{1\leq i\leq n} \tau_i$. By setting the score processes in (3.12), evaluated at $s^{\star}$, equal to $\mathbf{0}$; that is, $\mathbf{U}(s^{\star};\boldsymbol{\theta},\beta) = \mathbf{0}$ and solving for $\boldsymbol{\theta}$ and β, we obtain the ML estimators, $\hat{\boldsymbol{\theta}}$ and $\hat{\beta}$ of $\boldsymbol{\theta}$ and β, respectively. Unfortunately, closed-form expressions for the ML estimators may not be possible to obtain. However, numerical methods such as the Newton-Raphson (NR) algorithm or Nelder-Mead simplex algorithm could be used to obtain the maximum likelihood estimators $\hat{\boldsymbol{\theta}}$ and $\hat{\beta}$. For convenience, with $\boldsymbol{\delta} = (\boldsymbol{\theta},\beta)^t$, then the observed Fisher information could be obtained by taking the partial derivative of $\mathbf{U}(s^{\star};\boldsymbol{\delta})$ with respect to the transpose of

$\boldsymbol{\delta}$ by

$$\mathbf{I}(s^\star;\boldsymbol{\delta}) = -\nabla_{\boldsymbol{\delta}^t}\mathbf{U}(s^\star;\boldsymbol{\delta}) = -\nabla_{\boldsymbol{\delta}^t}\nabla_{\boldsymbol{\delta}} l(s^\star;\boldsymbol{\delta})$$

when $\mathbf{I}(s^\star;\boldsymbol{\delta})$ is invertible. The NR algorithm uses the iteration

$$\boldsymbol{\delta}^{new} \leftarrow \boldsymbol{\delta}^{old} + \mathbf{I}(s^\star;\boldsymbol{\delta}^{old})^{-1}\mathbf{U}(s^\star;\boldsymbol{\delta}^{old}).$$

This iteration procedure will be implemented in our simulation studies.

3.2 Asymptotic Properties of the MLE

3.2.1 Preliminaries

We shall now turn our attention to the asymptotic properties of the ML estimators. Borgan [8], using Lenglart's inequality and Rebolledo's [34] martingale central limit theorem, studied consistency and asymptotic normality of ML estimators in parametric counting process models. The setting of Borgan's [8] paper applies to our situation, and subject to the regularity conditions in his paper, we obtain asymptotic properties in this recurrent events setting. We will consider the case where the number of subjects is increasing [that is $n \to \infty$] in contrast to the situation where only one subject is considered and the time of monitoring increases to infinity [that is $\tau \to \infty$] as in Sellke [36]. If stated in calendar time as in Borgan's [8] paper, one of the conditions in our situation would contain random arguments, the $R_i(w)$s. Consequently, verifying that condition and obtaining closed-form expressions for the limit of the covariation

process will be difficult. To overcome this difficulty, those conditions will be reformulated in terms of gap-time. In so doing, we can obtain closed-form expressions for the limit of the covariation process, thereby enabling theoretical efficiency comparison.

3.2.2 Reformulated Process

Following Sellke's [36] idea, for $(s,t) \in [0,s^\star]^2$, we define the doubly-indexed process in which the first index pertains to the calendar time and the second to the gap-time by

$$Z_i(s,t) = I\{R_i(s) \leq t\} \quad \text{for } i = 1,2,\ldots,n. \tag{3.13}$$

Note that, for fixed t, $Z_i(s,t)$ indicates whether, at calendar time s, at most t units of time have elapsed since the last event. Moreover, $Z_i(\cdot,t)$ is predictable w.r.t. $\mathbf{F}$, bounded, and has left-continuous paths. We now define the generalized likelihood process, say L_G, for each fixed t and calendar time s via

$$\begin{aligned} L_G(s,t;\boldsymbol{\delta}) &= \prod_{i=1}^{n}\Bigg[\left\{\prod_{w=0}^{s} dA_i^{\tau}(w,\boldsymbol{\delta})^{\triangle N_i^{\tau}(w)}\right\} \times \exp\left\{-\int_0^s Y_i^{\dagger}(w)\lambda_G(w;\boldsymbol{\theta},\beta)dw\right\} \\ &\times \left\{\prod_{w=0}^{s} dA_i^{\dagger}(w,\boldsymbol{\delta})^{Z_i(w,t)\triangle N_i^{\dagger}(w)}\right\} \exp\left\{-\int_0^s Z_i(w,t)Y_i^{\dagger}(w)\lambda_F(R_i(w),\boldsymbol{\delta})dw\right\}\Bigg]. \end{aligned}$$

The above generalized likelihood process, as well as it's maximizer, say $\hat{\boldsymbol{\delta}}'(s^\star,t^\star)$, for $t^\star \in [0,s^\star]$, score processes and Fisher information, all reduce to those defined in the previous section as $t \to \infty$, which correspond to $Z_i(s,\infty) = 1$. As said earlier and pointed out in Stocker and Peña [39], regularity conditions in terms of calendar time

in Borgan [8] contain the process given by

$$S_n(s,t;\boldsymbol{\delta}) = \frac{1}{n}\sum_{i=1}^{n}\int_0^s \mathbf{H}_i(R_i(w);\boldsymbol{\delta})Z_i(w,t)Y_i^{\dagger}(w)\lambda_F(R_i(w);\boldsymbol{\delta})dw. \tag{3.14}$$

We will now reformulate the process $S_n(s,t;\boldsymbol{\delta})$ in terms of gap-times. Toward this end, we define the doubly-indexed processes for the ith unit via

$$N_i(s,t) = \int_0^s Z_i(w,t)dN_i^{\dagger}(w) \quad \text{and} \quad A_i^{\dagger}(s,t) = \int_0^s Z_i(w,t)dA_i^{\dagger}(w). \tag{3.15}$$

Then, it can be shown that $A_i^{\dagger}(s,t) = \int_0^t Y_i(s,w)\lambda_F(w)dw$ [cf. Proposition 1 of Peña et al. [32], see also Chapter 6] where

$$Y_i(s,t) = \sum_{j=1}^{N_i^{\dagger}(s-)} I\{T_{ij} \geq t\} + I\left\{(s \wedge \tau_i) - S_{iN_i^{\dagger}(s-)} \geq t\right\}.$$

The process $N_i(s,t)$ represents the number of events that occurred on or before calendar time s for the ith subject and whose inter-event times are at most t, whereas $Y_i(s,t)$ represents the number of events in $[0,s]$ whose inter-event times are at least t, plus a count on whether or not the right-censored last inter-event time is also at least t [cf. Peña et al. [30]]. Observe that both these processes have a random number of summands since the $N_i^{\dagger}(s-)$s are random. These dependencies arise because of the sum-quota accrual scheme in the recurrent events setting. The $Y_i(s,t)$s can be interpreted as the gap-time formulation of the at-risk process. The next lemma gives an expression of $S_n(s,t;\boldsymbol{\delta})$ in terms of gap-time. See also Stocker and Peña [39] who gave a general version of the lemma.

Lemma 3.1

$$S_n(s,t;\boldsymbol{\delta}) = \frac{1}{n}\sum_{i=1}^{n}\int_0^t Y_i(s,v)\lambda_F(v;\boldsymbol{\delta})\boldsymbol{H}_i(v;\boldsymbol{\delta})dv.$$

In the above Lemma, it can be seen that the hazard rate, λ_F, and the matrix of predictable processes, $\mathbf{H}_i$, both now have an argument of v instead of $R_i(w)$.

Proof: We have

$$\begin{aligned} S_n(s,t;\boldsymbol{\delta}) &= \int_0^s \mathbf{H}_i(R_i(w),\boldsymbol{\delta})I(\tau_i \geq w)\lambda_F(R_i(w),\boldsymbol{\delta})Z(w,t)dw \\ &= \int_0^{s\wedge\tau_i} \mathbf{H}_i(R_i(w),\boldsymbol{\delta})\lambda_F(R_i(w),\boldsymbol{\delta})I(R_i(w) \leq t)dw. \end{aligned}$$

Now subdividing the interval $[0,s]$, we obtain

$$\begin{aligned} &\int_0^{s\wedge\tau_i} \mathbf{H}_i(R_i(w),\boldsymbol{\delta})\lambda_F(R_i(w),\boldsymbol{\delta})I(R_i(w) \leq t)dw = \\ &\sum_{j=1}^{N_i^\dagger(s-)} \int_{S_{ij-1}}^{S_{ij}} \mathbf{H}_i(R_i(w),\boldsymbol{\delta})\lambda_F(R_i(w),\boldsymbol{\delta})I(R_i(w) \leq t)dw \\ &+ \int_{S_{iN_i^\dagger(s-)}}^{s\wedge\tau_i} \mathbf{H}_i(R_i(w),\boldsymbol{\delta})\lambda_F(R_i(w),\boldsymbol{\delta})I(R_i(w) \leq t)dw. \end{aligned}$$

By a change of variable $v := R_i(w) = w - S_{iN_i^\dagger(w-)}$ in the above equation, we obtain

$$
\begin{aligned}
&\int_0^{s\wedge\tau_i} \mathbf{H}_i(R_i(w),\boldsymbol{\delta})\lambda_F(R_i(w),\boldsymbol{\delta})I(R_i(w)\le t)dw \\
&= \sum_{j=1}^{N_i^\dagger(s-)} \int_0^{T_{ij}} \mathbf{H}_i(v,\boldsymbol{\delta})\lambda_F(v,\boldsymbol{\delta})I(v\le t)dv + \int_0^{(s\wedge\tau_i)-S_{iN_i^\dagger(s-)}} \mathbf{H}_i(v,\boldsymbol{\delta})\lambda_F(v,\boldsymbol{\delta})I(v\le t)dv \\
&= \sum_{j=1}^{N_i^\dagger(s-)} \int_0^{t} \mathbf{H}_i(v,\boldsymbol{\delta})\lambda_F(v,\boldsymbol{\delta})I(0\le v\le T_{ij})dv + \\
&\int_0^{t} \mathbf{H}_i(v,\boldsymbol{\delta})\lambda_F(v,\boldsymbol{\delta})I((s\wedge\tau_i)-S_{iN_i^\dagger(s-)}\ge v)dv \\
&= \int_0^t \left\{ \sum_{j=1}^{N_i^\dagger(s-)} I(T_{ij}\ge v) + I((s\wedge\tau_i)-S_{iN_i^\dagger(s-)}\ge v) \right\} \mathbf{H}_i(v,\boldsymbol{\delta})\lambda_F(v,\boldsymbol{\delta})dv \\
&= \int_0^t Y_i(s,v)\mathbf{H}_i(v,\boldsymbol{\delta})\lambda_F(v,\boldsymbol{\delta})dv.
\end{aligned}
\tag{3.16}
$$

3.2.3 Regularity Conditions and Asymptotic Results

With this reformulation in place, we may now state the appropriate regularity conditions in terms of gap-times which are needed for the consistency and asymptotic normality of the ML estimators. Let $\boldsymbol{\delta}_0 = (\boldsymbol{\theta}_0, \beta_0)$ be the true parameter value, $s^\star$ be the maximum monitoring time for all units, and $t^\star > 0$.

A: There exists a neighborhood Γ_0 of $\boldsymbol{\delta}_0$ such that for all $t \in [0, t^\star]$, the first, second, and third order partial derivative with respect to $\boldsymbol{\delta}$ of $\lambda_F(t,\boldsymbol{\delta})$ and $\log\lambda_F(t,\boldsymbol{\delta})$ exist and are continuous in $\boldsymbol{\delta} \in \boldsymbol{\Gamma_0}$. The log-likelihood equation may be differentiated three times with respect to $\boldsymbol{\delta} \in \boldsymbol{\Gamma_0}$ and the order of integration may be interchanged.

B: There exist finite functions $\sigma_{jk}(s,t;\boldsymbol{\delta})$ on Γ_0 such that for all j,k as $n\to\infty$,

$$\frac{1}{n}\sum_{i=1}^{n}\int_0^{t^\star} Y_i(s^\star,w)\left\{\frac{\partial}{\partial\boldsymbol{\delta_j}}\log\lambda_F(w,\boldsymbol{\delta}_0)\right\}\left\{\frac{\partial}{\partial\boldsymbol{\delta_k}}\log\lambda_F(w,\boldsymbol{\delta}_0)\right\}\lambda_F(w,\boldsymbol{\delta}_0)dw \xrightarrow{p} \sigma_{jk}(s^\star,t^\star;\boldsymbol{\delta_0}).$$

C: The matrix $\Sigma(s^\star,t^\star;\boldsymbol{\delta}_0)$ with entries $\sigma_{jk}(s^\star,t^\star;\boldsymbol{\delta}_0)$ is positive definite.

D: As $n\to\infty$, $\frac{1}{n}\sum_{i=1}^{n}\int_0^{t^\star}\{\nabla_{\boldsymbol{\delta}\boldsymbol{\delta}^t}\log\lambda_F(w,\boldsymbol{\delta}_0)\}^{\otimes 2}Y_i(s^\star,w)\lambda_F(w,\boldsymbol{\delta}_0)dw \xrightarrow{p}$ finite quantity; component by component.

E: There exists predictable processes $G_1(t)$ and $G_2(t)$, such that

$$\sup_{\boldsymbol{\delta}\in\Gamma_0}\left|\frac{\partial^3}{\partial\delta_j\partial\delta_k\partial\delta_l}\lambda_F(t,\boldsymbol{\delta})\right| \le G_1(t) \quad \textit{and} \quad \sup_{\boldsymbol{\delta}\in\Gamma_0}\left|\frac{\partial^3}{\partial\delta_j\partial\delta_k\partial\delta_l}\log\lambda_F(t,\boldsymbol{\delta})\right| \le G_2(t).$$

In addition, $\frac{1}{n}\sum_{i=1}^{n}\int_0^{t^\star} G_1(w)Y_i(s^\star,w)dw$ and $\frac{1}{n}\sum_{i=1}^{n}\int_0^{t^\star} G_2(w)Y_i(s^\star,w)\lambda_F(w,\boldsymbol{\delta})dw$ converge in probability to finite quantities as $n\to\infty$.

Subject to the above regularity conditions, the following theorem follows from Borgan's [8] asymptotic results.

Theorem 3.1 *Assume conditions (A)-(E) hold for $t^\star>0$ and $s^\star>0$. As $n\to\infty$, there exists a sequence of solutions $(\hat{\boldsymbol{\theta}}_n,\hat{\beta}_n)^t(s^\star,t^\star)$ to the sequence of ML estimating equations $\boldsymbol{U}_n(s^\star,t^\star;\boldsymbol{\theta},\beta)=\mathbf{0}$ such that $(\hat{\boldsymbol{\theta}}_n,\hat{\beta}_n)^t(s^\star,t^\star)\xrightarrow{p}(\boldsymbol{\theta}_0,\beta_0)^t$ and*

$$\sqrt{n}(\hat{\boldsymbol{\delta}}_n(s^\star,t^\star)-\boldsymbol{\delta}_0)\xrightarrow{d}N(\mathbf{0},\Sigma(s^\star,t^\star;\boldsymbol{\delta}_0)^{-1}).$$

In addition, $\frac{1}{n}\boldsymbol{I}_n(s^\star,t^\star;\hat{\boldsymbol{\delta}}_n)$ is a consistent estimator of $\Sigma(s^\star,t^\star;\boldsymbol{\delta}_0)$, where $\boldsymbol{I}_n(s^\star,t^\star;\hat{\boldsymbol{\delta}}_n)$ is the observed Fisher information based on the n subjects up to time $s^\star$, for $t^\star>0$.

3.2.4 Expression of the Asymptotic Covariance Matrix

For the purpose of theoretical efficiency comparison, it is desirable to obtain closed-form expressions for the asymptotic covariance matrix. Note first that, from the score process in (3.12), the components of $\mathbf{H}_i$ are predictable. Moreover, $\mathbf{M}_i(w;\boldsymbol{\theta},\beta)$ is a vector of square integrable martingales. Hence, by stochastic integration theory, $\mathbf{U}(w;\boldsymbol{\delta})$ is a vector of $(p+1)$ square integrable martingales with a predictable covariation process given by

$$\langle \mathbf{U}_{\boldsymbol{\theta}}(\boldsymbol{\theta},\beta), \mathbf{U}_{\beta}(\boldsymbol{\theta},\beta)\rangle(s) = \sum_{i=1}^{n} \int_0^s \mathbf{H}_i(w;\boldsymbol{\theta},\beta) d\langle \mathbf{M}_i, \mathbf{M}_i^t\rangle(w) \mathbf{H}_i(w;\boldsymbol{\theta},\beta)^t; \tag{3.17}$$

where

$$d\langle \mathbf{M}_i, \mathbf{M}_i^t\rangle(w) = \begin{pmatrix} Y_i^{\dagger}(w)\beta\lambda_F(w,\boldsymbol{\theta}) & 0 \\ 0 & Y_i^{\dagger}(w)\lambda_F(R_i(w),\boldsymbol{\theta}) \end{pmatrix} dw. \tag{3.18}$$

Now, using (3.12) and (3.18), (3.17) becomes

$$\begin{aligned} \langle \mathbf{U}_{\boldsymbol{\theta}}(\boldsymbol{\theta},\beta), \mathbf{U}_{\beta}(\boldsymbol{\theta},\beta)\rangle(s) &= \sum_{i=1}^{n} \int_0^s \begin{pmatrix} \varphi(w,\boldsymbol{\theta}) & \varphi(R_i(w),\boldsymbol{\theta}) \\ \frac{1}{\beta} & 0 \end{pmatrix} \\ &\quad \times \begin{pmatrix} Y_i^{\dagger}(w)\beta\lambda_F(w,\boldsymbol{\theta}) & 0 \\ 0 & Y_i^{\dagger}(w)\lambda_F(R_i(w),\boldsymbol{\theta}) \end{pmatrix} \\ &\quad \times \begin{pmatrix} \varphi(w,\boldsymbol{\theta})^t & \frac{1}{\beta} \\ \varphi(R_i(w),\boldsymbol{\theta})^t & 0 \end{pmatrix} dw. \end{aligned} \tag{3.19}$$

For brevity, let

$$\mathbf{H}_i(w;\boldsymbol{\theta},\beta) = \left(\begin{smallmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{smallmatrix}\right) \quad \text{and} \quad d\langle \mathbf{M}_i, \mathbf{M}_i^t\rangle(w) = \left(\begin{smallmatrix} V_{i1} & 0 \\ 0 & V_{i2} \end{smallmatrix}\right) dw.$$

By some simple matrix manipulations, (3.19) becomes

$$\begin{aligned}
&\langle \mathbf{U}_{\boldsymbol{\theta}}(\boldsymbol{\theta},\beta), \mathbf{U}_{\beta}(\boldsymbol{\theta},\beta)\rangle(s) \\
&= \sum_{i=1}^{n} \int_0^s \begin{pmatrix} V_{i1}H_{11}^{\otimes 2} + V_{i2}H_{12}^{\otimes 2} & V_{i1}H_{11}H_{21} + V_2H_{12}H_{22} \\ V_{i1}H_{11}H_{21} + V_2H_{12}H_{22} & V_{i1}H_{21}^2 + V_{i2}H_{22}^2 \end{pmatrix} dw \\
&= \sum_{i=1}^{n} \int_0^s \begin{pmatrix} V_{i1}\varphi(w,\boldsymbol{\theta})^{\otimes 2} + V_{i2}\varphi(w,\boldsymbol{\theta})^{\otimes 2} & V_{i1}\frac{\varphi(w,\boldsymbol{\theta})}{\beta} \\ V_{i1}\frac{\varphi(w,\boldsymbol{\theta})}{\beta} & \frac{V_{i1}}{\beta^2} \end{pmatrix} dw \\
&= \sum_{i=1}^{n} \int_0^s \left\{ \begin{pmatrix} \varphi(w,\boldsymbol{\theta})^{\otimes 2} & \frac{\varphi(w,\boldsymbol{\theta})}{\beta} \\ \frac{\varphi(w,\boldsymbol{\theta})^t}{\beta} & \frac{1}{\beta^2} \end{pmatrix} V_{i1} + \begin{pmatrix} \varphi(R_i(w),\boldsymbol{\theta})^{\otimes 2} & 0 \\ 0 & 0 \end{pmatrix} V_{i2} \right\} dw.
\end{aligned} \tag{3.20}$$

Next, using expressions for V_{i1} and V_{i2}, (3.20) yields

$$\begin{aligned}
\langle \mathbf{U}_{\boldsymbol{\theta}}(\boldsymbol{\theta},\beta), \mathbf{U}_{\beta}(\boldsymbol{\theta},\beta)\rangle(s) &= \sum_{i=1}^{n} \int_0^s \left\{ \begin{pmatrix} \varphi(w,\boldsymbol{\theta}) \\ \frac{1}{\beta} \end{pmatrix}^{\otimes 2} \beta Y_i^{\dagger}(w)\lambda_F(w,\boldsymbol{\theta}) \right. \\
&\quad \left. + \begin{pmatrix} \varphi(R_i(w),\boldsymbol{\theta}) \\ 0 \end{pmatrix}^{\otimes 2} Y_i^{\dagger}(w)\lambda_F(R_i(w),\boldsymbol{\theta}) \right\} dw.
\end{aligned} \tag{3.21}$$

Now, we need to find the limit of $\frac{1}{n}\langle \mathbf{U}_{\boldsymbol{\theta}}(\boldsymbol{\theta},\beta), \mathbf{U}_{\beta}(\boldsymbol{\theta},\beta)\rangle(s)$ as $n \to \infty$. Since the $Y_i^{\dagger}(w)$s is an i.i.d. sequence, by the strong law of large numbers, we have for each

$w \in [0, s^\star]$, $\frac{1}{n}\sum_{i=1}^n Y_i^\dagger(w) \stackrel{a.s.}{\to} E\{Y_1^\dagger(w)\}$. Since $Y_1^\dagger(w) = I(\tau_1 \geq w)$, we have

$$\frac{1}{n}\sum_{i=1}^n Y_i^\dagger(w) \stackrel{a.s.}{\to} E\{Y_1^\dagger(w)\} = P(\tau_1 \geq w) = \exp\{-\beta\Lambda_F(w, \boldsymbol{\theta})\} \equiv y(w; \boldsymbol{\theta}, \beta). \quad (3.22)$$

Note that the right-hand side of (3.21) contains two terms. We need to show that each term converges uniformly to a matrix with finite components. For a matrix $\mathbf{A} = (a_{ij})$, let $\|\mathbf{A}\| = \sup |a_{ij}|$. The following lemma pertains to the uniform convergence of the first term.

Lemma 3.2 *Assume $\nabla_{\theta_j} \log\{\lambda_F(w, \boldsymbol{\theta})\}$, $j = 1, 2, ..., p$, and $\lambda_F(w, \boldsymbol{\theta})$ are bounded on $[0, t^\star] \times \Theta_0$ and are continuous functions of $\boldsymbol{\theta}$. Then, for each $s \in [0, s^\star]$, we have*

$$\sup_{s\in[0,s^\star]} \left\| \int_0^s \begin{bmatrix} \varphi(w, \boldsymbol{\theta}) \\ \frac{1}{\beta} \end{bmatrix}^{\otimes 2} \beta\lambda_F(w, \boldsymbol{\theta}) \left\{ \frac{\sum_{i=1}^n Y_i^\dagger(w)}{n} - y(w; \boldsymbol{\theta}, \beta) \right\} dw \right\| = o_p(1)$$

as $n \to \infty$.

Remarks: Notice here that the assumptions on the lemma or regularity Condition A may be too restrictive in some situations. In that case, some additional conditions need to be imposed on the derivative of $\log\{\lambda_F(t, \boldsymbol{\theta})\}$ [cf. also Borgan [8]].

Proof: From the almost sure convergence in (3.22) and the Glivenko-Cantelli theorem, we have $\sup_{w\in[0,s^\star]} \left| \frac{\sum_{i=1}^n Y_i^\dagger(w)}{n} - y(w; \boldsymbol{\theta}, \beta) \right| \stackrel{a.s.}{\to} 0$. That is, $\forall \epsilon > 0$, $\exists N_0$ such that, for $n > N_0$, $\sup_{w\in[0,s^\star]} \left| \frac{\sum_{i=1}^n Y_i^\dagger(w)}{n} - y(w; \boldsymbol{\theta}, \beta) \right| < \epsilon$. As a consequence, the latter holds for

all $s \in [0, s^\star]$. Now, let $B = \sup_{w\in[0,s^\star]} |\nabla_{\theta_i} \log \lambda(w, \boldsymbol{\theta})|$, for $i = 1, ..., p$ then

$$\sup_{s\in[0,s^\star]} \left\| \int_0^s \begin{pmatrix} \varphi(\boldsymbol{\theta}, w) \\ \frac{1}{\beta} \end{pmatrix}^{\otimes 2} \beta\lambda_F(w, \boldsymbol{\theta}) \left\{ \frac{\sum_{i=1}^n Y_i^\dagger(w)}{n} - y(w, \boldsymbol{\theta}, \beta) \right\} dw \right\|$$

$$\leq \sup_{s\in[0,s^\star]} \int_0^s \left\| \begin{pmatrix} \varphi(\boldsymbol{\theta}, w) \\ \frac{1}{\beta} \end{pmatrix}^{\otimes 2} \right\| \beta\lambda_F(w, \boldsymbol{\theta}) \left| \frac{\sum_{i=1}^n Y_i^\dagger(w)}{n} - y(w, \boldsymbol{\theta}, \beta) \right| dw$$

$$\leq \beta B \sup_{s\in[0,s^\star]} \int_0^s \epsilon dw$$

$$\leq \beta B \epsilon s^\star < \infty.$$

Since ϵ is arbitrary, the result follows.

Now, turning to the second term in the right-hand side of (3.21), we see that it contains the random quantity $R_i(w)$. The gap-time reformulation in Lemma 3.1 will be useful in obtaining the limit of this second term, which will finally give asymptotic covariance in Condition B. This reformulation contains the process $Y_i(s, t)$. To obtain closed-form expressions for the asymptotic covariance in Condition B, we need to find the limit of $\frac{1}{n}\sum_{i=1}^n Y_i(s, t)$ as $n \to \infty$. The next proposition, whose proof is in Peña et al. [32] gives the limit of the sample mean of the $Y_i(s, t)$s.

Proposition 3.1 *For each $s \in [0, s^\star]$, $t \in [0, t^\star]$,*

$$\sup_{s\in[0,t^\star]} \left| \frac{1}{n} \sum_{i=1}^n Y_i(s, t) - y(s, t) \right| = o_p(1);$$

as $n \to \infty$, where $y(s, t) = E\{Y_1(s, t)\} = \bar{F}(t)\bar{G}(t) + \bar{F}(t)I\{t \leq s\} \int_t^s \rho(w - t)dG(w)$ and where $\rho(t)$ is the renewal function of F.

Note in the above proposition that if $t > s$, then $y(s,t) = 0$.

In the case of the KG model, we have

$$y(s,t) = \bar{F}^{\beta+1}(t) - \beta\bar{F}(t)\int_t^s \rho(w-t)d\bar{F}(w)\bar{F}^{\beta-1}(w).$$

From the above proposition, we can conclude that $y(s,t)$ may be estimated via $n^{-1}\sum_{i=1}^n Y_i(s,t)$. Now letting $\varphi(R_i(w),\boldsymbol{\theta})^{\otimes 2} = \mathbf{H}(R_i(w))$, where $\mathbf{H}$ is a $p \times p$ matrix of functions, an application of the Glivenko-Cantelli theorem together with Lemma 3.1 and Proposition 3.1 give explicit forms of the asymptotic covariance matrix in Condition B as

$$\frac{1}{n}\langle \mathbf{U}_\theta(s;\boldsymbol{\theta},\beta), \mathbf{U}_\beta(s;\boldsymbol{\theta},\beta)\rangle(t^\star) \xrightarrow{p}$$
$$\int_0^{t^\star} \begin{pmatrix} \varphi(w,\boldsymbol{\theta}) \\ \frac{1}{\beta} \end{pmatrix}^{\otimes 2} \beta\lambda_F(w,\boldsymbol{\theta})\exp\{-\beta\Lambda_F(w,\boldsymbol{\theta})\}dw + \begin{pmatrix} \int_0^{t^\star} y(s,t)\mathbf{H}(t)\lambda_F(t,\boldsymbol{\theta})dt \\ \mathbf{0} \end{pmatrix}^{\otimes 2}. \tag{3.23}$$

Note that each of the terms in the right-hand side of (3.23) is a $(p+1)\times(p+1)$ matrix. In addition, the first is the extra variance obtained by exploiting the KG structure and the second term, as we will see later, is the only term that would have been obtained if we ignored the informative censoring.

3.2.5 Weak Convergence Result: Parametric Case

In this section, we give asymptotic properties pertaining to a sequence of estimators of the gap-time distribution defined on $C[0,t^\star]$. As an aside, we point out that an

estimator of the gap-time distribution is obtained by plugging in the estimator of $\boldsymbol{\theta}$ in the expression of $\bar{F}$, that is $\hat{\bar{F}}(t) = \bar{F}(t, \hat{\theta}_n(s^\star, t^\star))$. The result is given next.

Theorem 3.2 *The process* $\{W_n = \sqrt{n}[\hat{\bar{F}}(t) - \bar{F}(t, \boldsymbol{\theta})] : t \in [0, t^\star]\}$ *converges weakly on* $C[0, t^\star]$ *to a zero-mean Gaussian process* $\{W(t) : t \in [0, t^\star]\}$ *whose covariance function is* $cov\{W(t_1), W(t_2)\} = \boldsymbol{\nabla}_{\boldsymbol{\theta}} \bar{F}(t_1, \boldsymbol{\theta}) \Sigma_1^{-1} \boldsymbol{\nabla}_{\boldsymbol{\theta}} \bar{F}(t_2, \boldsymbol{\theta})^t$ *for* $t_1, t_2 \in [0, t^\star]$*, where* Σ_1^{-1} *is the inverse of the* $p \times p$ *matrix obtained by taking the first* p *rows and* p *columns of the matrix in the right-hand side of* (3.23).

Proof: The proof of the above theorem relies heavily on the continuous mapping theorem and the functional delta-method. Define the mapping $H : \mathbb{R}^p \to C[0, t^\star]$ by $H(\boldsymbol{\theta}) = \{\bar{F}(t, \boldsymbol{\theta}); t \in [0, t^\star]\}$. The proof of the theorem will be done in two steps. We first need to show that the function H is Hadamard differentiable at a point $\boldsymbol{\theta}$. Then, the result will follow by invoking the functional delta-method and the continuous mapping theorem. Define the linear map $dH(\boldsymbol{\theta}) : \mathbb{R}^p \to C[0, t^\star]$, such that for every $\mathbf{h}$ in $\mathbb{R}^p$, we have

$$\mathbf{h} \mapsto dH(\boldsymbol{\theta}).\mathbf{h}.$$

Notice here that $dH(\boldsymbol{\theta})$ is an operator such that, for every $\mathbf{h} \in \mathbb{R}^p$,

$$dH(\boldsymbol{\theta}).\mathbf{h} = \left\{ \boldsymbol{\nabla}_{\boldsymbol{\theta}} \bar{F}(t, \boldsymbol{\theta})^t.\mathbf{h} = \langle \boldsymbol{\nabla}_{\boldsymbol{\theta}} \bar{F}(t, \boldsymbol{\theta})^t; \mathbf{h} \rangle = \sum_{j=1}^{p} \nabla_{\theta_j} \bar{F}(t; \boldsymbol{\theta}) h_j, 0 \leq s \leq t^\star \right\}. \tag{3.24}$$

That is, for $s \in [0, t^\star]$, $\{dH(\boldsymbol{\theta}).\mathbf{h}\}(s) = \{\boldsymbol{\nabla}_{\boldsymbol{\theta}} \bar{F}(s, \boldsymbol{\theta})^t.\mathbf{h}\}$. We want to show that $H(\boldsymbol{\theta})$ is Hadamard differentiable. Equip $C[0, t^\star]$ with uniform topology and consider a real

sequence $\{a_n\} \to \infty$ and a sequence $\mathbf{h}_n \to \mathbf{h}$ in $\mathbb{R}^p$. We need to show that

$$a_n\{\bar{F}(t, \boldsymbol{\theta} + a_n^{-1}\mathbf{h}_n) - \bar{F}(t, \boldsymbol{\theta})\} \to \boldsymbol{\nabla}_{\boldsymbol{\theta}}\bar{F}(s, \boldsymbol{\theta})^t.\mathbf{h} \quad as \quad n \to \infty. \tag{3.25}$$

Now, Taylor expansion around $\boldsymbol{\theta}$ of $\bar{F}(t, \boldsymbol{\theta} + a_n^{-1}\mathbf{h}_n)$ yields

$$a_n\{\bar{F}(t, \boldsymbol{\theta} + a_n^{-1}\mathbf{h}_n) - \bar{F}(t, \boldsymbol{\theta}) - \boldsymbol{\nabla}_{\boldsymbol{\theta}}\bar{F}(t, \boldsymbol{\theta})^t.\mathbf{h}\} =$$

$$\{a_n(\bar{F}(t, \boldsymbol{\theta}) + a_n^{-1}\mathbf{h}_n.\boldsymbol{\nabla}_{\boldsymbol{\theta}}\bar{F}(t, \boldsymbol{\theta_n}^{\star})^t - \bar{F}(t, \boldsymbol{\theta})) - \boldsymbol{\nabla}_{\boldsymbol{\theta}}\bar{F}(t, \boldsymbol{\theta})^t.\mathbf{h}\};$$

where $\boldsymbol{\theta}_{\boldsymbol{n}}^{\star} \in [\boldsymbol{\theta}, \boldsymbol{\theta} + a_n^{-1}\mathbf{h}_n]$. Adding and subtracting $\boldsymbol{\nabla}_{\boldsymbol{\theta}}\bar{F}(t, \boldsymbol{\theta_n}^{\star})^t.\mathbf{h}$; letting $\boldsymbol{\nabla}_{\boldsymbol{\theta}}\bar{F}(t, \boldsymbol{\theta})^t = dH(\boldsymbol{\theta})$ and using the notation $\|\mathbf{a}\| = \sum_{i=1}^n |a_i|$, $\langle \mathbf{a}, \mathbf{b} \rangle = \mathbf{a}.\mathbf{b} = \sum_{i=1}^n a_i.b_i$ for vector $\mathbf{a}$ and $\mathbf{b}$, we have

$$\begin{aligned}
|dH(\boldsymbol{\theta}_n^{\star}).\mathbf{h}_n - dH(\boldsymbol{\theta}).\mathbf{h}| &= |dH(\boldsymbol{\theta}_n^{\star}).\mathbf{h}_n - dH(\boldsymbol{\theta}_n^{\star}).\mathbf{h} + dH(\boldsymbol{\theta}_n^{\star}).\mathbf{h} - dH(\boldsymbol{\theta}).\mathbf{h}| \\
&\leq |dH(\boldsymbol{\theta}_n^{\star}).\mathbf{h}_n - dH(\boldsymbol{\theta}_n^{\star}).\mathbf{h}| + |dH(\boldsymbol{\theta}_n^{\star}).\mathbf{h} - dH(\boldsymbol{\theta}).\mathbf{h}| \\
&= |\langle dH(\boldsymbol{\theta}_n^{\star}); \mathbf{h}_n - \mathbf{h}\rangle| + |\langle dH(\boldsymbol{\theta}_n^{\star}) - dH(\boldsymbol{\theta}); \mathbf{h}\rangle| \\
&\leq \|dH(\boldsymbol{\theta}_n^{\star})\|.\|\mathbf{h}_n - \mathbf{h}\| + \|dH(\boldsymbol{\theta}_n^{\star}) - dH(\boldsymbol{\theta})\|.\|\mathbf{h}\|;
\end{aligned}$$

where the last inequality above is obtained by using a property of the dot product $\langle a, b\rangle \leq \|a\|\|b\|$; for vectors $\mathbf{a}$ and $\mathbf{b}$. Next, as $n \to \infty$, $\boldsymbol{\theta}_n^{\star} \to \boldsymbol{\theta}$, and since dH is continuous in $\boldsymbol{\theta}$, $dH(\boldsymbol{\theta}_n^{\star}) \to dH(\boldsymbol{\theta})$. Moreover, since $\mathbf{h}_n \to \mathbf{h}$ in $\mathbb{R}^p$, then $\|\mathbf{h}_n - \mathbf{h}\| \to \mathbf{0}$, and the result in (3.25) follows. As a consequence, H is Hadamard differentiable. Now, from Theorem 3.1, we have $\sqrt{n}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}) \Rightarrow Z$, where Z is the limiting random variable given in the theorem. By the functional delta-method and the mapping

theorem, it thus follows that

$$\sqrt{n}[\hat{\bar{F}}(t, \boldsymbol{\theta_n}) - \bar{F}(t, \boldsymbol{\theta})] \Rightarrow \boldsymbol{\nabla_\theta} \bar{F}(t, \boldsymbol{\theta})^t.Z \equiv W(t);$$

whose covariance function is given by $cov(W(t_1), W(t_2)) = \boldsymbol{\nabla_\theta} \bar{F}(t_1, \boldsymbol{\theta}) \Sigma_1^{-1} \boldsymbol{\nabla_\theta} \bar{F}(t_2, \boldsymbol{\theta})^t$. This finally proves the theorem.

A special case of the above theorem is the one that gives asymptotic results of the estimator of the gap-time for a fixed t.

Corollary 3.1 *Let $\Sigma_1(s^\star, t^\star)$ be the $p \times p$ sub-matrix with components equal the first p rows and p columns of $\Sigma(s^\star, t^\star)$, then*

$$\sqrt{n}\Big(\bar{F}(t, \hat{\boldsymbol{\theta}}_n) - \bar{F}(t, \boldsymbol{\theta})\Big) \xrightarrow{d} N\Big\{\mathbf{0}, \boldsymbol{\nabla_\theta} \bar{F}(t, \boldsymbol{\theta})^t \Sigma_1(s^\star, t^\star; \boldsymbol{\theta})^{-1} \boldsymbol{\nabla_\theta} \bar{F}(t, \boldsymbol{\theta})\Big\}.$$

Proof: This follows from Theorem 3.2 by projection.

3.2.6 Estimation Ignoring Informative Censoring

In this section, we investigate the asymptotic properties of the estimator of the gap-time distribution ignoring the KG structure. From (3.5), a misspecified likelihood process is obtained by ignoring the contribution of the censoring to the likelihood and is given by

$$L(\boldsymbol{\theta}, s) = \prod_{i=1}^{n} \left\{ \prod_{w=0}^{s} [dA_i^\dagger(w, \boldsymbol{\theta})^{\triangle N_i^\dagger(w)}] \times [1 - dA_i^\dagger(w, \boldsymbol{\theta})]^{1 - \triangle N_i^\dagger(w)} \right\}; \tag{3.26}$$

and the associated score process is

$$U_{\boldsymbol{\theta}}(s, \boldsymbol{\theta}) = \int_0^s \varphi(R_i(w), \boldsymbol{\theta}) dM_i^{\dagger}(w; \boldsymbol{\theta}). \tag{3.27}$$

Setting the score process at $s = s^\star$ to zero yields the ML estimators $\hat{\boldsymbol{\theta}}_n$ of $\boldsymbol{\theta}$ as before. The score process is again a martingale and its quadratic variation process is

$$\langle \mathbf{U}(\boldsymbol{\theta}) \rangle(s) = \sum_{i=1}^{n} \int_0^s \varphi(R_i(w), \boldsymbol{\theta})^t Y_i^{\dagger}(w) \lambda_F(R_i(w), \boldsymbol{\theta}) \varphi(R_i(w), \boldsymbol{\theta}) dw. \tag{3.28}$$

Using the same approach as before by reformulating the process using Lemma 3.1, we have, as $n \to \infty$

$$\frac{1}{n} \langle \mathbf{U}(\boldsymbol{\theta}) \rangle(s, t^\star) \xrightarrow{p} \int_0^{t^\star} y(s,t) \mathbf{H}(t) \lambda_F(t, \boldsymbol{\theta}) dt. \tag{3.29}$$

For brevity, let $\Sigma_2(s, t^\star; \boldsymbol{\theta}) = \int_0^{t^\star} y(s,t) \mathbf{H}(t) \lambda_F(t, \boldsymbol{\theta}) dt$. $\Sigma_2(s, t^\star; \boldsymbol{\theta})$ is a $p \times p$ matrix, which is a sub-matrix of the second term in the right-hand side of (3.23). Then, we have the following corollary.

Corollary 3.2 *Assume conditions (A)-(E) hold for $s^\star > 0$. Then, as $n \to \infty$, there exists a consistent solution $\hat{\boldsymbol{\theta}}_n = \hat{\boldsymbol{\theta}}_n(s^\star, t^\star)$ to the ML estimating equation $\boldsymbol{U}(s^\star, t^\star; \boldsymbol{\theta}) = \mathbf{0}$. Moreover,*

$$\sqrt{n}(\hat{\boldsymbol{\theta}}_n - \boldsymbol{\theta}_0) \xrightarrow{d} N(\mathbf{0}, \Sigma_2(s^\star, t^\star; \boldsymbol{\theta}_0)^{-1}).$$

Proof: Since conditions (A)-(E) hold for $\boldsymbol{\theta}$ and β jointly, they also hold for $\boldsymbol{\theta}$ alone and hence the limiting distribution is obtained. Now, since our covariation process converges to a matrix with finite components as shown in (3.23), the corollary follows from Theorem 3.1.

The next corollary deals with the limiting distribution of the estimator of the gap-time distribution when we ignore informative censoring.

Corollary 3.3 *For a fixed t, we have,*

$$\sqrt{n}\Big(\bar{F}(t,\hat{\boldsymbol{\theta}}_n) - \bar{F}(t,\boldsymbol{\theta})\Big) \xrightarrow{d} N\Big\{0, \boldsymbol{\nabla}_{\boldsymbol{\theta}}\bar{F}(t,\boldsymbol{\theta})^t\Sigma_2(s^\star,t^\star;\boldsymbol{\theta})^{-1}\boldsymbol{\nabla}_{\boldsymbol{\theta}}\bar{F}(t,\boldsymbol{\theta})\Big\}.$$

3.3 Efficiency Comparison

In this section, comparisons between the estimators derived under the KG assumption with recurrent event data and those derived ignoring the KG structure will be investigated. Cheng and Lin [12] have shown that the product limit estimator is less efficient than their proposed estimator under the single event KG model. They have also shown that the heavier the censoring, which occurs when the censoring parameter β increases, the greater the asymptotic efficiency of their proposed estimator.

Let $\hat{\bar{F}}(t)$ and $\tilde{\bar{F}}(t)$ denote the estimators, derived under informative censoring and ignoring informative censoring, respectively. To measure the asymptotic relative efficiency (ARE) of $\hat{\bar{F}}(t)$ relative to $\tilde{\bar{F}}(t)$, one can use the function

$$ARE\{\hat{\bar{F}}(t) : \tilde{\bar{F}}(t)\} = \frac{\boldsymbol{\nabla}_{\boldsymbol{\theta}}\bar{F}(t,\boldsymbol{\theta})^t\Sigma_1(s^\star,t;\boldsymbol{\theta})^{-1}\boldsymbol{\nabla}_{\boldsymbol{\theta}}\bar{F}(t,\boldsymbol{\theta})}{\boldsymbol{\nabla}_{\boldsymbol{\theta}}\bar{F}(t,\boldsymbol{\theta})^t\Sigma_2(s^\star,t;\boldsymbol{\theta})^{-1}\boldsymbol{\nabla}_{\boldsymbol{\theta}}\bar{F}(t,\boldsymbol{\theta})}, \tag{3.30}$$

since, asymptotically, the aforementioned estimators are unbiased.

The next theorem on efficiency, is one of the results of this research.

Theorem 3.3 *Under an informative monitoring period, $\hat{\bar{F}}(t)$ is always more efficient than $\tilde{\bar{F}}(t)$, that is, for each $t > 0$,*

$$ARE\{\tilde{\bar{F}}(t) : \hat{\bar{F}}(t)\} = \frac{\boldsymbol{\nabla_\theta}\bar{F}(t,\boldsymbol{\theta})^t\Sigma_1(s^\star,t;\boldsymbol{\theta})^{-1}\boldsymbol{\nabla_\theta}\bar{F}(t,\boldsymbol{\theta})}{\boldsymbol{\nabla_\theta}\bar{F}(t,\boldsymbol{\theta})^t\Sigma_2(s^\star,t;\boldsymbol{\theta})^{-1}\boldsymbol{\nabla_\theta}\bar{F}(t,\boldsymbol{\theta})} \leq 1$$

Proof: It suffices to show that

$$\begin{aligned}&\{\boldsymbol{\nabla_\theta}\bar{F}(t,\boldsymbol{\theta})^t\Sigma_1(s^\star,t)^{-1}\boldsymbol{\nabla_\theta}\bar{F}(t,\boldsymbol{\theta}) - \boldsymbol{\nabla_\theta}\bar{F}(t,\boldsymbol{\theta})^t\Sigma_2(s^\star,t)^{-1}\boldsymbol{\nabla_\theta}\bar{F}(t,\boldsymbol{\theta})\}\\ &\equiv \boldsymbol{\nabla_\theta}\bar{F}(t,\boldsymbol{\theta})^t\{\Sigma_1(s^\star,t)^{-1} - \Sigma_2(s^\star,t)^{-1}\}\boldsymbol{\nabla_\theta}\bar{F}(t,\boldsymbol{\theta})\end{aligned} \tag{3.31}$$

is negative definite. This is equivalent to showing that $\{\Sigma_1(s^\star,t)^{-1} - \Sigma_2(s^\star,t)^{-1}\}$ is negative definite. To prove this, note first by (3.23) that $\Sigma_1(s^\star,t)$ and $\Sigma_2(s^\star,t)$ are both positive definite and that $\Sigma_1(s^\star,t)$ is the sum of two positive definite matrices where the second is nothing but $\Sigma_2(s^\star,t)$. Let $\Sigma_1(s^\star,t) = \Upsilon_1(s^\star,t) + \Upsilon_2(s^\star,t)$ and $\Sigma_2(s^\star,t) = \Upsilon_2(s^\star,t)$. For brevity, we will drop the argument $(s^\star,t)$. Now, since Σ_1 and Σ_2 are positive definite, there exists positive definite matrices A and B such that

$\Upsilon_1 = A^2$ and $\Upsilon_2 = B^2$, [cf. Hoffman and Kunze [21]]. Then,

$$\begin{aligned}
(\Upsilon_1 + \Upsilon_2)^{-1} - \Upsilon_2^{-1} &= (A^2 + B^2)^{-1} - B^{-1^2} \\
&= B^{-1}\{B(A^2 + B^2)^{-1}B - I\}B^{-1} \\
&= B^{-1}\{(B^{-1})^{-1}(A^2 + B^2)^{-1}(B^{-1})^{-1} - I\}B^{-1} \\
&= B^{-1}\{(B^{-1}A^2B^{-1} + I)^{-1} - I\}B^{-1}. \qquad (3.32)
\end{aligned}$$

It is seen that the expression given in (3.32) is negative definite if and only if $(B^{-1}A^2B^{-1} + I)^{-1} - I$ is negative definite, [cf. Hoffman and Kunze [21]]. Now, if $\lambda_1, ..., \lambda_p$ are eigenvalues of $B^{-1}A^2B^{-1}$, then $(B^{-1}A^2B^{-1})^{-1} - I$ has eigenvalues $\frac{1}{1+\lambda_1} - 1, ..., \frac{1}{1+\lambda_p} - 1$. Therefore,

$$(B^{-1}A^2B^{-1} + I)^{-1} - I = \begin{bmatrix} \frac{1}{1+\lambda_1} - 1 & 0 & \dots & 0 \\ 0 & \frac{1}{1+\lambda_2} - 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \frac{1}{1+\lambda_p} - 1 \end{bmatrix}. \qquad (3.33)$$

Since the $\lambda_i > 0$, for all i, (3.33) implies that $(B^{-1}A^2B^{-1} + I)^{-1} - I$ is negative definite.

Remarks: Although our proposed estimators are shown to be more efficient than those derived ignoring informative censoring, more needs to be done in terms of quantifying the degree of efficiency gained in this general parametric estimation. In the next chapter we will be able to quantify the efficiency gained in the case of HPP.

Chapter 4

Estimation in the Homogeneous Poisson Process (HPP) Case

In this chapter, we will consider parametric estimation of the event-time distribution when assuming that T_{ij}, $j = 1, 2, \ldots,$ $i = 1, 2, \ldots, n$, are i.i.d. exponentially distributed with constant failure rate θ; that is, $\lambda_F(w, \theta) = \theta$. This is also known as the homogeneous Poisson process (HPP) model with observation window dependent on the gap-time distribution. Now, because of the memoryless property of the exponential distribution, $R_i(w, \theta)$ is nullified in the hazard. Therefore, $\lambda_F(w, \theta) = \lambda_F(R_i(w), \theta) = \theta$ and $\lambda_G(w, \theta) = \beta\theta$, and as a consequence, Borgan's [8] regularity conditions need not be reformulated in terms of gap-times as was done in the general case. We will give closed-form expressions for the ML estimators of θ and β and obtain their asymptotic properties. Lastly, our proposed parametric estimator of $\bar{F}(t)$ will be compared to the nonparametric estimator proposed in Peña et al. [30].

4.1 Maximum Likelihood Estimator of θ and β

Following the derivations in the previous chapters, we automatically obtain the log-likelihood process as

$$\begin{aligned} l(s;\theta,\beta) &= \sum_{i=1}^{n} N_i^{\tau}(w)\log\beta + \sum_{i=1}^{n}\int_0^s \log\{Y_i^{\dagger}(w)\theta\}\{dN_i^{\dagger}(w)+dN_i^{\tau}(w)\} \\ &\quad -\sum_{i=1}^{n}\int_0^s Y_i^{\dagger}(w)\{\theta+\theta\beta\}dw. \end{aligned} \tag{4.1}$$

To obtain closed-form expressions of the ML estimators of θ and β, we first need to obtain the score processes. For this special case, it is clear that the order of integration can be interchanged and the following score processes for the two parameters obtain

$$\begin{aligned} U_{\theta}(s;\theta,\beta) &= \sum_{i=1}^{n}\left\{\int_0^s \frac{Y_i^{\dagger}(w)}{Y_i^{\dagger}(w)\theta}dN_i^{\dagger}(w) + \int_0^s \frac{Y_i^{\dagger}(w)}{Y_i^{\dagger}(w)\theta}dN_i^{\tau}(w)\right\} \\ &\quad -\sum_{i=1}^{n}\int_0^s Y_i^{\dagger}(w)(1+\beta)dw \\ &= \frac{1}{\theta}\sum_{i=1}^{n}\left\{\int_0^s \{dN_i^{\dagger}(w)+dN_i^{\tau}(w)\}\right\} - (1+\beta)\sum_{i=1}^{n}\int_0^s Y_i^{\dagger}(w)dw. \end{aligned}$$

Now, using the facts that $\int_0^s dN_i^{\dagger}(w) = N_i^{\dagger}(s)$ and $\int_0^s dN_i^{\tau}(w) = N_i^{\tau}(s)$, we obtain

$$U_{\theta}(s;\theta,\beta) = \frac{1}{\theta}\sum_{i=1}^{n}\{N_i^{\dagger}(s)+N_i^{\tau}(s)\} - (1+\beta)\sum_{i=1}^{n}\int_0^s Y_i^{\dagger}(w)dw; \tag{4.2}$$

$$U_{\beta}(s;\theta,\beta) = \frac{\sum_{i=1}^{n} N_i^{\tau}(s)}{\beta} - \theta\sum_{i=1}^{n}\int_0^s Y_i^{\dagger}(w)dw. \tag{4.3}$$

Setting (4.2) and (4.3) evaluated at $s^\star$, respectively to zeros, yield closed-form expressions for the ML estimators given by

$$\hat{\theta}_n(s^\star) = \frac{\sum_{i=1}^n \{N_i^\dagger(s^\star) + N_i^\tau(s^\star)\}}{(1+\hat{\beta}_n(s^\star))\sum_{i=1}^n \int_0^{s^\star} Y_i^\dagger(w)dw}; \tag{4.4}$$

$$\hat{\theta}_n(s^\star)\hat{\beta}_n(s^\star) = \frac{\sum_{i=1}^n N_i^\tau(s^\star)}{\sum_{i=1}^n \int_0^{s^\star} Y_i^\dagger(w)dw}. \tag{4.5}$$

Now, if we allow $s^\star \to \infty$, $N_i^\dagger(\infty) = K_i$ [the total number of observations for each subject] and $\int_0^\infty I(\tau_i \geq w)dw = \int_0^{\tau_i} I(\tau_i \geq w)dw = \tau_i$, [the total time on test for each subject], then (4.4) and (4.5) further yield

$$\hat{\theta}_n(s^\star) = \frac{\sum_{i=1}^n K_i}{\sum_{i=1}^n \tau_i} \quad \text{and} \quad \hat{\beta}_n(s^\star) = \frac{n}{\sum_{i=1}^n K_i}. \tag{4.6}$$

Thus, as is to be expected, the ML estimator of θ is the *occurrence/exposure* rate.

4.2 Asymptotic Properties in the HPP case

In this section, we shall derive the large sample properties of the ML estimators of θ and β jointly. To obtain these asymptotic properties, we just need to verify that Borgan's [8] conditions hold in this special case. Condition A follows immediately since the derivatives of all orders of θ, $\beta\theta$, $\log(\theta)$ and $\log(\beta\theta)$ exist and are continuous in $\boldsymbol{\delta} = (\theta, \beta)^t$. In addition, the order of differentiation can be interchanged. To prove Conditions B and D, we need to show that our covariation process converges to a positive definite matrix with finite elements. First, observe that, from (4.2) and

(4.3), we have

$$\begin{pmatrix} U_\theta(s;\theta,\beta) \\ U_\beta(s;\theta,\beta) \end{pmatrix} = \sum_{i=1}^{n} \int_0^s \begin{pmatrix} \frac{1}{\theta}\{dM_i^\dagger(w;\theta) + dM_i^\tau(w;\theta,\beta)\} \\ \frac{1}{\beta} dM_i^\tau(w;\theta,\beta) \end{pmatrix}$$

$$= \sum_{i=1}^{n} \int_o^s \begin{pmatrix} \frac{1}{\theta} & \frac{1}{\theta} \\ 0 & \frac{1}{\beta} \end{pmatrix} \begin{pmatrix} dM_i^\dagger(w;\theta) \\ dM_i^\tau(w;\theta,\beta) \end{pmatrix}. \tag{4.7}$$

Notice here that the score processes do not involve $t^\star$ because the backward recurrence time is nullified in the hazard due to the memoryless property of the exponential distribution. Now, letting $\mathbf{H}_i(w;\theta,\beta) = \left(\begin{smallmatrix} \frac{1}{\theta} & \frac{1}{\theta} \\ 0 & \frac{1}{\beta} \end{smallmatrix}\right)$ and $\mathbf{M}_i(w;\theta,\beta) = (M_i^\dagger(w;\theta), M_i^\tau(w;\theta,\beta))^t$, (4.7) becomes

$$\begin{pmatrix} U_\theta(s;\theta,\beta) \\ U_\beta(s;\theta,\beta) \end{pmatrix} = \sum_{i=1}^{n} \int_0^s \mathbf{H}_i(w;\theta,\beta) d\mathbf{M}_i(w;\theta,\beta). \tag{4.8}$$

Using stochastic integration theory, $\mathbf{H}_i$ is a matrix whose components are deterministic functions and bounded, hence predictable. Moreover, the components of $\mathbf{M}_i(w;\theta,\beta)$ are square integrable martingales with respect to the filtration $\mathcal{F}_s = \sigma\{N_i^\dagger(w), Y_i^\dagger(w+),\ w \leq s\}$. As a consequence, $(U_\theta(s;\theta,\beta), U_\beta(s;\theta,\beta))^t$ is a square integrable martingale with predictable covariation process given by

$$
\begin{aligned}
\langle U_\theta(\theta,\beta), U_\beta(\theta,\beta)\rangle(s) &= \sum_{i=1}^{n}\int_0^s \left\{ \begin{pmatrix} \frac{1}{\theta} & \frac{1}{\theta} \\ 0 & \frac{1}{\beta} \end{pmatrix} \right. \\
&\quad \times \begin{pmatrix} \theta Y_i^\dagger(w) & 0 \\ 0 & \beta\theta Y_i^\dagger(w) \end{pmatrix} \left. \begin{pmatrix} \frac{1}{\theta} & 0 \\ \frac{1}{\theta} & \frac{1}{\beta} \end{pmatrix} \right\} dw \\
&= \sum_{i=1}^{n}\int_0^s \begin{pmatrix} \frac{1+\beta}{\theta} & 1 \\ 1 & \frac{\theta}{\beta} \end{pmatrix} Y_i^\dagger(w)dw. \qquad (4.9)
\end{aligned}
$$

We now need to find the limit of $\frac{1}{n}\langle U_\theta(\theta,\beta), U_\beta(\theta,\beta)\rangle(s)$ as $n \to \infty$. From Lemma 3.2 and the fact that the cumulative hazard of the exponential distribution up to time w is θw, it follows that

$$
\int_0^s \frac{1}{n}\sum_{i=1}^{n} Y_i^\dagger(w)dw \xrightarrow{p} \int_0^s P(\tau_1 \geq w)dw = \frac{1-\exp\{-\beta\theta s\}}{\beta\theta} \qquad (4.10)
$$

as $n \to \infty$. In fact, Lemma 3.2 gives a stronger form of convergence, the uniform convergence. Therefore, from (4.9) and (4.10), we obtain for $s = s^\star$ and as $n \to \infty$

$$
\begin{aligned}
&\frac{1}{n}\langle U_\theta(\theta,\beta), U_\beta(\theta,\beta)\rangle(s^\star) \xrightarrow{p} \\
&\begin{pmatrix} \frac{1+\beta}{\theta} & 1 \\ 1 & \frac{\theta}{\beta} \end{pmatrix} \frac{1-\exp\{-\beta\theta s^\star\}}{\beta\theta} \equiv \Xi(s^\star;\theta,\beta). \qquad (4.11)
\end{aligned}
$$

It is clear from this that Conditions B and C are satisfied. For Condition E, we

have

$$\sup_{\boldsymbol{\delta}\in\Gamma_0}\left|\frac{\partial^3}{\partial\theta^3}\log(\theta)\right| \leq \sup_{\boldsymbol{\delta}\in\Gamma_0}\left|\frac{2}{\theta^3}\right| \quad \text{and} \quad \sup_{\boldsymbol{\delta}\in\Gamma_0}\left|\frac{\partial^3}{\partial\beta^3}\log(\beta\theta)\right| \leq \sup_{\boldsymbol{\delta}\in\Gamma_0}\left|\frac{2}{\theta^3}\right|.$$

The right-hand side of the above two inequalities are bounded by a positive constant in Γ_0 not depending on θ and β. The others parts of Condition E follow in a similar manner. Since Borgan's [8] conditions hold, the following theorem obtains

Theorem 4.1 *With probability tending to one,* $(U_\theta(s^\star;\theta,\beta);U_\beta(s^\star;\theta,\beta))^t = \mathbf{0}$ *has a unique solution* $(\hat{\theta}_n(s^\star),\hat{\beta}_n(s^\star))^t$ *with* $(\hat{\theta}_n(s^\star),\hat{\beta}_n(s^\star))^t \xrightarrow{p} (\theta_0,\beta_0)^t$ *as* $n\to\infty$*. Moreover,*

$$\sqrt{n}\big((\hat{\theta}_n(s^\star),\hat{\beta}_n(s^\star))^t - (\theta_0,\beta_0)^t\big) \xrightarrow{d} N(0,\Xi^{-1}(s^\star;\theta_0,\beta_0));$$

where $\Xi(s^\star;\theta_0,\beta_0)$ *is the matrix given in* (4.11) *and which could be estimated by* $I_n^{-1}(s^\star;\hat{\theta}_n,\hat{\beta}_n)$*, the Fisher information matrix.*

4.3 Efficiency Comparison

In this section, we will compare the nonparametric estimator of the gap-time distribution derived in Peña et al. [30] to the estimator obtained in the previous section which exploits the KG structure. It is worth mentioning here that, because of the memoryless property of the exponential distribution, the ML estimator of θ, when ignoring the KG structure is the same as the one derived by exploiting the KG structure. In Peña et al. [30], a nonparametric estimator of the survivor function, $\bar{F}(t)$,

under the recurrent event setting was derived and its asymptotic variance is

$$\sigma^2(s,t) = \bar{F}(t)^2 \int_0^t \frac{\Lambda(dv)}{y(s,v)}, \tag{4.12}$$

where $y(s,v)$ is as in Proposition 3.1. Now, assuming (1.1) holds and noting that $\Lambda(t) = \theta t$ and $\bar{F}(t) = \bar{F}(t,\theta) = \exp(-\theta t)$, so that $y(\infty, v) = \{(\beta+1)/\beta\}\exp\{-\theta(\beta+1)v\}$, therefore (4.12) becomes, if we allow $s \to \infty$,

$$\sigma^2(\infty,t) = \frac{\beta}{(1+\beta)^2}\exp(-2\theta t)\big\{\exp\{(\beta+1)\theta t\} - 1\big\}. \tag{4.13}$$

Meanwhile, by (4.11), it follows that

$$\Xi^{-1}(s^\star;\theta,\beta) = \frac{\beta^2\theta}{1-\exp(-\beta\theta s^\star)}\begin{pmatrix} \frac{\theta}{\beta} & -1 \\ -1 & \frac{1+\beta}{\theta} \end{pmatrix}. \tag{4.14}$$

Now, Theorem 3.1 together with (4.14), gives the asymptotic distribution of the properly standardized version of $\hat{\theta}_n(s^\star)$, where the asymptotic variance is obtained by taking the first component of the matrix in (4.14) as

$$\sqrt{n}(\hat{\theta}_n(s^\star) - \theta) \xrightarrow{d} N\left(0, \frac{\beta\theta^2}{\{1-\exp(-\theta\beta s^\star)\}}\right). \tag{4.15}$$

An estimator of the gap-time distribution is automatically obtained by plugging in the estimator of θ in the expression of $F(t)$, that is, $\hat{\bar{F}}(t) = F(t,\hat{\theta}_n)$. Next, since $\frac{\partial}{\partial\theta}\bar{F}(t,\theta) = t\exp(-\theta t)$, by the δ-method and (4.15), we easily obtain the following

asymptotic distribution of the survivor function:

$$\sqrt{n}(\bar{F}(t;\hat{\theta}(s^\star)) - \bar{F}(t;\theta)) \xrightarrow{d} N\left[0, \frac{\beta(t\theta)^2\exp(-2\theta t)}{1-\exp\{-\beta(\theta s^*)\}}\right]. \tag{4.16}$$

Let $\tilde{\bar{F}}(t)$ be the estimator in Peña et al. [30]. By allowing $s^\star \to \infty$, the ARE of $\tilde{\bar{F}}(t)$ relative to $\hat{\bar{F}}(t)$ is given by

$$ARE\{\tilde{\bar{F}}(t) : \hat{\bar{F}}(t)\} = \frac{\{t\theta(\beta+1)\}^2}{\{\exp(\beta+1)\theta t - 1\}}. \tag{4.17}$$

The above derivations, all together lead to the following efficiency results.

Theorem 4.2 *Under the KG model, for each $t > 0$, the ARE of $\tilde{\bar{F}}(t)$ relative to $\hat{\bar{F}}(t)$ satisfies the following:*

1. *For fixed t, as $\beta \to \infty$, $ARE\{\tilde{\bar{F}}(t) : \hat{\bar{F}}(t)\} \to 0$;*

2. *For fixed t, as $\beta \to 0$, $ARE\{\tilde{\bar{F}}(t) : \hat{\bar{F}}(t)\} \to \frac{(t\theta)^2}{\exp(t\theta)-1}$;*

3. *Let $p = \bar{F}(t,\theta)^{\beta+1}$ and $f(p) = e^{p-1} - \sqrt{p}$. Then*

$$ARE\{\tilde{\bar{F}}(t) : \hat{\bar{F}}(t)\} \leq \frac{p_0\{log(p_0)\}^2}{1-p_0} \approx .65, for\ all\ t$$

where p_0 is the solution to $f(p) = 0$.

Proof: 1) This is clear from (4.17). 2) This also follows from (4.17). To prove 3), note that $\bar{F}(t,\theta) = \exp(-\theta t)$ and $\bar{G}(t;\theta,\beta) = \exp(-\theta\beta t)$. Therefore, the distribution of $Z_i = T_{ij} \wedge \tau_i$ for $i = 1, ..., n$, is $\bar{H}(t;\theta,\beta) = \bar{F}(t,\theta)\bar{G}(t;\theta,\beta) = \bar{F}(t;\beta,\theta)^{\beta+1} =$

$\exp\{-(\beta+1)\theta t\}$. Next, from (4.17) and letting $p = \bar{F}(t,\theta)^{\beta+1}$ we have:

$$\begin{aligned}
\frac{\{t\theta(\beta+1)\}^2}{\{\exp(\beta+1)\theta t - 1\}} &= \frac{\{\log(\bar{H}(t,\theta,\beta))^{-1}\}^2}{\bar{H}(t;\theta,\beta)^{-1} - 1} \\
&= \frac{p\{\log(p)\}^2}{1-p} \\
&\equiv \text{Eff}(p). \qquad (4.18)
\end{aligned}$$

The graph of Eff(p) shown in Figure 4.1 reveals that this function increases to a global maximum and then decreases. From the expression of the ARE in (4.17), let $e(p) \equiv \text{Eff}(p)$. We have

$$e'(p) = \frac{\log p\{\log p + 2(1-p)\}}{(1-p)^2}, \qquad (4.19)$$

and setting (4.19) to 0 yields the solution to $f(p) = 0$. The global maximum of ≈ 0.65 can be seen in the graph of Eff(p) in Figure 4.1 which proves 3).

Remarks: Surprisingly, the results obtained in the preceding theorem parallel those obtained in Cheng and Lin [12] in the single event setting. Intuitively, the result in (1) means that the heavier the censoring, the better the ARE of our proposed estimator. If $\beta \to \infty$, the monitoring period τ approaches zero and the information about the gap-time distribution contained in the τ_is becomes more important since we will most likely observe at most one complete observation per unit. On the other hand, if $\beta \to 0$, then $\tau \to \infty$. In this situation, it is intuitive that the impact of the right-censored observations will disappear because more and more complete observations will be observed. Indeed, this is manifested by the easily-verified result that the

expression of the ARE in (4.17) coincides with the ARE of the parametric estimator $\hat{\bar{F}}(t) = \exp(-t/\bar{T})$ relative to the nonparametric estimator $\tilde{\bar{F}}(t) = \sum_{i=1}^{n} I\{T_i \geq t\}/n$ of the gap-time survivor function based on an i.i.d. sample $T_1, T_2, ..., T_n$. The graph of efficiency shown in Figure 4.1 describes the behavior of the ARE for different values of p. Figure 4.2 shows that, as $\beta \to 0$ and $t \to \infty$, the ARE decreases to 0, which shows that the parametric estimator of the survivor function of the gap-time is more efficient then the nonparametric estimator obtained when using a sequence of n i.i.d. exponentially distributed random variables as the observation window becomes unbounded.

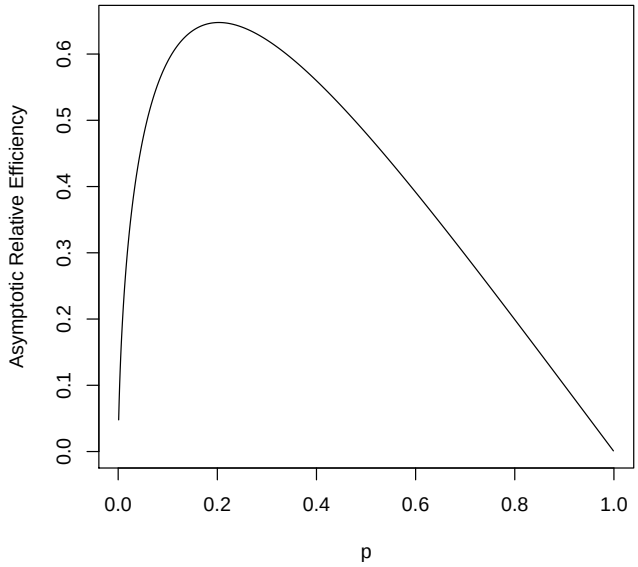

Figure 4.1: Efficiency versus p in the HPP case.

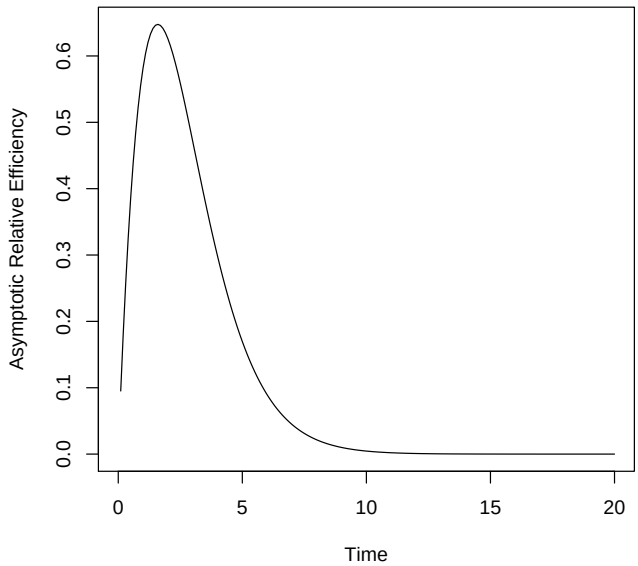

Figure 4.2: Efficiency versus time when $\beta \to 0$ in the HPP case.

Chapter 5

Simulation Study in the Parametric Case

Simulation studies were conducted using the statistical software R. They were performed to examine the small to moderate sample size properties of the estimator $\hat{\bar{F}}(\infty, t)$ and that of $\tilde{\bar{F}}(\infty, t)$ and to ascertain efficiency for small to moderate sample sizes. The goals were to examine the effect of samples size n taken from the set $\{10, 15, 20, 25, 30, 50\}$; to examine the effect of using an increasing or decreasing baseline intensity function and the censoring parameter β which was taken from $\{0.14, 0.17, 0.2, 0.25, 0.33, 0.5\}$ in order to also examine the effect of an increasing proportion of censored observations. For simulation in the general case, the true distribution was chosen to be Weibull with scale parameter $\alpha = 1$ and shape parameter μ taken from the set $\{.5, 1, 2\}$. The two values of the shape parameter .5 and 2 are taken in order to assess the effect of increasing and decreasing failure rate. The data

created were a combination of n, $\alpha = 1$, μ, and β. A total of $1,000$ simulated samples were generated for each (n, μ, α, β), hence a total of $1000 \times 6 \times 6 \times 2$ data sets were generated. Simulated biases, variances, and mean square-errors (MSE) were then obtained for values of $t \in [0.2, 2]$ by an increment of 0.1 when taking into account informative censoring and the case when it is ignored.

5.1 Simulation Results in the General Case

5.1.1 Results by exploiting the KG model in the general case

Figures 5.1-5.3, respectively, show graphs of biases, variances and MSEs for different values of β and n when we exploit the KG structure to estimate the survivor function $\bar{F}(t)$. The graphs are shown for the Weibull distribution with scale parameter $\alpha = 1$ and shape parameter $\mu = 2$ [which corresponds to an increasing failure rate]. For brevity, we have suppressed the graph for $\alpha = 1$ and shape parameter $\mu = .5$ [which corresponds to a decreasing failure rate] but will provide discussion of them. For both values of the shape parameter, the estimators appear to have little bias and the biases, the MSEs and variances are decreasing as the sample size increases, which supports our theoretical results. However, these results are less noticeable for the case where $\mu = 2$. It is noted that in both cases, as β increases [that is, as the proportion of censored observations increases], the MSEs, biases, and variances also increase, which is not surprising. The biases, MSEs and variances have a decreasing trend as time increases and are more stable when using a decreasing failure rate, while we

observed an increasing MSE and variance in $[0.2, 1]$ and then a decrease in $[1, 2]$ for the increasing failure rate. However, there is a lot of variability in the biases in the case of an increasing failure rate. This may be due to the fact that for increasing failure rate, the number of events occurrence increase and consequently the estimator of the survivor function is affected.

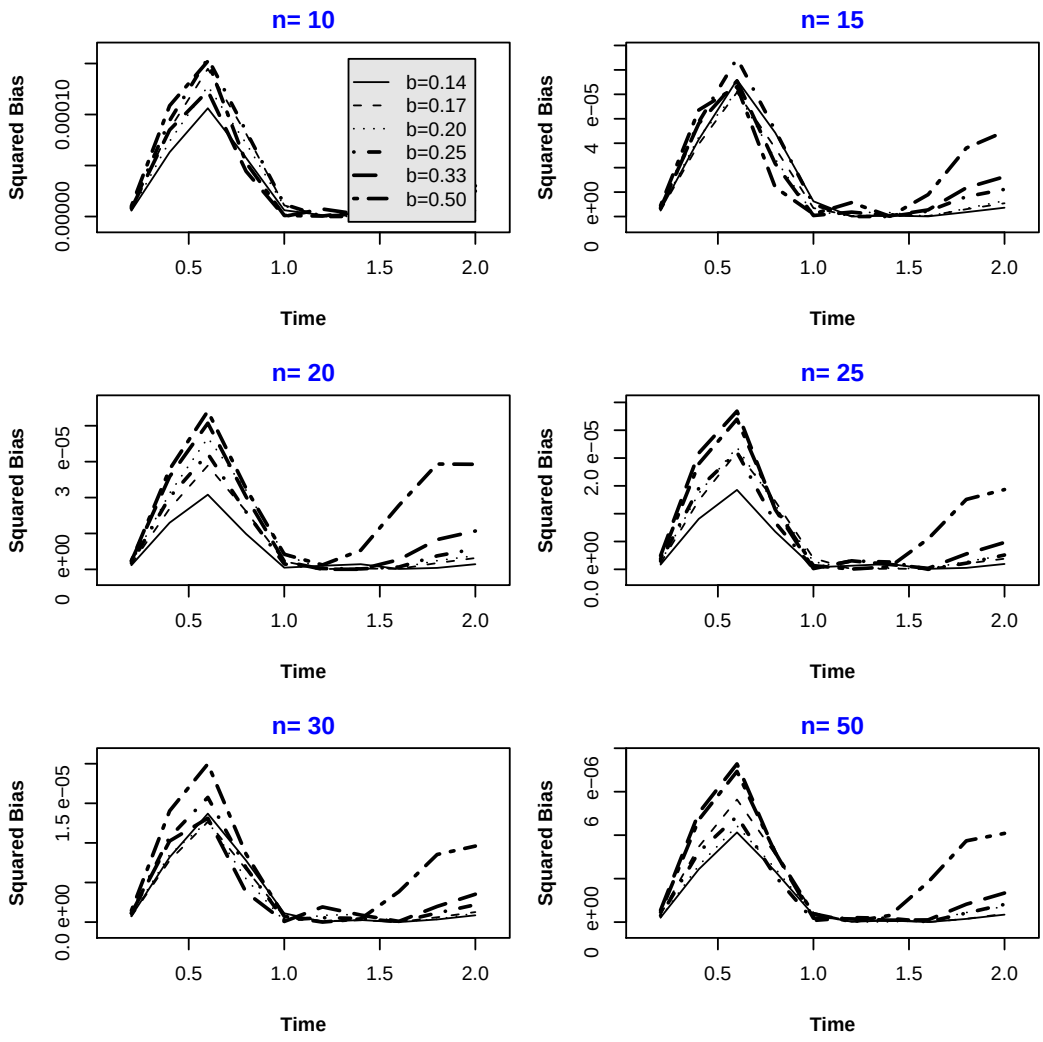

Figure 5.1: Biases in the estimator of $\bar{F}(t)$ for different values of n and β under the KG model. True distribution is Weibull (1, 2).

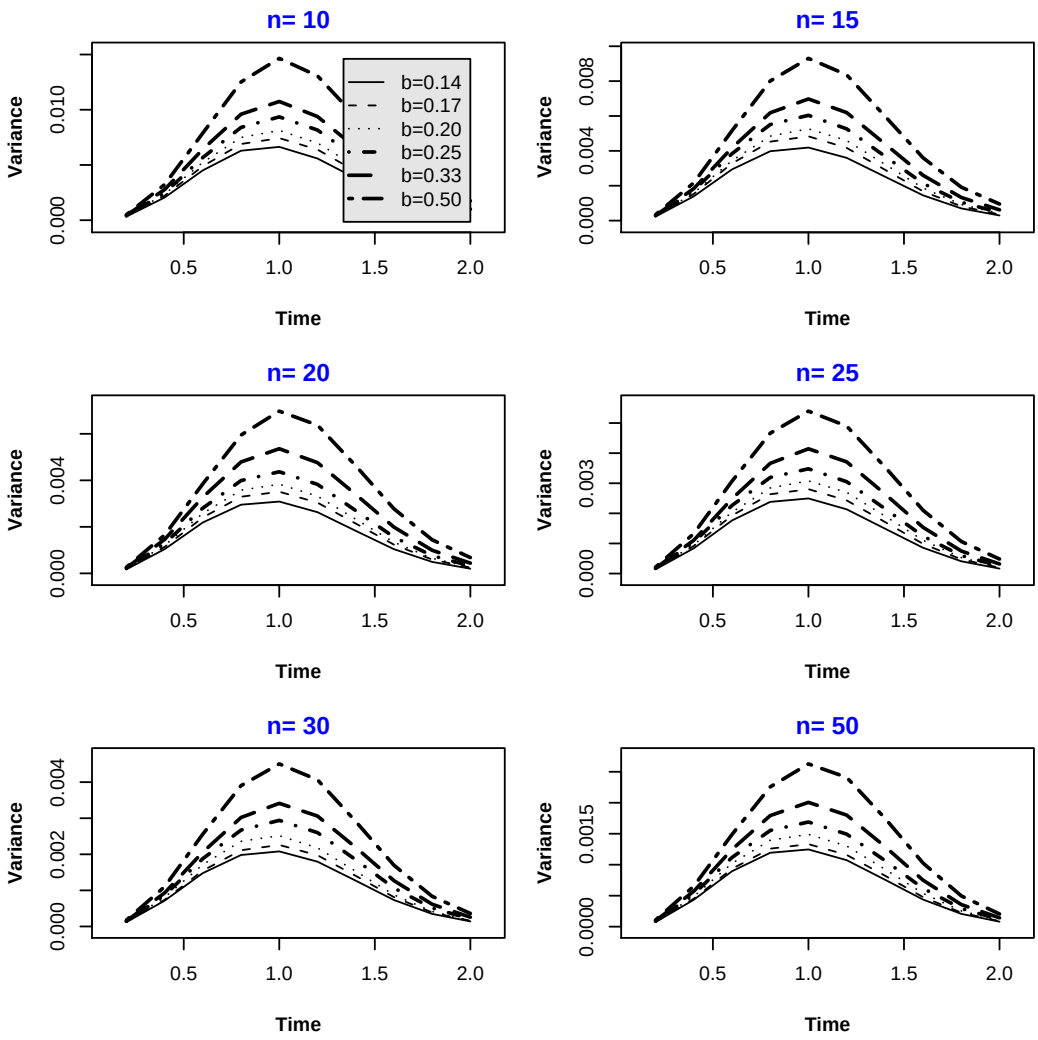

Figure 5.2: Variances in the estimator of $\bar{F}(t)$ for different values of n and β under the KG model. True distribution is Weibull (1, 2).

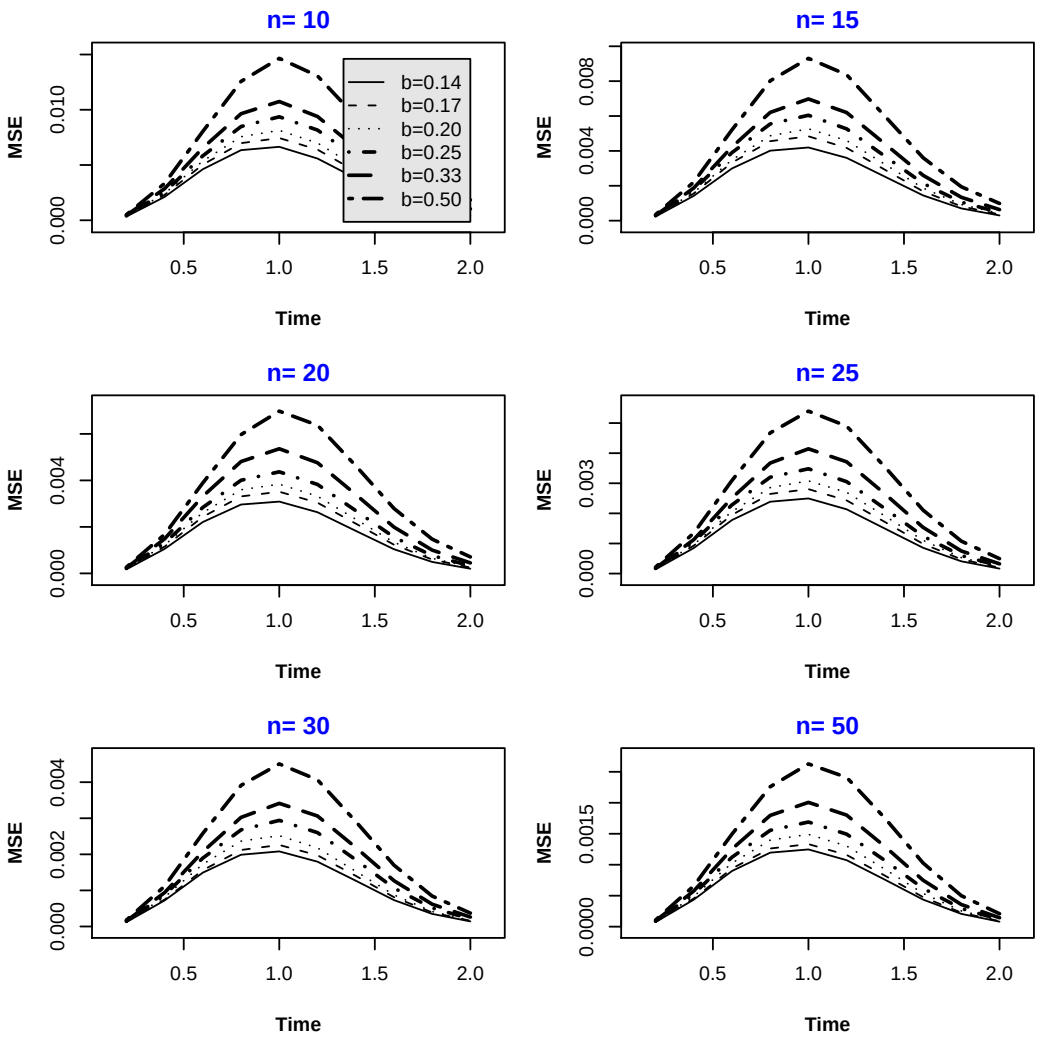

Figure 5.3: MSE in the estimator of $\bar{F}(t)$ for different values of n and β under the KG model. True distribution is Weibull (1, 2)

5.1.2 Results by ignoring the KG Model in the general case

Figures 5.4−5.6 give simulation results for small to moderate samples and for different values of β. The shape of the graphs in each case appears to be the same. It can also be seen that as sample size increases, the estimator appears to have little bias and the bias becomes smaller as the sample size increases. This supports the theoretical

results obtained in Chapter 3 on the asymptotic distribution of $\bar{F}(t)$ when ignoring the KG structure. Although the biases, variances and MSEs are all larger when we exploit the KG structure and when the true distribution is Weibull with scale parameter 1 and shape parameter .5 [decreasing failure rate] as compared to when we ignore it, it should be noted that their respective graphs have a decreasing trend as time increases, whereas we observe that the graphs increase to a maximum value, then decrease when a Weibull with shape parameter 2 [increasing failure rate] is used.

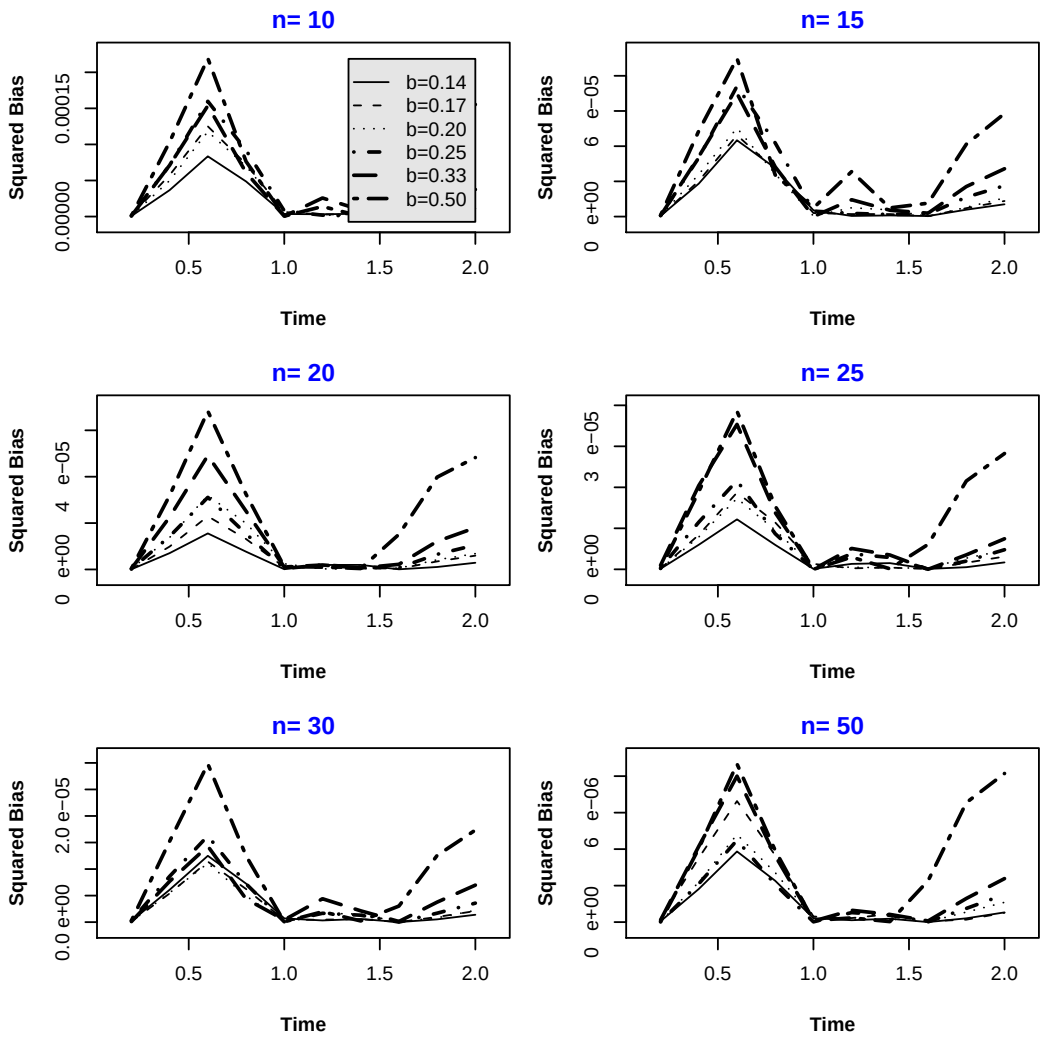

Figure 5.4: Biases in the estimator of $\bar{F}(t)$ for different values of n and β when ignoring the KG model. True distribution is Weibull (1, 2)

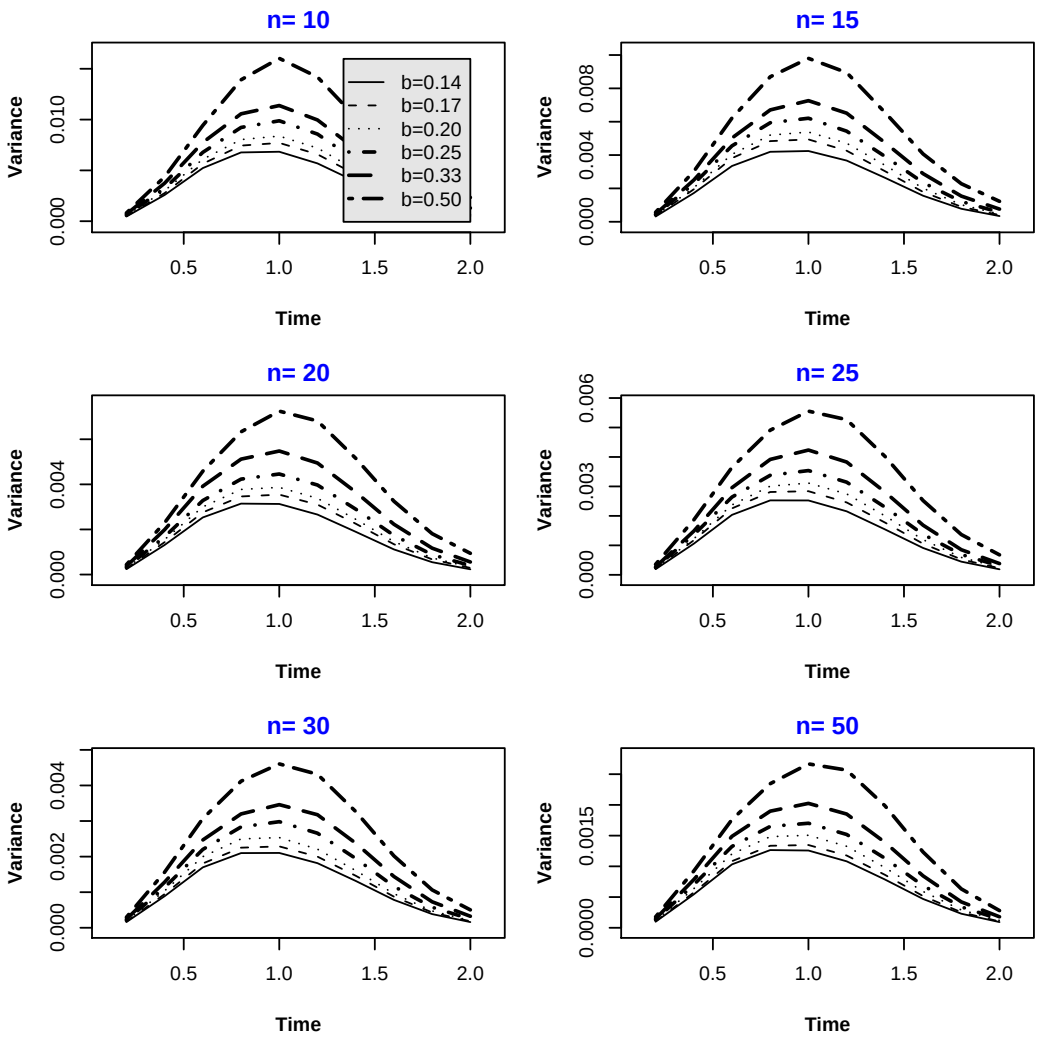

Figure 5.5: Variances in the estimator of $\bar{F}(t)$ for different values of n and β when ignoring the KG model. True distribution is Weibull (1, 2).

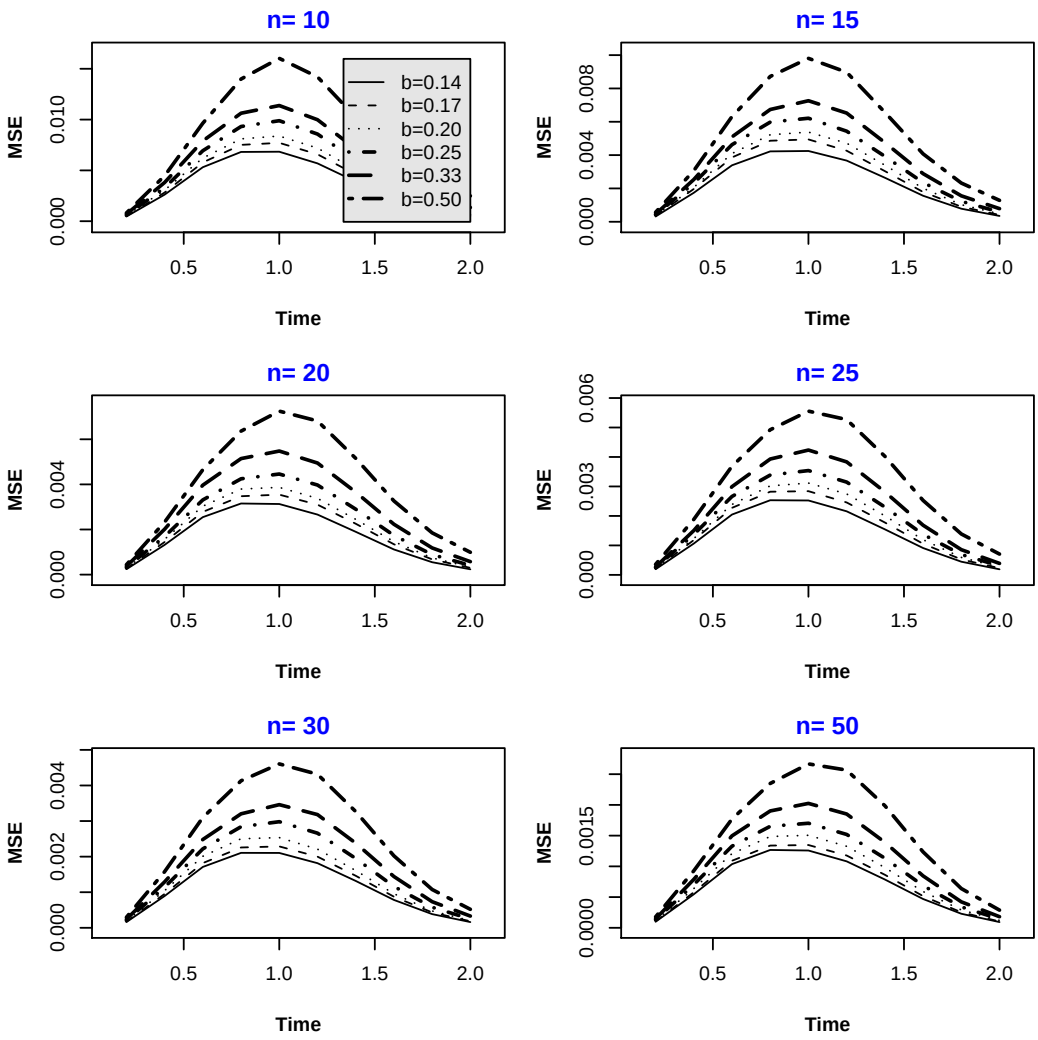

Figure 5.6: MSE in the estimator of $\bar{F}(t)$ for different values of n and β when ignoring the KG model. True distribution is Weibull (1, 2).

5.1.3 Efficiency comparison in the general case

Figures 5.7 and 5.8 show graphs of efficiency of the estimator of the gap-time distribution when the true distribution is Weibull (1, .5) and Weibull (1, 2), respectively. In both cases, we notice that the efficiency increases to some value and then decreases. However, the efficiency is larger than 1 for some values of t and for small sample

sizes when a decreasing failure rate is used. Nevertheless, as sample size increases, the estimator under the KG model becomes more efficient. In contrast, efficiency is always less than 1 for any sample size and any time t when using a Weibull $(1, 2)$.

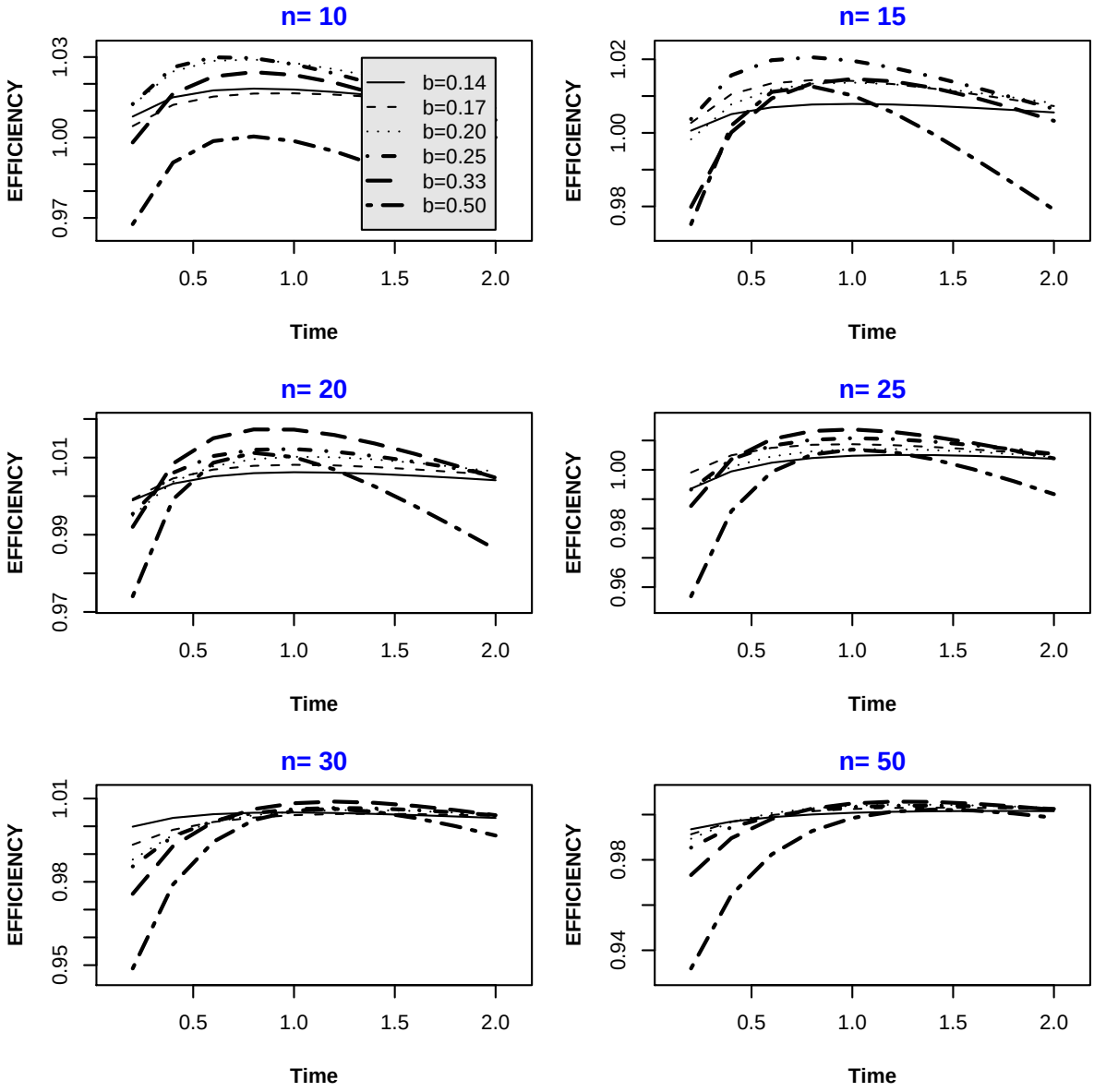

Figure 5.7: Efficiency of the estimator obtained ignoring informative censoring versus that obtained when it is taken into account for different values of n and β when the true distribution is assumed to be Weibull(1, .5).

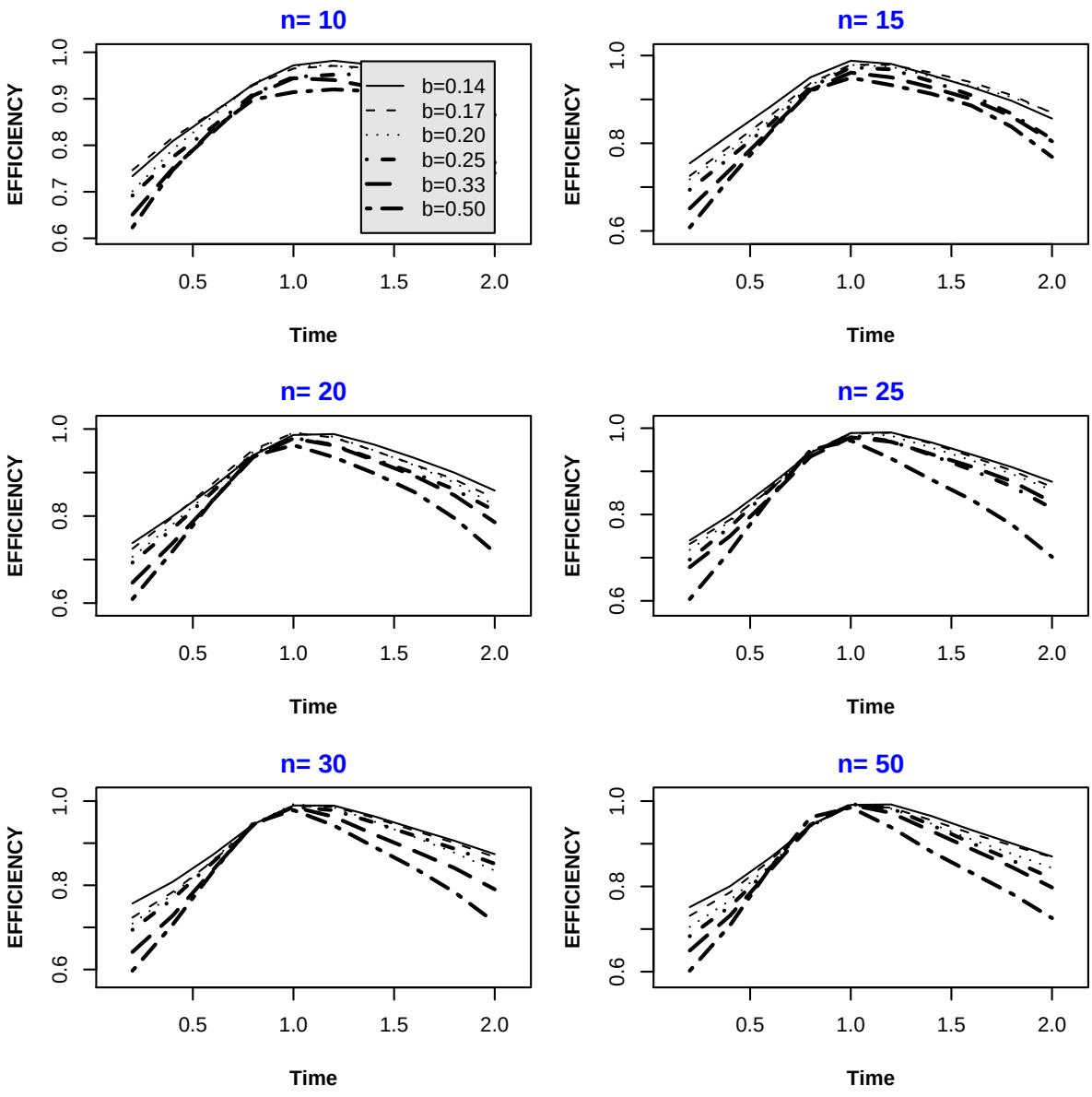

Figure 5.8: Efficiency of the estimator obtained ignoring informative censoring versus that obtained when it is taken into account for different values of n and β when the true distribution is assumed to be Weibull(1, 2).

5.2 Simulation Results in HPP Case

For this particular case of the HPP, data were generated in the same manner, but now using an exponential distribution with parameter $\theta = 1$. The τs were generated from an exponential distribution with parameter β. Graphs of biases, MSEs and variances were obtained but are not shown here. However, they support our theoretical results as the sample size increases. The Figures 5.7-5.9 of efficiency support our theoretical results of efficiency as being bounded by $\simeq 0.65$.

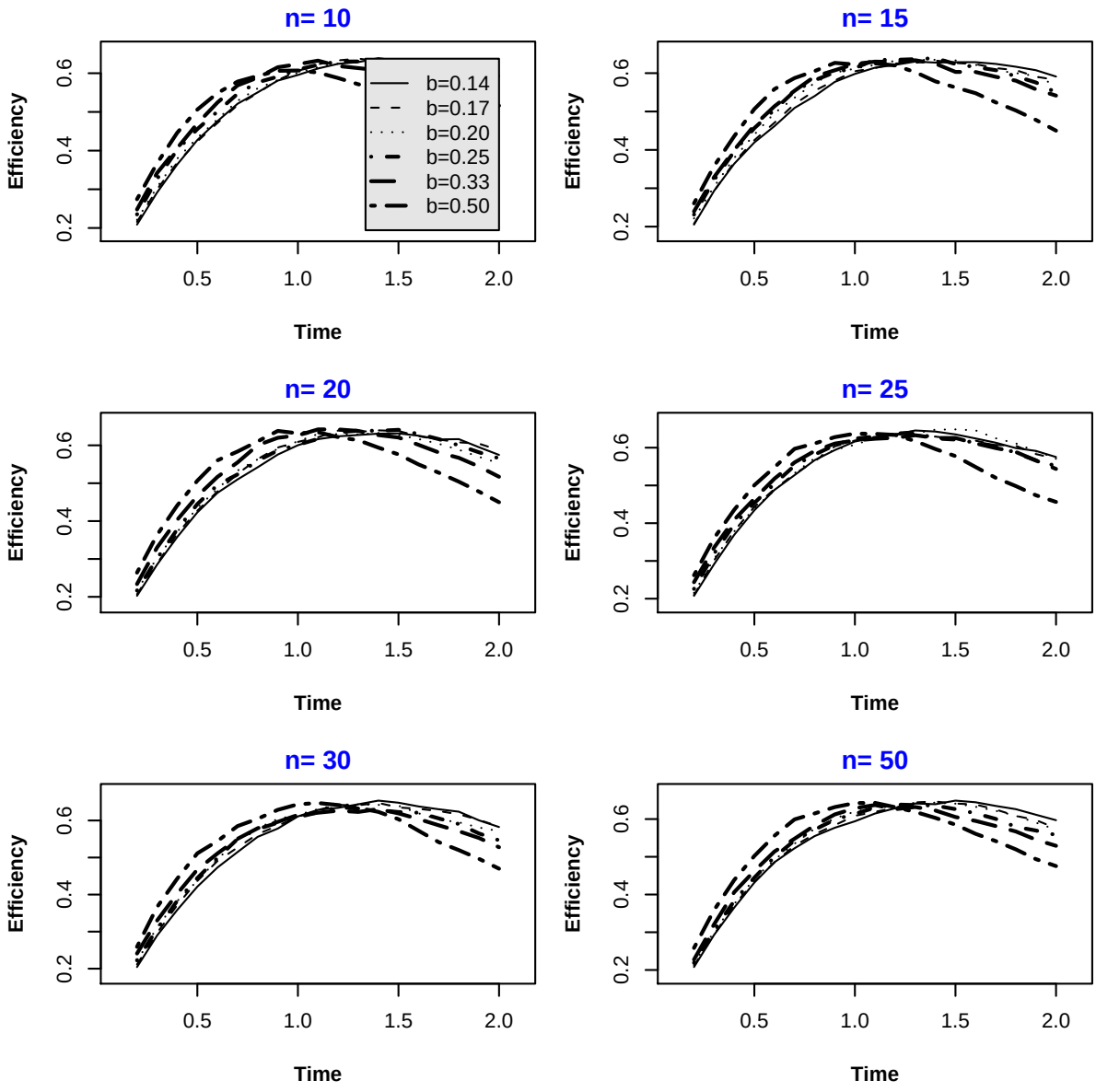

Figure 5.9: Efficiency of the estimator obtained ignoring informative censoring versus that obtained when it is taken into account for different values of n and β in the homogeneous Poisson process case.

Chapter 6

Semiparametric Estimation

6.1 Introduction

In this chapter, we discuss semiparametric estimation of the survivor function of the gap-time distribution under the KG model. Let us review some research that has been done on nonparametric estimation with recurrent event data. As mentioned in the introduction, this problem has been considered by Gill [[19], [20]], Wang and Chang [44] and, recently, Peña et al. [30], among others. For instance, Gill [[19], [20]] derived Nelson-Aalen and Kaplan-Meier type estimators and established their asymptotic properties using empirical process techniques. Wang and Chang [44] considered a weighted moment estimator of the gap-time distribution. Their estimator is based on all subjects who experience at least one event. Moreover, it ignores the last censored observation. As in Gill [[19], [20]], they also derived asymptotic properties using empirical process techniques. Peña et al. [30] derived analogs of Kaplan-Meier

and Nelson-Aalen estimators and obtained their asymptotic properties using counting process techniques which was done by embedding the problem into an appropriate counting process framework that utilizes both calendar time and gap-time.

Although our estimators and those in Peña et al. [30] are based on the same type of data, the most important difference is the way in which informative censoring is handled. In the fully nonparametric estimator in Peña et al. [30], the impact of the data accrual scheme, which is inherent in the monitoring of recurrent events was manifested through the occurrence of the renewal function in the asymptotic variance. In contrast, in our semiparametric setting, we further examine and explore the impact of an informative monitoring period. As in the estimation of preceding chapters, the KG model will be used as a model for informative censoring.

6.2 Doubly-Indexed Process

Estimating the cumulative hazard, and consequently, the gap-time distribution should be based on counting processes and at-risk processes that are associated with the failure times T_{ij}s. These processes are not adapted to the filtration on which the recurrent event is observed, which is a calendar time filtration. For this semiparametric estimation, the stochastic processes we should consider turn out to be those indexed by both gap-time and calendar time [cf. Sellke [36] and Peña et al. [30]]. In what follows, λ represents λ_F, the hazard of F. Recall from Chapter 2, $Z_i(s,t) = I(R_i(s) \leq t)$; $i = 1, ..., n$, indicates whether at most t time units have elapsed since the last event

occurrence. For $s, t \geq 0$ and $i = 1, ..., n$, define

$$\begin{aligned} N_i(s,t) &= \int_0^s Z_i(w,t) dN_i^\dagger(w); \\ A_i(s,t) &= \int_0^s Z_i(w,t) dA_i^\dagger(w); \\ M_i(s,t) &= \int_0^s Z_i(w,t) dM_i^\dagger(w); \end{aligned} \tag{6.1}$$

where $N_i^\dagger(w)$, $A_i^\dagger(w)$ and $M_i^\dagger(w)$ are the calendar time processes defined in Chapter 3. Notice here that, since $N_i^\dagger(s)$ is a jump process, $N_i(s,t)$ is a summation, therefore

$$\begin{aligned} N_i(s,t) &= \int_0^s Z_i(w,t) dN_i^\dagger(w) \\ &= \sum_{j=1}^{N_i^\dagger(s-)} I(R_i(S_{ij}) \leq t) \\ &= \sum_{j=1}^{N_i^\dagger(s-)} I(S_{ij} - S_{ij-1} \leq t) \\ &= \sum_{j=1}^{N_i^\dagger(s-)} I(T_{ij} \leq t). \end{aligned}$$

Hence, $N_i(s,t)$ counts the number of observed events occurring over the calendar period $[0, s]$ whose interoccurrence times are at most t. Next, consider $A_i(s,t)$. We

will now provide an alternative form for $A_i(s,t)$, [cf. also Peña et al. [30] and [32]].

$$\begin{aligned}
A_i(s,t) &= \int_0^s Z_i(w,t)dA_i^\dagger(w) \\
&= \int_0^s Z_i(w,t)Y_i^\dagger(w)\lambda(R_i(w))dw \\
&= \sum_{j=1}^{N_i^\dagger(s-)} \int_{S_{ij-1}}^{S_{ij}} I(R_i(w)\le t)I(\tau_i \ge w)\lambda(R_i(w))dw \\
&\quad + \int_{S_{iN_i^\dagger(s-)}}^{s} I(R_i(w)\le t)I(\tau_i \ge w)\lambda(R_i(w))dw.
\end{aligned}$$

A change of variable $v = w - S_{ij-1}$ in the previous equation gives

$$\begin{aligned}
A_i(s,t) &= \sum_{j=1}^{N_i^\dagger(s-)} \int_0^{T_{ij}} I(v\le t)I(\tau_i \ge v + S_{ij})\lambda(v)dv \\
&\quad + \int_0^{s-S_{iN_i^\dagger(s-)}} I(v\le t)I(\tau_i \ge v + S_{iN_i^\dagger(s-)})\lambda(v)dv \\
&= \int_0^t \sum_{j=1}^{N_i^\dagger(s-)} I(T_{ij}\ge w)I(\tau_i \ge v + S_{ij-1}) \\
&\quad + \int_0^t I(s - S_{iN_i^\dagger(s-)} \ge w) \times I(\tau_i \ge v + S_{iN_i^\dagger(s-)})\lambda(v)dv \\
&= \int_0^t \Big\{ \sum_{j=1}^{N_i^\dagger(s-)} I(T_{ij}\ge w) + I(s\wedge\tau_i - S_{iN_i^\dagger(s-)} \ge w\Big\}\lambda(v)dv \\
&= \int_0^t Y_i(s,w)\lambda(w)dw;
\end{aligned}$$

where $Y_i(s,t)$ is the generalized at-risk process defined in Chapter 3. It can be seen that $A_i(s,t)$ now has a multiplicative form. Moreover, $Y_i(\cdot,t)$ is the at-risk process corresponding to the process $N_i(\cdot,t)$. Now, from the above two expressions, we have

$$M_i(s,t) = N_i(s,t) - A_i(s,t) \tag{6.2}$$

is, for fixed $t \geq 0$, a square-integrable martingale with respect to $\mathcal{F}_s$, $s > 0$. However, the process $M_i(s, \cdot)$ is not a martingale for a fixed s. It can be shown that, as $s \to \infty$, the doubly-indexed and at-risk processes obtained correspond to the gap-time processes. [cf. Peña et al. [30] for details].

6.3 Method of Moment Estimators

We address, in this section, the problem of estimating the model parameters $\Lambda(\cdot) = \int_0^{\cdot} \lambda(w)dw$. An estimator of the survivor function of the gap-time distribution $\bar{F}(t)$ will follow by virtue of the relationship between $\bar{F}(t)$ and $\Lambda(t)$; that is, $\bar{F}(t) = \prod_{w=0}^{t}\{1 - \Lambda(dw)\}$, where $\prod$ represents product integration. Toward this end, we define the following aggregated processes:

$$\begin{aligned} N(s,t) &= \sum_{i=1}^{n} N_i(s,t); \\ A(s,t) &= \sum_{i=1}^{n} A_i(s,t) = \int_0^t \left\{ \sum_{i=1}^{n} Y_i(s,t) \right\} \lambda(w)dw; \\ M(s,t) &= \sum_{i=1}^{n} M_i(s,t). \end{aligned} \tag{6.3}$$

Additionally, letting $Y(s,t) = \sum_{i=1}^{n} Y_i(s,t)$, we have

$$M(s,t) = N(s,t) - A(s,t) = N(s,t) - \int_0^t Y(s,w)\lambda(w)dw. \tag{6.4}$$

Notice that $M(\cdot, t)$ is a zero-mean square integrable martingale with predictable variation process given by

$$\langle M(\cdot, t), M(\cdot, t)\rangle(s) = \int_0^t Y(s, w)\lambda(w)dw. \tag{6.5}$$

In order to derive the semiparametric estimator of $\Lambda(\cdot)$, we will need a result that connects stochastic integration over calendar time to those over gap-time. This result is stated here in a specialized form [see Proposition 1 of Peña et al. [32]] adapted to our needs.

Proposition 6.1 *Let $R_i(w)$ be as before and let $H_i(s, R_i(w))$ be a bounded $\mathcal{F}$-predictable process. Then, we have*

$$\int_0^s H_i(s, R_i(w))M_i(dw, t) = \int_0^t H_i(s, v)M_i(s, dv).$$

Before continuing on with the derivation of the estimator of $\Lambda(t)$, we define

$$N^\tau(s) = \sum_{i=1}^n N_i^\tau(s), \quad Y^\tau(s) = \sum_{i=1}^n Y_i^\tau(s), \quad A_i^\tau(s) = \int_0^s Y_i^\tau(w)\beta\lambda(w)dw; \tag{6.6}$$

where we $N_i^\tau(s) = I\{\tau_i \leq s\}$ and $Y_i^\tau(s) = I\{\tau_i \geq s\}$. Since $Y^\tau(s)$ is the at-risk process associated with $N^\tau(s)$, we have that $M^\tau(s) = N^\tau(s) - A^\tau(s)$ is a zero-mean square integrable martingale with respect to $\mathcal{F}_0 \vee \sigma\{N^\tau(s), Y^\tau(s); s > 0\}$. Let $s^\star = \tau_{(n)}$, the maximum of the $\tau_i s$. Since $s^\star$ is fixed, one can take $t = s^\star = s \geq \tau_{(n)}$. Hence, from

(6.2) and (6.6), we have

$$
\begin{aligned}
M(s,dt) &= N(s,dt) - Y(s,t)\Lambda(dt); && (6.7)\\
M^{\tau}(dt) &= N^{\tau}(dt) - \beta Y^{\tau}(t)\Lambda(dt); && (6.8)\\
M(s,dt) + M^{\tau}(dt) &= N(s,dt) + N^{\tau}(dt) - \{Y(s,t) + \beta Y^{\tau}(t)\}\Lambda(dt). && (6.9)
\end{aligned}
$$

Let $J(s,w;\beta) = I\{Y(s,w)+\beta Y^{\tau}(w) > 0\}$ and note that the processes $Y_i(s,w)$, $Y_i^{\dagger}(s)$ and $J(s,w;\beta)$ are bounded and left-continuous for each fixed w, hence predictable. It follows from (6.9) that

$$
\begin{aligned}
\int_0^t \frac{J(s,w;\beta)\{M(s,dw) + M^{\tau}(dw)\}}{Y(s,w) + \beta Y^{\tau}(w)} &= \int_0^t \frac{J(s,w;\beta)\{N(s,dw) + N^{\tau}(dw)\}}{Y(s,w) + \beta Y^{\tau}(w)} \\
&\quad - \int_0^t J(s,w;\beta)\Lambda(dt). \qquad (6.10)
\end{aligned}
$$

Using Proposition 6.1, we have

$$
\begin{aligned}
\int_0^t \frac{J(s,w;\beta)\{M(s,dw) + M^{\tau}(dw)\}}{Y(s,w) + \beta Y^{\tau}(w)} &= \sum_{i=1}^{n} \int_0^t \frac{J(s,w;\beta)\{M_i(s,dw) + M_i^{\tau}(dw)\}}{Y(s,w) + \beta Y^{\tau}(w)} \\
&= \sum_{i=1}^{n} \int_0^s \frac{J(s,R_i(v);\beta)\{M_i(dv,t) + M_i^{\tau}(dv)\}}{Y(s,R_i(v)) + \beta Y^{\tau}(R_i(v))}.
\end{aligned}
$$

It then follows that

$$
\left\{ \sum_{i=1}^{n} \int_0^s \frac{J(s,R_i(v);\beta)}{Y(s,R_i(v)) + \beta Y^{\tau}(R_i(v))} \{M_i(dv,t) + M_i^{\tau}(dv)\}; s \geq 0 \right\}
$$

is a zero-mean square integrable martingale for every fixed $t \geq 0$. Hence, we have the following moment identity:

$$E\left\{\int_0^t \frac{J(s,w;\beta)\{N(s,dw)+N^\tau(dw)\}}{Y(s,w)+\beta Y^\tau(w)}\right\} = E\left\{\int_0^t J(s,w;\beta)\Lambda(dw)\right\}. \tag{6.11}$$

The above moment identity motivates a Nelson-Aalen type estimator for $\Lambda(\cdot)$ since the numerator of $\frac{N(s,dw)+N^\tau(dw)}{Y(s,w)+\beta Y^\tau(w)}$ is the total number of event occurrences, including all the censoring, while the denominator is the generalized at-risk process, plus the number of subjects still at risk. Notice in the previous equation that β is unknown. Using (6.11), an 'estimator' of Λ for a fixed t given the value of β in the interval $[0,s]$ [which we denote by $\widetilde{\Lambda}(s,t)|\beta$] is subsequently obtained and given by

$$\widetilde{\Lambda}(s,t|\beta) = \int_0^t J(s,w;\beta)\left\{\frac{N(s,dw)+N^\tau(dw)}{Y(s,w)+\beta Y^\tau(w)}\right\}; \tag{6.12}$$

and the corresponding product-limit "estimator" of $\bar{F}$ is

$$\widetilde{\bar{F}}(s,t;\beta) = \prod_{w=0}^{t}\{1-\widetilde{\Lambda}(s,dw|\beta)\} = \prod_{w=0}^{t}\left\{1-\frac{N(s,dw)+N^\tau(dw)}{Y(s,w)+\beta Y^\tau(w)}\right\}; \tag{6.13}$$

where the symbol '$\prod$' represents product integration. The next section deals with the estimation of β.

6.4 Estimation of β

An estimator $\hat{\Lambda}(\cdot)$ of $\Lambda(\cdot)$ is obtained by first obtaining an estimator of β which will be plugged in $\widetilde{\Lambda}(s,t|\beta)$ to finally obtain $\hat{\Lambda}(\cdot)$. To obtain an estimator of β, we need to maximize the generalized log-likelihood process given next over $\widetilde{\Lambda}(s,t|\beta)$. This approach was also taken by Andersen and Gill [2], Peña et al. [31] and Kvam and Peña [28]. For instance, Andersen and Gill [2] derived an estimator of the cumulative hazard function under a Cox's regression model for counting process by first estimating the unknown regression coefficient $\boldsymbol{\beta}$. Asymptotic properties of the estimators were also derived. Using a counting processes approach in their 2005 paper, Kvam and Peña [28] proposed a semiparametric estimator for the component baseline cumulative hazard in an equal load-share model. The semiparametric estimator was obtained by first obtaining an estimator of the load-share parameters via a likelihood profile approach. Asymptotic properties of the cumulative hazard, as well as those of the load-share parameter were also derived. Peña et al. [31], using similar techniques, developed procedures for estimating the parameters of the general class of semiparametric models for recurrent events. We now develop the profile likelihood for β. Let $s^\star = \max \tau_i$. Following Jacod [24], the full generalized likelihood process is given by

$$\begin{aligned} L(s^\star, t|\beta) &= \prod_{i=1}^{n}\prod_{v=0}^{s^\star}\{Y_i^\dagger(v)\lambda(R_i(v))dv\}^{Z_i(v,t)\triangle N_i^\dagger(v)}\{Y_i^\tau(v)\beta\lambda(v)dv\}^{\triangle N_i^\tau(v)} \\ &\quad \times \exp\left\{-\sum_{i=1}^{n}\int_0^{s^\star} Z_i(v,t)Y_i^\dagger(v)\lambda(R_i(v))dv \right. \\ &\quad \left. -\sum_{i=1}^{n}\int_0^{s^\star} Y_i^\tau(v)\beta\lambda(v)dv\right\}. \end{aligned} \tag{6.14}$$

To obtain the likelihood profile, the "estimator" $\widetilde{\Lambda}(s,t|\beta)$ will be substituted in the generalized likelihood in (6.14) and the resulting expression will yield the profile likelihood $L_p(s^\star,t;\beta)$. Toward this end, we will consider the different portions in the right-hand side of (6.14). Consider first the argument in the exponential function. By taking $t^\star = s^\star \geq \max \tau_i$, and noting that

$$\widetilde{\Lambda}(s,dt|\beta) = \frac{N(s,dt)+N^\tau(dt)}{Y(s,t)+\beta Y^\tau(t)};$$

it follows that

$$\begin{aligned}&\sum_{i=1}^n \int_0^{s^\star} Z_i(v,t)Y_i^\dagger(v)\lambda(R_i(v))dv + \sum_{i=1}^n \int_0^{s^\star} Y_i^\tau(v)\beta\lambda(v)dv \\ &= \int_0^{s^\star} \{Y(s^\star,w)+\beta Y^\tau(w)\}\lambda(w)dw.\end{aligned} \tag{6.15}$$

Now, using the expression of $\widetilde{\Lambda}(s,t|\beta)$ in (6.12), it is easily seen that

$$\begin{aligned}\int_0^{s^\star} \{Y(s^\star,w)+\beta Y^\tau(w)\}\lambda(w)dw &= \int_0^{s^\star} \{Y(s^\star,w)+\beta Y^\tau(w)\}\frac{N(s,dw)+N^\tau(dw)}{Y(s,w)+\beta Y^\tau(w)} \\ &= \int_0^{s^\star} \{N(s,dw)+N^\tau(dw)\}. \end{aligned} \tag{6.16}$$

It can be seen that the expression in the right-hand side of (6.16) is independent of β and therefore, the exponential part does not contribute to the likelihood profile. Next, consider the term $\prod_{i=1}^n \prod_{v=0}^{s^\star} \{Y_i^\tau(v)\beta\lambda(v)dv\}^{\triangle N_i^\tau(v)}$. Again, substituting

$\widetilde{\Lambda}(s, dv|\beta)$ for $\lambda(v)dv$, we obtain

$$\prod_{i=1}^{n}\prod_{v=0}^{s^\star}\left\{Y_i^\tau(v)\beta\frac{N(s^\star, dv)+N^\tau(dv)}{Y(s^\star, v)+\beta Y^\tau(v)}\right\}^{\triangle N_i^\dagger(v)} \propto \prod_{i=1}^{n}\prod_{v=0}^{s^\star}\left\{\frac{\beta}{Y(s^\star, v)+\beta Y^\tau(v)}\right\}^{\triangle N_i^\tau(v)}. \tag{6.17}$$

For the last term, let $\triangle f(t)$ represent the jump of $f(t)$ at t, thus

$$\begin{aligned}
&\prod_{i=1}^{n}\prod_{v=0}^{s^\star}\{Y_i^\dagger(v)\lambda(R_i(v))dv\}^{Z_i(v,t)\triangle N_i^\dagger(v)} \\
&= \prod_{i=1}^{n}\prod_{v=0}^{s^\star}\left\{Y_i^\dagger(v)\triangle\Lambda(R_i(v))dv\right\}^{Z_i(v,t)\triangle N_i^\dagger(v)} \\
&= \prod_{i=1}^{n}\prod_{v=0}^{s^\star}\left\{Y_i^\dagger(v)\frac{\triangle N(s^\star, R_i(v))+\triangle N^\tau(R_i(v))}{Y(s^\star, R_i(v))+\beta Y^\tau(R_i(v))}\right\}^{Z_i(v,t)\triangle N_i^\dagger(v)} \\
&\propto \prod_{i=1}^{n}\prod_{v=0}^{s^\star}\left\{\frac{1}{Y(s^\star, R_i(v))+\beta Y^\tau(R_i(v))}\right\}^{Z_i(v,t^\star)\triangle N_i^\dagger(v)} \\
&= \prod_{i=1}^{n}\prod_{j=1}^{N_i^\dagger(s^\star)}\left\{\frac{1}{Y(s^\star, R_i(S_{ij}))+\beta Y^\tau(R_i(S_{ij}))}\right\}^{Z_i(S_{ij},t^\star)\triangle N_i^\dagger(S_{ij})}.
\end{aligned} \tag{6.18}$$

Combining (6.17) and the last equation in (6.18), the relevant portion of the profile likelihood is:

$$\begin{aligned}
L_p(\beta; s^\star, t^\star) \;=\; &\prod_{i=1}^{n}\prod_{v=0}^{s^\star}\left\{\frac{1}{Y(s^\star, R_i(v))+\beta Y^\tau(R_i(v))}\right\}^{Z_i(v,t^\star)\triangle N_i^\dagger(v)} \\
&\times\left\{\frac{\beta}{Y(s^\star, v)+\beta Y^\tau(v)}\right\}^{\triangle N_i^\tau(v)};
\end{aligned} \tag{6.19}$$

and the log-likelihood profile is therefore given by

$$
\begin{aligned}
l_p(\beta; s^\star, t^\star) &= -\sum_{i=1}^{n} \int_0^{s^\star} \log\{Y(s^\star, R_i(v)) + \beta Y^\tau(R_i(v))\} Z_i(v, t^\star) dN_i^\dagger(v) \\
&\quad + \sum_{i=1}^{n} \int_0^{s^\star} \{\log \beta - \log\{Y(s^\star, v) + \beta Y^\tau(v)\}\} dN_i^\tau(v) \\
&= -\sum_{i=1}^{n} \int_0^{s^\star} \log\{Y(s^\star, R_i(v)) + \beta Y^\tau(R_i(v))\} N_i(dv, t^\star) \\
&\quad + \sum_{i=1}^{n} \int_0^{s^\star} \{\log \beta - \log\{Y(s^\star, v) + \beta Y^\tau(v)\}\} dN_i^\tau(v). \qquad (6.20)
\end{aligned}
$$

From this profile log-likelihood, the estimator $\hat{\beta}$ of β will be obtained. By taking the derivative of the profile log-likelihood, we obtain

$$
\begin{aligned}
\frac{\partial}{\partial \beta} l_p(\beta; s^\star, t^\star) &= \sum_{i=1}^{n} \int_0^{s^\star} \frac{-Y^\tau(R_i(v))}{Y(s^\star, R_i(v)) + \beta Y^\tau(R_i(v))} N_i(dv, t^\star) \\
&\quad + \frac{n}{\beta} - \sum_{i=1}^{n} \int_0^{s^\star} \frac{Y^\tau(v)}{Y(s^\star, v) + \beta Y^\tau(v)} dN_i^\tau(v) \\
&= \int_0^{t^\star} \frac{-Y^\tau(R_i(v))}{Y(s^\star, R_i(v)) + \beta Y^\tau(R_i(v))} N(s^\star, dv) \\
&\quad + \frac{n}{\beta} - \int_0^{t^\star} \frac{Y^\tau(v)}{Y(s^\star, v) + \beta Y^\tau(v)} N^\tau(dv) \\
&= \int_0^{s^\star} \frac{-Y^\tau(v)}{Y(s^\star, v) + \beta Y^\tau(v)} \{N(s^\star, dv) + N^\tau(dv)\} \\
&\quad + \frac{1}{\beta} \int_0^{s^\star} N^\tau(dv). \qquad (6.21)
\end{aligned}
$$

Setting (6.21) to zero yields the estimate $\hat{\beta}$ of β. It is clear that a closed-form expression of $\hat{\beta}$ will be difficult to obtain. However, numerical techniques, such as the Newton-Raphson algorithm can be used to obtain the estimate of $\hat{\beta}$. Upon obtaining the estimator $\hat{\beta}$, the estimator $\hat{\Lambda}(s^\star, t)$ based on data collected on all subjects on

$[0, s^\star]$ is automatically obtained by plugging in $\hat{\beta}$ into the expression of $\widetilde{\Lambda}(s, t|\beta)$. As a consequence, we have

$$\hat{\Lambda}(s^\star, t) = \int_0^t J(s^\star, w; \hat{\beta}) \left\{ \frac{N(s^\star, dw) + N^\tau(dw)}{Y(s^\star, w) + \hat{\beta} Y^\tau(w)} \right\}. \tag{6.22}$$

Notice here that if we remove the terms $N^\tau(dw)$ and $\hat{\beta} Y^\tau(w)$, respectively, in the numerator and denominator of the expression given in (6.22), we obtain the full nonparametric estimator of the cumulative hazard in Peña et al. [30]. The next section deals with the asymptotic properties of our proposed estimators; that is, their consistency and convergence to normal distribution.

6.5 Asymptotic Properties

In this section, we study the large sample properties [i.e $n \to \infty$] of the estimators $\hat{\beta}$ and $\hat{\Lambda}(s, t)$. We start with the consistency of the proposed estimators.

6.5.1 Consistency of the Proposed Estimators

We first deal with the consistency of $\hat{\beta}_n$.

Theorem 6.1 *There exists a sequence of solutions $\hat{\beta}_n$ to the equation $U(\beta; s^\star, t^\star) \equiv \frac{\partial l_p(\beta; s^\star, t^\star)}{\partial \beta} = 0$ with $\hat{\beta}_n \xrightarrow{p} \beta_0$ as $n \to \infty$.*

The proof relies heavily on Proposition 3.1 and Theorem 1 of Peña et al. [32], a result that pertains to weak- convergence to a zero-mean Gaussian process of a sequence of

martingale processes. To facilitate reading of the dissertation, we recall Theorem 1 of Peña et al. [32].

Let $\mathcal{S}$ be a compact subset of $\mathbb{R}$. For $i = 1, ..., n$ and $(s,t) \in \mathcal{S}^2$, let $\mathbf{H}_i(s,t) = (H_{1i}, ..., H_{pi})^t$ be a vector-valued process on $(\Omega, \mathcal{F}, P)$, some probability space; with $H_{ij}(\cdot,\cdot)$ bounded, and for each $t \in \mathcal{S}$, the process $\{\mathbf{H}_i(s,t) : s \in \mathcal{S}\}$ is $\mathbf{F}$-predictable. Next, consider a sequence of processes of the form

$$\mathbf{W}(s,t) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \int_0^t \mathbf{H}_i(s,w) M_i(s,dw), \quad (s,t) \in \mathcal{S}^2. \tag{6.23}$$

These types of processes arise in statistical models concerning renewal processes and recurrent events. Of interest are the asymptotic properties of $\mathbf{W}(s,t)$. For this purpose, consider a triangular sequence of processes of forms given in (6.23), so that for each $n = 1, 2, ...$, we have

$$\begin{aligned} \mathbf{W}^{(n)}(s,t) &= \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \int_0^t \mathbf{H}_i^{(n)}(s,w) dM_i^{(n)}(s,dw); \\ \mathbf{V}^{(n)}(s,t) &= \frac{1}{n} \sum_{i=1}^{n} \int_0^t [\mathbf{H}_i^{(n)}(s,w)]^{\otimes 2} Y_i^{(n)}(s,w) \lambda(w) dw. \end{aligned}$$

The following theorem [for proof of the theorem, cf. Peña et al. [32]] pertains to the weak convergence of the sequence of $\mathbf{W}$-processes.

Theorem 6.2 (Peña et al. [32]) *Fix an $s \in \mathcal{S}$. Suppose the following conditions are satisfied for $t, t_1, t_2 \in [0, t^\star]$ where $t^\star \in \mathcal{S}$:*

1. *The processes $\{\boldsymbol{H}_i(v,w) : 0 \le v \le s; 0 \le w \le t^\star\}$ are left-continuous in (v,w), and there exists a deterministic function $\boldsymbol{h}(v,w)$ on $[0,s] \times [0,t^\star]$ which*

is continuous in (v, w) *and bounded and such that*

$$\max_{1\le i\le n}\sup_{0\le w\le t^\star}|\boldsymbol{H}_i(s,w)-\boldsymbol{h}(s,w)|\xrightarrow{p}0;$$

2. *For each* $s\in\mathcal{S}$, $\inf_{0\le w\le t^\star} y(s,w)>0$ *where* $y(s,w)=E[Y_1(s,w)]$*;*

3. *The matrix function*

$$\boldsymbol{\Sigma}(s,t)=\int_0^t \boldsymbol{h}(s,w)^{\otimes 2}y(s,w)\lambda(w)dw,$$

is such that for each $t_1,t_2\in[0,t^\star]$ *with* $t_1<t_2$, $0<det\{\boldsymbol{\Sigma}(s,t_2)-\boldsymbol{\Sigma}(s,t_1)\}<\infty$*;*

and for each t, *as* $n\to\infty$, $\|\boldsymbol{V}^{(n)}(s,t)-\boldsymbol{\Sigma}(s,t)\|\xrightarrow{p}0$.

Then, as $n\to\infty$, *the sequence of processes* $\{\boldsymbol{W}^{(n)}(s,t):t\in[0,t^\star],n=1,2,...\}$ *converges weakly on Skorohod's space* $\mathcal{D}[0,t^\star]$ *to a Gaussian process* $\{\boldsymbol{W}^{(\infty)}(s,t):t\in[0,t^\star]\}$ *with a zero-mean and covariance matrix function given by,*

$$\boldsymbol{Cov}\{\boldsymbol{W}^{(\infty)}(s,t_1),\boldsymbol{W}^{(\infty)}(s,t_2)\}=\begin{bmatrix}\boldsymbol{\Sigma}(s,t_1) & \boldsymbol{\Sigma}(s,t_1)\\ \boldsymbol{\Sigma}(s,t_1) & \boldsymbol{\Sigma}(s,t_2)\end{bmatrix}.$$

We now proceed with the proof of Theorem 6.1.

Proof of Theorem 6.1: Consider the function of β, $D(\beta;t^\star,s^\star)$, given by

$$D(\beta;t^\star,s^\star)=\frac{1}{n}\{l_p(\beta;t^\star,s^\star)-l_p(\beta_0,t^\star,s^\star)\}\equiv(1/n)X(\beta;t^\star,s^\star),$$

where $l_p(\beta; t^\star, s^\star)$ is the profile likelihood of β given in (6.20), β_0 the true value of β, and β is the free parameter belonging to $\mathcal{B}$, a compact subset of $\mathbb{R}_+^\star$. The proof will follow the same route of consistency proof for a nuisance parameter such us in Andersen and Gill [2]. We need to show that $D(\beta; t^\star, s^\star)$ is a concave function which converges in probability to some concave function of β and then use a concave function argument stated in Corollary 2.2 of the Mathematical Preliminaries to conclude that $\hat{\beta}_n \xrightarrow{p} \beta_0$. [see also Andersen and Gill [2] who used the same approach to show consistency of the vector $\boldsymbol{\beta}$ of unknown regression coefficients in the Cox regression model for counting processes.] Now, after some straightforward algebra, it can be shown that

$$\begin{aligned} X(\beta; t, s) &= -\int_0^t \log\left\{\frac{Y(s,w) + \beta Y^\tau(w)}{Y(s,w) + \beta_0 Y^\tau(w)}\right\} N(s, dw) \\ &\quad - \int_0^t \log\left\{\frac{Y(s,w) + \beta Y^\tau(w)}{Y(s,w) + \beta_0 Y^\tau(w)}\right\} N^\tau(dw) - \log\frac{\beta}{\beta_0}\int_0^t N^\tau(dw) \\ &= -\int_0^t \log\left\{\frac{Y(s,w) + \beta Y^\tau(w)}{Y(s,w) + \beta_0 Y^\tau(w)}\right\} \{N(s, dw) + N^\tau(dw)\} \\ &\quad - \log\frac{\beta}{\beta_0}\int_0^t N^\tau(dw). \end{aligned} \tag{6.24}$$

Since $\log(x)$ is a concave function $[\log(a/b) = -\log(b/a)]$ and both $N_i(s, dw)$ and $N_i^\tau(w)$ are jump processes with a finite number of jumps, each of the terms in (6.24) is a linear combination of concave functions and therefore, $X(\beta; t^\star, s^\star)$ is a concave function. We now need to show that $(1/n)X(\beta; t^\star, s^\star) \equiv D(\beta; t^\star, s^\star)$ converges in

probability to a concave function. Using (6.9), we have

$$\begin{aligned} D(\beta;t,s) &= -\frac{1}{n}\int_0^t \log\left\{\frac{Y(s,w)+\beta Y^\tau(w)}{Y(s,w)+\beta_0 Y^\tau(w)}\right\}\{M(s,dw)+M^\tau(dw)\} \\ &\quad -\frac{1}{n}\int_0^t \log\left\{\frac{Y(s,w)+\beta Y^\tau(w)}{Y(s,w)+\beta_0 Y^\tau(w)}\right\}\{Y(s,w)+\beta Y^\tau(w)\}\lambda(w)dw \\ &\quad -\frac{1}{n}\int_0^t \log\frac{\beta}{\beta_0}M^\tau(dw) - \frac{1}{n}\log\frac{\beta}{\beta_0}\int_0^t \beta Y^\tau(w)\lambda(w)dw. \end{aligned} \tag{6.25}$$

Now, by Proposition 3.1 we have $\sup_{0\le w\le t^\star}|Y(s,w)/n - y(s,w)| = o_p(1)$ and by the weak law of large numbers we have $\sup_{0\le w\le t^\star}|Y^\tau(w)/n - \bar{G}(w)| = o_p(1)$. As a consequence

$$\sup_{0\le w\le t^\star}\left|\frac{\sum_{i=1}^n Y_i(s,w)+\beta\sum_{i=1}^n Y_i^\tau(w)}{n} - \{y(s,w)+\beta\bar{G}(w)\}\right| = o_p(1). \tag{6.26}$$

Next, the result in (6.26) along with the continuity of the logarithm function and Theorem 1 of Peña et al. [32] [or equivalently Theorem 6.2] can be used to show that the first and the third terms in the right-hand side of (6.25) are asymptotically negligible. Assuming $y(s^\star,w)+\beta_0\bar{G}(w) > 0$ and, using $\sup_{0\le w\le t^\star}|Y^\tau(w)/n - \bar{G}(w)| = o_p(1)$ and (6.26), we obtain the limiting expression for $D(\beta;t^\star,s^\star)$ as

$$\begin{aligned} &D(\beta;t^\star,s^\star) \xrightarrow{p} \\ &-\int_0^{t^\star}\log\left\{\frac{y(s^\star,w)+\beta\bar{G}(w)}{y(s^\star,w)+\beta_0\bar{G}(w)}\right\}\{y(s^\star,w)+\beta\bar{G}(w)\}\lambda(w)dw \\ &-\log\frac{\beta}{\beta_0}\int_0^{t^\star}\beta\bar{G}(w)\lambda(w)dw. \end{aligned} \tag{6.27}$$

Denote by $f(\beta)$ the limiting function in (6.27). For $\beta\in\mathcal{B}$, and applying Corollary 5.9 of Bartle [4] [which allows interchanging order of integration for continuous and

bounded integrand], we have that $\frac{\partial f(\beta)}{\partial \beta}$ is zero at $\beta = \beta_0$ and that $\frac{\partial^2 f(\beta)}{\partial \beta^2}$ is always negative for any $\beta \in \mathcal{B}$ and is also negative at $\beta = \beta_0$. Thus, $D(\beta; t^\star, s^\star)$ converges in probability for any $\beta \in \mathcal{B}$ to a concave function with unique maximum at $\beta = \beta_0$. Since $D(\beta; t^\star, s^\star)$, as a function of β, is maximized at $\beta = \hat{\beta}$, when $\hat{\beta}$ exists, it then follows from Corollary 2.2 in the Mathematical Preliminaries that $\hat{\beta} \xrightarrow{p} \beta_0$, which completes the proof of consistency of β.

The next Theorem establishes uniform consistency of $\hat{\Lambda}(s,t)$ and $\hat{\bar{F}}(s,t)$.

Theorem 6.3 *If $t^\star > 0$, $y(s,t^\star) > 0$ and $\Lambda(t^\star)$ is finite, then, as $n \to \infty$, we have:*

1. $\sup_{t \in [0,t^\star]} \left| \hat{\Lambda}(s,t) - \Lambda(t) \right| \xrightarrow{p} 0$;
2. $\sup_{t \in [0,t^\star]} \left| \hat{\bar{F}}(s,t) - \bar{F}(t) \right| \xrightarrow{p} 0$.

Proof: Proof of 1. To this end, define $\Lambda^\star(t) = \int_0^t J(s,w;\beta)\Lambda(dw)$ and by triangle inequality

$$\begin{aligned} &\sup_{0 \le t \le t^\star} |\hat{\Lambda}(s,t) - \Lambda(t)| \le \\ &\sup_{0 \le t \le t^\star} |\hat{\Lambda}(s,t) - \widetilde{\Lambda}(s,t|\beta)| + \sup_{0 \le t \le t^\star} |\widetilde{\Lambda}(s,t|\beta) - \Lambda^\star(t)| + \sup_{0 \le t \le t^\star} |\Lambda^\star(t) - \Lambda(t)|. \end{aligned} \tag{6.28}$$

It suffices to show that each term in the right-hand side of (6.28) is asymptotically negligible. For the last term, using Markov's inequality, we have, for any $\epsilon in (0,1)$

$$\begin{aligned} P\left\{ \sup_{0 \le t \le t^\star} |\Lambda^\star(t) - \Lambda(t)| > \epsilon \right\} &\le E\left\{ \sup_{0 \le t \le t^\star} |\Lambda^\star(t) - \Lambda(t)| \right\} / \epsilon \\ &\le E\left\{ \sup_{0 \le t \le t^\star} \int_0^t \{I(Y(s,w) + \beta Y^\tau(w) = 0)\} \lambda(w) dw \right\} / \epsilon \\ &\le \int_0^{t^\star} \{P(Y_1(s,w) + \beta Y_1^\tau(w) = 0)\}^n \Lambda(t^\star)/\epsilon. \end{aligned}$$

Using the dominated convergence theorem, since $P(Y_1(s,w)+\beta Y_1^\tau(w)=0)$ is bounded, we have

$$\sup_{0\le t\le t^\star}|\Lambda^\star(t)-\Lambda(t)|\le\{P(Y_1(s,t^\star)+\beta Y_1^\tau(t^\star)=0)\}^n\Lambda(t^\star)/\epsilon. \tag{6.29}$$

Since $P(Y_1(s,t^\star)+\beta Y_1^\tau(t^\star)=0)<1$ and $\Lambda(t^\star)<\infty$, as $n\to\infty$, it follows that

$$\sup_{0\le t\le t^\star}|\Lambda^\star(t)-\Lambda(t)|=o_p(1).$$

We now deal with the second term in the right-hand side of (6.28). Note that

$$\begin{aligned}&|\widetilde{\Lambda}(s,t|\beta)-\Lambda^\star(t)|\le\\&\frac{1}{n}\left|\int_0^t\left[\frac{J(s,w;\beta)}{(Y(s,w)+\beta Y^\tau(w))/n}-\frac{1}{y(s,w)+\beta\bar{G}(w)}\right]\{M(s,dw)+M^\tau(dw)\}\right|\\&+\frac{1}{n}\left|\int_0^t\frac{M(s,dw)+M^\tau(dw)}{y(s,w)+\beta\bar{G}(w)}\right|.\end{aligned} \tag{6.30}$$

Using Theorem 6.2, (6.26) and continuity considerations, we have

$$\sup_{0\le t\le t^\star}\frac{1}{n}\left|\int_0^t\left[\frac{J(s,w;\beta)}{(Y(s,w)+\beta Y^\tau(w))/n}-\frac{1}{y(s,w)+\beta\bar{G}(w)}\right]\{M(s,dw)+M^\tau(dw)\}\right|=o_p(1). \tag{6.31}$$

Next, consider the second term in the right-hand side of (6.30). Because $1/[y(s,w)+\beta\bar{G}(w)]$ is positive, deterministic and bounded, hence predictable; then, Theorem 6.2 implies that the process

$$\left\{\frac{1}{\sqrt{n}}\int_0^t\frac{M(s,dw)+M^\tau(dw)}{y(s,w)+\beta\bar{G}(w)};t\in[0,t^\star]\right\} \tag{6.32}$$

converges weakly to a zero-mean Gaussian process, and as a consequence,

$$\sup_{0\le t\le t^\star} |\widetilde{\Lambda}(s,t|\beta) - \Lambda^\star(t)| = o_p(1). \tag{6.33}$$

Consider finally the first term in the right-hand side of (6.28). For brevity, let $H(\beta; s, w) = 1/\{Y(s,w)+\beta Y^\tau(w)\}$. Using this notation, we have $|\hat{\Lambda}(s,t)-\widetilde{\Lambda}_0(s,t|\beta)| = |\int_0^t J(s,w;\beta)\{H(\hat{\beta}; s, w) - H(\beta_0; s, w)\}\{N(s,dw) + N^\tau(dw)\}|$, where β is either $\hat{\beta}$ or β_0. Next, a Taylor expansion of $H(\beta; s, w)$ yields the quantity

$$H(\hat{\beta}; s, w) - H(\beta_0; s, w) = (\hat{\beta} - \beta_0)H'(\beta^\star; s, w);$$

where $\beta^\star$ is in $[\hat{\beta}, \beta_0]$. Therefore, that first term becomes

$$|\hat{\Lambda}(s,t) - \widetilde{\Lambda}(s,t|\beta)| = (\hat{\beta} - \beta_0)\left|\int_0^t \frac{J(s,w;\beta^\star)Y^\tau(w)}{\{Y(s,w) + \beta^\star Y^\tau(w)\}^2}\{N(s,dw) + N^\tau(dw)\}\right|. \tag{6.34}$$

To show that $\sup_{0\le t\le t^\star} |\hat{\Lambda}(s,t) - \widetilde{\Lambda}(s,t|\beta)| = o_p(1)$, it suffices to show that

$$A(s,t;\beta^\star) \equiv \int_0^t \frac{J(s,w;\beta^\star)Y^\tau(w)}{\{Y(s,w) + \beta^\star Y^\tau(w)\}^2}\{N(s,dw) + N^\tau(dw)\} \tag{6.35}$$

converges in probability for any $\beta^\star$ such that $\beta^\star \xrightarrow{p} \beta_0$ and the result will follow using the consistency of $\hat{\beta}$. Using (6.9) and some boundedness conditions, notice that

$$\begin{aligned}
&\left| \int_0^t \frac{J(s,w;\beta^\star)Y^\tau(w)\{N(s,dw)+N^\tau(dw)\}}{\{Y(s,w)+\beta^\star Y^\tau(w)\}^2} \right| \\
&\leq \frac{1}{\beta^\star} \int_0^t \frac{J(s,w;\beta^\star)\{N(s,dw)+N^\tau(dw)\}}{\{Y(s,w)+\beta^\star Y^\tau(w)\}} \\
&\leq \frac{1}{\beta^\star}\frac{1}{n} \int_0^t \frac{J(s,w;\beta^\star)\{M(s,dw)+M^\tau(dw)\}}{\{Y(s,w)+\beta^\star Y^\tau(w)\}/n} + \frac{1}{\beta^\star}\int_0^t \lambda(w)dw.
\end{aligned} \tag{6.36}$$

Furthermore, as $\beta^\star \to \beta_0$ and $n \to \infty$, we have, for $y(s,w)+\beta_0\bar{G}(w) > 0$

$$\left| \frac{1}{\beta^\star}\int_0^t \frac{J(s,w,\beta^\star)[M(s,dw)+M^\tau(dw)]}{\{Y(s,w)+\beta^\star Y^\tau(w)\}/n} - \frac{1}{\beta_0}\int_0^t \frac{[M(s,dw)+M^\tau(dw)]}{y(s,w)+\beta_0\bar{G}(w)} \right| \to 0.$$

Next, using Theorem 6.2 and since $\frac{1}{\sqrt{n}}\int_0^t \frac{M(s,dw)+M^\tau(dw)}{y(s,w)+\beta_0\bar{G}(w)}$ is bounded, we obtain

$$\frac{1}{n}\frac{1}{\beta^\star}\int_0^t \frac{M(s,dw)+M^\tau(dw)}{\{Y(s,w)+\beta^\star Y^\tau(w)\}/n} \xrightarrow{p} 0$$

as $n \to \infty$. The second term in the right-hand side of (6.36) is bounded since $\beta^\star$ is finite and by assumption in the statement of the theorem, $\Lambda(t^\star)$ is finite. It follows that, as $n \to \infty$, and for any $\beta^\star$ such that $\beta^\star \xrightarrow{p} \beta_0$,

$$\left| \int_0^t \frac{J(s,w)Y^\tau(w)}{\{Y(s,w)+\beta^\star Y^\tau(w)\}^2}\{N(s,dw)+N^\tau(dw)\} \right| = O_p(1). \tag{6.37}$$

Now, since $\hat{\beta}_n$ is consistent, that is, $\hat{\beta}_n \xrightarrow{p} \beta_0$, we obtain that

$$\sup_{0\leq t\leq t^\star} |\hat{\Lambda}(s,t) - \widetilde{\Lambda}(s,t|\beta)| = o_p(1); \tag{6.38}$$

and this completes the proof of uniform consistency of $\hat{\Lambda}(s,t)$.

Proof of 2. With the consistency of $\hat{\Lambda}(s,t)$ established, the uniform consistency of $\hat{\bar{F}}(s,t)$ follows using the product integral representation of $\hat{\bar{F}}$ and $\bar{F}$ and the Duhamel equation [see Andersen et al., equation 2.6.5, page 91.] See for instance the proof of uniform consistency of $\hat{\bar{F}}(s,t)$ in Peña et al. [30].

6.5.2 Weak Convergence Result: Semiparametric case

In this section, we address the problem of joint convergence of $\hat{\Lambda}(t)$ and $\hat{\beta}$. From this, we obtain the weak convergence to a zero-mean Gaussian process of the process $\{W(s,t) = \sqrt{n}[\hat{\Lambda}(s,t) - \Lambda(t)] : t \in [0,t^\star]\}$ on the Skorohod space $\mathcal{D}[0,t^\star]$, the space of right continuous functions with left hand limit, also called cadlag functions. The weak convergence of the process $\{V(s,t) = \sqrt{n}[\hat{\bar{F}}(s,t) - \bar{F}(t)] : t \in [0,t^\star]\}$ will easily follow on the same space. We start with the joint convergence.

Theorem 6.4 *For a fixed s, the process*

$$\left\{ \boldsymbol{W}(s,t) = \sqrt{n} \begin{pmatrix} \hat{\Lambda}(s,t) - \Lambda(t) \\ \hat{\beta} - \beta_0 \end{pmatrix} : t \in [0,t^\star] \right\}$$

converges weakly on $\mathcal{D}[0,t^\star] \times \mathbb{R}_+^\star$ to a zero-mean Gaussian process $\boldsymbol{W}(s,t)$ whose variance function is given by $\Upsilon(s,t)$. [expression given on Page 100*]*

Before proceeding with the proof of Theorem 6.4, we require a generalization of Theorem 6.2. For that purpose, replace the vector-valued process $\mathbf{H}_i(s,t)$ by $p \times q$ matrix $\mathbf{H}_i(s,t)$ whose components are bounded and the process $\{\mathbf{H}_i(s,t) : s \in \mathcal{S}\}$ is

$\mathbf{F}$-predictable. Further, let $\mathbf{M}(s,t)$ be a p rows vector of martingales for each fixed t. Of interest are the asymptotic properties of processes of the form

$$\mathbf{W}(s,t) = \frac{1}{\sqrt{n}} \sum_{i=1}^{n} \int_0^t \mathbf{H}_i(s,w)\mathbf{M}_i(s,dw), \quad (s,t) \in \mathcal{S}^2. \tag{6.39}$$

Consider a triangular sequence of processes of type (6.39) and assume the following conditions are satisfied:

1. The components of $\mathbf{H}_i(s,t)$ are left-continuous in (s,t) and converge uniformly on $[0,s] \times [0,t^\star]$ to some deterministic matrix $\mathbf{h}(s,t)$ of bounded functions on $[0,s] \times [0,t^\star]$;

2. Assume Conditions 2 and 3 of Theorem 6.2 are satisfied

Then, the result in Theorem 6.2 holds for this generalized version of processes of form given in (6.39).

Proof of Theorem 6.4**:** The norm of a vector $\mathbf{a} = (a_i)$ or a matrix $\mathbf{A} = (a_{ij})$ will respectively be $\|\mathbf{a}\| = \sup_i |a_i|$ and $\|\mathbf{A}\| = \sup_{ij} |a_{ij}|$. In particular, for a vector $\mathbf{a} = (a_1, 0)^t$, $\|\mathbf{a}\| = |a_1|$.

With $\Lambda^\star(t)$, $\hat{\Lambda}(s,t)$ and $\widetilde{\Lambda}(s,t|\beta)$ as before, we have the decomposition

$$\begin{aligned}\sqrt{n}\begin{pmatrix} \hat{\Lambda}(s,t) - \Lambda(t) \\ \hat{\beta} - \beta_0 \end{pmatrix} &= \sqrt{n}\begin{pmatrix} \hat{\Lambda}(s,t) - \widetilde{\Lambda}(s,t|\beta) \\ 0 \end{pmatrix} + \sqrt{n}\begin{pmatrix} \widetilde{\Lambda}(s,t|\beta) - \Lambda^\star(t) \\ \hat{\beta} - \beta_0 \end{pmatrix} \\ &\quad + \sqrt{n}\begin{pmatrix} \Lambda^\star(t) - \Lambda(t) \\ 0 \end{pmatrix}. \end{aligned} \tag{6.40}$$

Here, the norm of the third term in (6.40) converges to 0 asymptotically as shown earlier in the consistency proof of $\hat{\Lambda}(s,t)$. Using Slutsky's theorem [cf. Resnick [35] Theorem 8.6.1, page 268], the joint asymptotic distribution of $\hat{\Lambda}(s,t)$ and $\hat{\beta}$ will now depend on the first and second term in the right-hand side of (6.40). Recall from (6.34) that $\hat{\Lambda}(s,t) - \widetilde{\Lambda}(s,t|\beta) = (\hat{\beta} - \beta_0)A(s,t;\beta^\star)$. We now deal with $\sqrt{n}(\hat{\beta} - \beta_0)$. From (6.21), we have

$$\begin{aligned} U(\beta; s^\star, t^\star) \equiv \frac{\partial}{\partial \beta} l_p(\beta; s^\star, t^\star) &= \int_0^{t^\star} \frac{-Y^\tau(v)}{Y(s^\star, v) + \beta Y^\tau(v)} \{N(s^\star, dv) + N^\tau(dv)\} \\ &\quad + \frac{1}{\beta} \int_0^{t^\star} N^\tau(dv). \end{aligned} \tag{6.41}$$

Using (6.8) and (6.9) along with some algebra, we obtain

$$\begin{aligned} U(\beta; s^\star, t^\star) &= \frac{1}{\beta} \int_0^{t^\star} M^\tau(dw) - \int_0^{t^\star} \frac{Y^\tau(w)\{M(s^\star, dw) + M^\tau(w)\}}{Y(s^\star, w) + \beta Y^\tau(w)} \\ &= \int_0^{t^\star} \frac{Y(s^\star, w) M^\tau(dw)}{\beta\{Y(s^\star, w) + \beta Y^\tau(w)\}} - \int_0^{t^\star} \frac{Y^\tau(w) M(s^\star, dw)}{Y(s^\star, w) + \beta Y^\tau(w)}. \end{aligned} \tag{6.42}$$

Taylor expansion of the score process around β_0 yields the quantity

$$U(\beta; s^\star, t^\star) - U(\beta_0; s^\star, t^\star) = (\beta - \beta_0) H(\beta^\star; s^\star, t^\star), \tag{6.43}$$

where the function H is given by

$$\begin{aligned} H(\beta^\star; s^\star, t^\star) &= \int_0^{t^\star} \left\{ \frac{Y^\tau(w)}{Y(s^\star, w) + \beta^\star Y^\tau(w)} \right\}^2 \{N(s^\star, dw) + N^\tau(dw)\} \\ &\quad - \int_0^{t^\star} \frac{N^\tau(dw)}{{\beta^\star}^2}; \end{aligned} \tag{6.44}$$

and where $\beta^\star$ is on the line segment between β and β_0. Inserting $\hat{\beta}$ for β in (6.43) and since $U(\hat{\beta}; s^\star, t^\star) = 0$ by definition, we get

$$\sqrt{n}(\hat{\beta} - \beta_0) = \left\{ -\frac{H(\beta^\star; s^\star, t^\star)}{n} \right\}^{-1} \frac{U(\beta_0; s^\star, t^\star)}{\sqrt{n}}. \tag{6.45}$$

For the convergence of $H(\beta^\star; s^\star, t^\star)/n$, notice that

$$\begin{aligned}
H(\beta^\star; s^\star, t^\star) &= \int_0^{t^\star} \left\{ \frac{Y^\tau(w)}{Y(s^\star, w) + \beta^\star Y^\tau(w)} \right\}^2 \{M(s^\star, dw) + M^\tau(dw)\} \\
&\quad + \int_0^{t^\star} \left\{ \frac{Y^\tau(w)}{Y(s^\star, w) + \beta^\star Y^\tau(w)} \right\}^2 \{Y(s^\star, w) + \beta^\star Y^\tau(w)\}\lambda(w)dw \\
&\quad - \int_0^{t^\star} \frac{M^\tau(dw)}{\beta^{\star 2}} - \int_0^{t^\star} \frac{\beta^\star Y^\tau(w)\lambda(w)dw}{\beta^{\star 2}} \\
&= \int_0^{t^\star} \left\{ \frac{Y^\tau(w)}{Y(s^\star, w) + \beta^\star Y^\tau(w)} \right\}^2 \{M(s^\star, dw) + M^\tau(dw)\} \\
&\quad - \int_0^{t^\star} \frac{Y^\tau(w) Y(s^\star, w)\lambda(w)dw}{\beta^\star\{Y(s^\star, w) + \beta^\star Y^\tau(w)\}} - \int_0^{t^\star} \frac{M^\tau(dw)}{\beta^{\star 2}}.
\end{aligned}$$

Again, using Theorem 6.2 and (6.26), we have for any $\beta^\star \xrightarrow{p} \beta_0$ and as $n \to \infty$,

$$\frac{H(\beta^\star; s^\star, t^\star)}{n} \xrightarrow{p} -\int_0^{t^\star} \frac{\lambda(w) y(s^\star, w)\bar{G}(w)dw}{\beta_0\{y(s^\star, w) + \beta_0 \bar{G}(w)\}} \equiv h(\beta_0; s^\star, t^\star). \tag{6.46}$$

Using (6.34) and (6.35), the first two terms of (6.40) now become

$$\begin{aligned}
&\sqrt{n}\begin{pmatrix} \hat{\Lambda}(s,t) - \widetilde{\Lambda}(s,t|\beta) \\ 0 \end{pmatrix} + \sqrt{n}\begin{pmatrix} \widetilde{\Lambda}(s,t|\beta) - \Lambda^\star(t) \\ \hat{\beta} - \beta_0 \end{pmatrix} \\
&= \sqrt{n}\begin{pmatrix} \widetilde{\Lambda}(s,t|\beta) - \Lambda^\star(t) + (\hat{\beta} - \beta_0)A(s,t;\beta) \\ \hat{\beta} - \beta_0 \end{pmatrix} \\
&= \sqrt{n}\begin{pmatrix} 1 & A(s,t;\beta) \\ 0 & 1 \end{pmatrix}\begin{pmatrix} \widetilde{\Lambda}(s,t|\beta) - \Lambda^\star(t) \\ \hat{\beta} - \beta_0 \end{pmatrix}.
\end{aligned} \tag{6.47}$$

Whereas

$$\widetilde{\Lambda}(s,t|\beta) - \Lambda^\star(t) = \int_0^t J(s,w;\beta)\frac{M(s,dw) + M^\tau(dw)}{Y(s,w) + \beta Y^\tau(w)}, \tag{6.48}$$

and using the compact notation $S^{(0)} \equiv S^{(0)}(s,w;\beta) = Y(s,w) + \beta Y^\tau(w)$ and $S^{(1)}(w) = Y^\tau(w)$, from (6.42), (6.45) and (6.48), we have

$$\begin{aligned}
&\sqrt{n}\begin{pmatrix} (\widetilde{\Lambda}(s,t|\beta) - \Lambda^\star(t)) \\ \hat{\beta} - \beta_0 \end{pmatrix} \\
&= \frac{1}{\sqrt{n}}\int_0^t \begin{pmatrix} \frac{nJ(s,w;\beta)}{S^{(0)}} & \frac{nJ(s,w;\beta)}{S^{(0)}} \\ -\left\{-\frac{H(\beta^\star;s,t^\star)}{n}\right\}^{-1}\frac{S^{(1)}(w)}{S^{(0)}} & \left\{-\frac{H(\beta^\star;s,t^\star)}{n}\right\}^{-1}\frac{Y(s,w)}{\beta S^{(0)}} \end{pmatrix}\begin{pmatrix} M(s,dw) \\ M^\tau(dw) \end{pmatrix} \\
&= \frac{1}{\sqrt{n}}\int_0^t \mathbf{H}(\beta;s,w)\mathbf{M}(s,dw).
\end{aligned} \tag{6.49}$$

The last equation in (6.49) is of the form given in (6.39) since $J(s,w;\beta)$ and $S^{(0)}$ are both left-continuous, hence predictable. Furthermore, using previous conver-

gence results, the components of $\mathbf{H}(\beta; s, w)$ all converge to some deterministic and bounded functions. Using stochastic integration theory, the variance process of $\frac{1}{\sqrt{n}} \int_0^t \mathbf{H}(\beta; s, w)\mathbf{M}(s, dw)$ is given by

$$\begin{aligned} &\frac{1}{n} \left\langle \int_0^{\cdot} \mathbf{H}(\beta; s, w)\mathbf{M}(s, dw) \right\rangle (t) = \\ &\frac{1}{n} \int_0^t \mathbf{H}(\beta; s, w) \begin{pmatrix} Y(s, w)\lambda(w) & 0 \\ 0 & \beta Y^{\tau}(w)\lambda(w) \end{pmatrix} \mathbf{H}^t(\beta; s, w) dw. \end{aligned} \tag{6.50}$$

Then, simple matrix manipulations yield

$$\frac{1}{n} \left\langle \int_0^{\cdot} \mathbf{H}(\beta; s, w)\mathbf{M}(s, dw) \right\rangle (t) = \frac{1}{n} \int_0^t \begin{pmatrix} \frac{\lambda(w)}{S^{(0)}} & 0 \\ 0 & \left\{-\frac{H(\beta^{\star}; s, t^{\star})}{n}\right\}^{-2} \frac{S^{(1)}}{\beta S^{(0)}} \lambda(w) Y(s, w) \end{pmatrix} dw. \tag{6.51}$$

Next, using (6.26) and (6.46), it is immediately seen that

$$\begin{aligned} &\frac{1}{n} \left\langle \int_0^{\cdot} \mathbf{H}(\beta; s, w)\mathbf{M}(s, dw) \right\rangle (t) \xrightarrow{p} \\ &\int_0^t \begin{pmatrix} \frac{\lambda(w)}{y(s,w)+\beta\bar{G}(w)} & 0 \\ 0 & h(\beta; s^{\star}, t^{\star})^{-2} \frac{\lambda(w) y(s,w) \bar{G}(w)}{\beta\{y(s,w)+\beta\bar{G}(w)\}} \end{pmatrix} dw \equiv \Sigma(s, t; \beta). \end{aligned} \tag{6.52}$$

Before proceeding with the asymptotic variance matrix of (6.40), we need to find the limit of $A(s,t;\beta^\star)$ as $n \to \infty$. Observe that

$$\begin{aligned}
A(s,t;\beta^\star) &= \int_0^t \frac{J(s,w;\beta^\star)Y^\tau(w)}{\{Y(s,w;\beta)+\beta^\star Y^\tau(w)\}^2}\{N(s,dw)+N^\tau(dw)\} \\
&= \int_0^t \frac{J(s,w;\beta^\star)Y^\tau(w)}{\{Y(s,w)+\beta^\star Y^\tau(w)\}^2}\{M(s,dw)+M^\tau(dw)\} \\
&\quad + \int_0^t \frac{J(s,w;\beta^\star)Y^\tau(w)\lambda(w)dw}{Y(s,w)+\beta^\star Y^\tau(w)} \\
&= \frac{1}{n}\int_0^t \frac{J(s,w;\beta)Y^\tau(w)/n}{\big\{\{Y(s,w)+\beta^\star Y^\tau(w)\}/n\big\}^2}\{M(s,dw)+M^\tau(dw)\} \\
&\quad + \int_0^t \frac{\{Y^\tau(w)/n\}\lambda(w)dw}{\{Y(s,w)+\beta^\star Y^\tau(w)\}/n}. \qquad (6.53)
\end{aligned}$$

That the first term in the right-hand side of (6.53) converges to 0 as $n \to \infty$ is guaranteed by Theorem 6.2 and (6.26). It therefore follows that, for any $\beta^\star \xrightarrow{p} \beta_0$

$$A(s,t;\beta^\star) \xrightarrow{p} \int_0^t \frac{\bar{G}(w)\lambda(w)dw}{y(s,w)+\beta_0\bar{G}(w)} \equiv a(s,t;\beta_0) \qquad (6.54)$$

as $n \to \infty$. Now (6.47), (6.52) and (6.54), altogether lead to the following asymptotic variance matrix for the joint limiting distribution of (6.40)

$$\begin{aligned}
&\left\langle \begin{pmatrix} 1 & A(s,\cdot;\beta^\star) \\ 0 & 1 \end{pmatrix} \frac{1}{\sqrt{n}}\int_0^{\cdot} \mathbf{H}(\beta;s,w)\mathbf{M}(s,dw) \right\rangle (t) \xrightarrow{p} \\
&\begin{pmatrix} 1 & a(s,t;\beta_0) \\ 0 & 1 \end{pmatrix} \Sigma(\beta_0;s,t) \begin{pmatrix} 1 & 0 \\ a(s,t;\beta_0) & 1 \end{pmatrix} \equiv \Upsilon(s,t;\beta_0). \qquad (6.55)
\end{aligned}$$

By some simple matrix calculations, the components of $\Upsilon(s,t;\beta_0)$ are given by, for $\beta_0 = \beta$

$$\begin{aligned}
\Upsilon_{11} &= \Sigma_{11} + a^2(s,t;\beta)\Sigma_{22}; \\
\Upsilon_{21} &= \Upsilon_{12} = a(s,t;\beta)\Sigma_{22}; \\
\Upsilon_{22} &= a^2(s,t;\beta)\Sigma_{22};
\end{aligned}$$

where

$$\begin{aligned}
\Sigma_{11} &= \int_0^t \frac{\lambda(w)dw}{y(s,w)+\beta\bar{G}(w)}; \\
\Sigma_{22} &= \left[\int_0^t \frac{\lambda(w)y(s,w)\bar{G}(w)dw}{\beta\{y(s,w)+\beta\bar{G}(w)\}}\right]^{-2} \int_0^t \frac{\lambda(w)y(s,w)\bar{G}(w)dw}{\beta\{y(s,w)+\beta\bar{G}(w)\}} \\
&= \left[\int_0^t \frac{\lambda(w)y(s,w)\bar{G}(w)dw}{\beta\{y(s,w)+\beta\bar{G}(w)\}}\right]^{-1}.
\end{aligned}$$

It is worth mentioning that in the above expression of $\Upsilon_{11}(s,t;\beta)$, the second term is the variance contribution arising from estimation of the parameter β. The next two corollaries give asymptotic properties of $\hat{\beta}$ and $\hat{\Lambda}(s,t)$ for $t^\star = t$.

Corollary 6.1 *As $n \to \infty$, $\sqrt{n}(\hat{\beta} - \beta_0) \xrightarrow{d} N(0, \Sigma_{22})$.*

Corollary 6.2 *Under the same conditions as in Theorem 2, we have:*

- *The process $\{W(s,t) = \sqrt{n}[\hat{\Lambda}(s,t) - \Lambda(t)] : t \in [0,t^\star]\}$ converges weakly on $\mathcal{D}[0,t^\star]$ to a zero-mean Gaussian process with variance function given by $\Upsilon_{11}(s,t;\beta)$;*

where

$$\Upsilon_{11}(s,t;\beta) = \int_0^t \frac{\lambda(w)dw}{y(s,w)+\beta\bar{G}(w)} + \left[\int_0^t \frac{\bar{G}(w)\lambda(w)dw}{y(s,w)+\beta\bar{G}(w)}\right]^2 \times \left[\int_0^t \frac{\lambda(w)y(s,w)\bar{G}(w)dw}{\beta\{y(s,w)+\beta\bar{G}(w)\}}\right]^{-1}.$$

- *The process* $\{V(s,t) = \sqrt{n}[\hat{\bar{F}}(s,t) - \bar{F}(t)] : t \in [0,t^\star]\}$ *converges weakly on* $\mathcal{D}[0,t^\star]$ *to a zero-mean Gaussian process with variance function given by*

$$\sigma^2_{\hat{\bar{F}}(s,t)} = \bar{F}^2(t)\Upsilon_{11}(s,t;\beta).$$

Remarks: Since the integrands in all the integrals in the expression of $\Upsilon_{11}(s,t;\beta)$ are bounded, if we let $\beta \to 0$, we have $\Upsilon_{11}(s,t;\beta) \to \int_0^t \frac{\lambda(w)dw}{y(s,w)}$, which is the asymptotic variance of the nonparametric estimator in Peña et al. [30]. Note also that, comparing the asymptotic variances of the survivor functions is equivalent to comparing their counterparts of the cumulative hazard since the two only differ by a multiplicative factor, $\bar{F}(t)^2$.

6.6 Efficiency Comparison

6.6.1 Comparison Semiparametric/Nonparametric Estimator

In this subsection, we compare the estimator of the gap-time (or equivalently the estimator of the cumulative hazard) obtained in Peña et al. [30] to our proposed estimator. We denote by $\tilde{\Lambda}(s^\star,t)$ [which is not to be confounded with the expression

in (6.12)] the estimator derived in Peña et al. [30] and by $\hat{\Lambda}(s^\star, t)$ the estimator proposed here. Since these two estimators are asymptotically unbiased, comparing the estimators is equivalent, as in the parametric case, to comparing their asymptotic variance pointwise using the ARE. We recall that the asymptotic variance of $\tilde{\Lambda}$ is given by

$$\sigma^2_{\tilde{\Lambda}}(s^\star, t) = \int_0^t \frac{\Lambda(dv)}{y(s^\star, v)}. \tag{6.56}$$

Let $\sigma^2_{\hat{\Lambda}}(s^\star, t)$ be the asymptotic variance of our proposed estimator. The next result pertains to the performance of this proposed estimator.

Theorem 6.5 *Assuming the KG model holds. The ARE of $\tilde{\Lambda}(t)$ to $\hat{\Lambda}(t)$ at time t satisfies*

$$ARE = \frac{\Upsilon_{11}(s^\star, t; \beta)}{\sigma^2_{\tilde{\Lambda}}(s^\star, t)} \leq 1;$$

with equality when $\beta = 0$. That is, at any fixed t, the semiparametric estimator is always more efficient than the fully nonparametric estimator.

Proof: Consider the second term in the right-hand side of $\Upsilon_{11}(s, t)$ and let

$$d\nu = \frac{\lambda(w)dw}{y(s^\star, w) + \beta\bar{G}(w)}; \tag{6.57}$$

so that $\bar{\nu} = \int_0^t d\nu = \nu(t)$. We have

$$\begin{aligned}\left[\int_0^t \frac{\bar{G}(w)\lambda(w)dw}{y(s^\star, w) + \beta\bar{G}(w)}\right]^2 &= \left[\int_0^t \sqrt{\frac{\bar{G}(w)}{y(s^\star, w)}}\sqrt{\bar{G}(w)y(s^\star, w)}d\nu\right]^2 \\ &= \bar{\nu}^2\left[\int_0^t \sqrt{\frac{\bar{G}(w)}{y(s^\star, w)}}\sqrt{\bar{G}(w)y(s^\star, w)}d\left\{\frac{\nu}{\bar{\nu}}\right\}\right]^2. \end{aligned} \tag{6.58}$$

Now, using Cauchy-Schwartz inequality, we obtain

$$\begin{aligned}\left[\int_0^t \frac{\bar{G}(w)\lambda(w)dw}{y(s^\star,w)+\beta\bar{G}(w)}\right]^2 &\leq \bar{\nu}^2\left[\int_0^t \frac{\bar{G}(w)}{y(s^\star,w)}d\left\{\frac{\nu}{\bar{\nu}}\right\}\right]\left[\int_0^t \bar{G}(w)y(s^\star,w)d\left\{\frac{\nu}{\bar{\nu}}\right\}\right] \\ &= \left[\int_0^t \frac{\bar{G}(w)}{y(s^\star,w)}d\nu\right]\left[\int_0^t \bar{G}(w)y(s^\star,w)d\nu\right]. \qquad (6.59)\end{aligned}$$

Consequently, we can obtain an upper bound for $\Upsilon_{11}(s^\star,t)$ by

$$\begin{aligned}\Upsilon_{11}(s,t;\beta) &\leq \int_0^t d\nu + \beta\frac{\left[\int_0^t \frac{\bar{G}(w)}{y(s^\star,w)}d\nu\right]\left[\int_0^t \bar{G}(w)y(s^\star,w)d\nu\right]}{\int_0^t \bar{G}(w)y(s^\star,w)d\nu} \\ &= \int_0^t \left[\frac{y(s^\star,w)+\beta\bar{G}(w)}{y(s^\star,w)}\right]d\nu. \qquad (6.60)\end{aligned}$$

Next, using the same notation as in (6.56), we have

$$\sigma^2_{\hat{\Lambda}}(s^\star,t) = \int_0^t \left[\frac{y(s^\star,w)+\beta\bar{G}(w)}{y(s^\star,w)}\right]d\nu; \qquad (6.61)$$

and therefore $ARE \leq 1$ which completes the proof of the theorem.

6.6.2 Efficiency Comparison in the HPP Case

In this section, comparison will be made between our proposed semiparametric estimator and the full parametric estimator under the assumption that the gap-time $\bar{F}(t)$ follows an exponential distribution with failure rate θ. Under the Koziol-Green model and after allowing $s^\star \to \infty$, an expression of $y(s^\star,w)$ [cf. Proposition 3.1 for

the general expression of $y(s,w)]$ is given by

$$y(\infty,t) = \left[\frac{\beta+1}{\beta}\right]\exp\{-\theta(1+\beta)t\}.$$

Next, using the fact that $\lambda(w)=\theta$ and $\bar{G}(w)=\exp\{-\theta\beta w\}$, it follows that

$$\begin{aligned}\Upsilon_{11}(\infty,t) &= \int_0^t \frac{\beta\theta dw}{\{\beta+1\}\exp\{-\theta(1+\beta)w\}+\beta^2\exp\{-\theta\beta w\}} \\ &\quad +\beta\frac{\left[\int_0^t \frac{\beta\theta\exp\{-\beta\theta w\}dw}{\{\beta+1\}\exp\{-\theta w(1+\beta)\}+\beta^2\exp\{-\theta w\beta\}}\right]^2}{\int_0^t \frac{\theta\{1+\beta\}\exp\{-\theta w[1+2\beta]dw\}}{\{\beta+1\}\exp\{-\theta w(1+\beta)\}+\beta^2\exp\{-\theta w\beta\}}}. \qquad (6.62)\end{aligned}$$

Meanwhile, from (4.13), under the exponential assumption above, we have

$$\sigma^2_{\tilde{\Lambda}}(\infty,t;\beta) = \frac{\beta}{(1+\beta)^2}\big\{\exp\{(\beta+1)\theta t\}-1\big\}. \qquad (6.63)$$

Hence (6.62) and (6.63) imply that the ARE, in this special case of the exponential distribution, is the ratio of the expressions in the right-hand side of (6.62) and (6.63). It is clear that closed form expression of the ARE will be difficult to obtain as in the parametric case. However, this can be computed numerically and the computed values of $\sigma^2_{\tilde{\Lambda}}$, $\sigma^2_{\hat{\Lambda}}$ as well as the ARE are summarized in Tables 6.1 to 6.4 for $\theta=1$ and different values of β, t.

t	.1	.2	.3	.4	.5	.6	.7	.8	.9	1
$\sigma^2_{\tilde{\Lambda}}(t)$	.0358	.0775	.1258	.1817	.2461	.3202	.4055	.5034	.6153	.7433
$\sigma^2_{\hat{\Lambda}}(t)$	.0359	.0777	.1262	.1826	.2481	.3243	.4127	.5155	.6349	.7736
ARE	.998	.9979	.9973	.9950	.9921	.9876	.9826	.9766	.9692	.9608

Table 6.1: Efficiency of the semiparametric estimator to the nonparametric estimator in the HPP case for $\theta=1$ and $\beta=.5$

t	.1	.2	.3	.4	.5	.6	.7	.8	.9	1
$\sigma^2_{\hat{\Lambda}}(t)$	.0551	.1223	.2032	.3003	.4158	.553	.715	.9048	1.126	1.385
$\sigma^2_{\hat{\Lambda}}(t)$	.0553	.1229	.2055	.3063	.4295	.58	.7637	.9882	1.262	1.599
ARE	.997	.9951	.989	.98	.968	.9534	.9362	.9156	.8924	.866

Table 6.2: Efficiency of the semiparametric estimator to the nonparametric estimator in the HPP case for $\theta = \beta = 1$

t	.1	.2	.3	.4	.5	.6	.7	.8	.9	1
$\sigma^2_{\hat{\Lambda}}(t)$	.0772	.1792	.3113	.4794	.6899	.9509	1.270	1.660	2.120	2.674
$\sigma^2_{\hat{\Lambda}}(t)$	.0777	.1826	.3243	.5155	.7736	1.122	1.592	2.227	3.080	4.24
ARE	.9938	.981	.9601	.9300	.8918	.847	.798	.745	.691	.630

Table 6.3: Efficiency of the semiparametric estimator to the nonparametric estimator in the HPP case for $\theta = 1$ and $\beta = 2$

t	.1	.2	.3	.4	.5	.6	.7	.8	.9	1
$\sigma^2_{\hat{\Lambda}}(t)$	.111	.292	.566	.959	1.508	2.265	3.309	4.773	6.844	9.881
$\sigma^2_{\hat{\Lambda}}(t)$	.114	.322	.700	1.391	2.649	4.941	9.117	16.726	30.592	55.860
ARE	.972	.908	.808	.689	.569	.458	.363	.285	.223	.176

Table 6.4: Efficiency in of the semiparametric estimator to the nonparametric estimator the HPP case for $\theta = 1$ and $\beta = 5$

Examining Tables 6.1 to 6.4, it is clearly seen that as β increases and for fixed t, our proposed estimator becomes more and more efficient. This is not surprising in light of reasons mentioned in the remarks on the HPP efficiency comparison in the parametric case. For a fixed value of β, as t increases, we also notice that the proposed estimator outperforms the estimator derived ignoring informative censoring. We have attempted to obtain specific bounds for all values of t in order to assess the overall efficiency gained, but failure to obtain a closed form expression for the ARE preclude this. In the next chapter, we present the results of our simulation study.

Chapter 7

Finite Sample Properties in the Semiparametric Case

We perform computer simulation studies to examine numerically the properties of the proposed estimators. The goals were (i) to examine the impact of moderate sample size on the properties of the estimators; (ii) to examine the biases, mean-square-error of the estimators and lastly (iii) to examine the performance of the semiparametric estimator with respect to the fully parametric estimator.

7.1 Simulation design

Sample size: To examine the effect of sample size, we took the sample size n from the set $\{10, 15, 20, 25, 30, 50\}$. These values were chosen to mimic realistic sample sizes encountered in fields such as biomedical, reliability and engineering.

True survivor function: The baseline distribution function was chosen to be the flexible and commonly used Weibull distribution since it gives rise to a decreasing and increasing failure rate. For instance, the scale parameter α was fixed at 1, but the shape parameter μ was allowed to take values in $\{.5, 1, 2\}$. The value .5 leads to a decreasing failure rate and 2 gives rise to an increasing failure rate, while the value 1 is just the exponential distribution.

Censoring mechanism: The τ_is are generated from a Weibull distribution with hazard rate equal to $\beta\mu t^{\mu-1}$.

Values of the censoring parameter β: To assess the impact of the censoring parameter, we took values of β from the set $\{.14, .17, .2, .25, .33, .5\}$. The choice of the values was motivated by the need to assess the impact of the proportion of censored observations. Recall that, from the review of literature, P(subject is censored)$=\frac{\beta}{\beta+1}$; so consequently, the set of values of β correspond respectively to about 12.3%, 14.5%, 16.6%, 20%, 25% and 33% censoring.

Computational formulas: For the estimator of the cumulative hazard proposed in (6.22), the computational formula is given by

$$\hat{\Lambda}(s^\star, t) = \sum_{i,j;T_{ij}\leq t} \frac{N(s^\star, T_{ij}) + N^\tau(T_{ij})}{Y(s^\star, T_{ij}) + \hat{\beta}Y^\tau(T_{ij})} + \sum_i \frac{N(s^\star, \tau_i) + N^\tau(\tau_i)}{Y(s^\star, \tau_i) + \hat{\beta}Y^\tau(\tau_i)}.$$

Having obtained an estimator of the cumulative hazard, the estimator of the survivor function of the gap-time is obtained by substituting the estimator of the cumulative hazard in the product integral form given in (6.13). As for the biases and MSE, the standard formulas are used. For the moderate sample size results, the ARE is

computed as the ratio of MSE, since, for small sample sizes, these estimators are not unbiased.

7.2 Discussion of Simulation Results

7.2.1 Estimating β

To obtain an estimate of β, there is a need to maximize the likelihood in (6.20) with respect to β. A convenient way to do this is to use a constrained optimization method available in the R-package and which is based on the Newton-Raphson algorithm. This constrained optimization is not sensitive to the starting value but we used 0.1 as a starting value. The true distribution was generated from the Weibull distribution with scale 1 but shape μ equal to $.5, 1, 2$. In the tables below, we summarize the results of estimating β for different values of μ and sample size n. Biases, standard errors and MSEs of $\hat{\beta}$ are evaluated based on 1000 replications. Examining Tables $7.1 - 7.3$, we note that as the sample size increases, the standard deviations of the $\hat{\beta}$s decrease steadily and the biases become negligible regardless of the value of the shape parameter. However, for $\mu = 1$, which corresponds to the exponential distribution, we notice a significant decreasing trend. We also observe that as the proportion of censored observations increases, the standard errors of the $\hat{\beta}$s increase also regardless of the shape parameter. This is to be expected because as β increases, that is, as the proportion of censored observations increases, fewer subjects are under study thereby providing less information.

	$n = 20, \alpha = .5$		$n = 50, \alpha = .5$	
True β	Bias	$\sigma^2_{\hat{\beta}}$	Bias	$\sigma^2_{\hat{\beta}}$
.14	.0035	.0021	.0018	.0006
.17	.0065	.0031	-.0003	.0009
.20	.0045	.0035	.0011	.0012
.25	.0058	.0057	.0028	.0020
.33	.0071	.0095	.0068	.0033
.50	.0077	.0218	.0094	.0086

Table 7.1: Simulated biases and standard errors for $\hat{\beta}$ for $\mu = .5$.

	$n = 20, \alpha = 2$		$n = 50, \alpha = 2$	
True β	Bias	$\sigma^2_{\hat{\beta}}$	Bias	$\sigma^2_{\hat{\beta}}$
.14	.0004	.0038	-.0000	.0013
.17	.0072	.0061	.0014	.0019
.20	.0023	.0060	.0013	.0024
.25	-.0009	.0095	-.0011	.0035
.33	.0042	.0016	.0025	.0066
.50	.0082	.0455	.0002	.0146

Table 7.2: Simulated biases and standard errors for $\hat{\beta}$ for $\mu = 2$.

7.2.2 Small Sample Properties

For each combination of n, β, μ, we performed 1000 replications of the simulation values. We then computed the empirical cumulative hazard, survivor function, biases and MSE for equally spaced values of t in the set $[.1, 2]$ by increments of 0.1 for both the estimators $\hat{\bar{F}}(t)$ and $\tilde{\bar{F}}(t)$. These are simulated when the KG model is assumed and when it is not for the purpose of efficiency comparison. As stated earlier, the case when it is not assumed coincides with the full nonparametric estimator derived in Peña et al. [30]. The simulated biases, MSEs and variances are reported for all values of the shape parameter when the KG model is assumed. It can clearly be seen that as the sample size increases, biases, MSEs and variances decrease without

	$n = 20, \alpha = 1$		$n = 50, \alpha = 1$	
True β	Bias	$\sigma^2_{\hat{\beta}}$	Bias	$\sigma^2_{\hat{\beta}}$
.14	.0038	.0027	.0007	.0010
.17	.0036	.0041	.0004	.0013
.20	.0023	.0052	-.0011	.0018
.25	.0074	.0082	.0029	.0027
.33	.0093	.0134	.0024	.0045
.50	.0184	.0310	.0067	.0011

Table 7.3: Simulated biases and standard errors for $\hat{\beta}$ for $\mu = 1$.

regard to the shape parameter. The results for the case where the KG is not assumed show similar patterns [not reported here, but only the MSEs are reported for $\mu = 2$] although in the former case, these empirical values are smaller [see for example Figures 7.10 and 7.11]. These are a lot smaller for the case of the exponential distribution ($\mu = 1$). We have also observed that, for $\mu = .5$ [decreasing failure rate distribution, DFR], the variances are smaller as compared to the increasing failure rate (IFR); $\mu = 2$. This may be explained by the fact that, for DFR distribution, we expect less variability in the survivor function. In terms of efficiency comparison, as β increases, the estimator $\hat{\bar{F}}(t)$ becomes more and more efficient. When we examine the efficiency figures, Figures $7.12 - 7.14$, we immediately notice that, for $\alpha = 1$ and $\mu = 2$, the ARE is always less then 1. However, for $\mu = .5$ (DFR), the ARE for some values of t are slightly larger than 1 for small values of β, say 0.14, 0.17, and 0.2, regardless of the sample size. The reason this is happening appears to be due to the slightly large values of MSEs for the corresponding values of β.

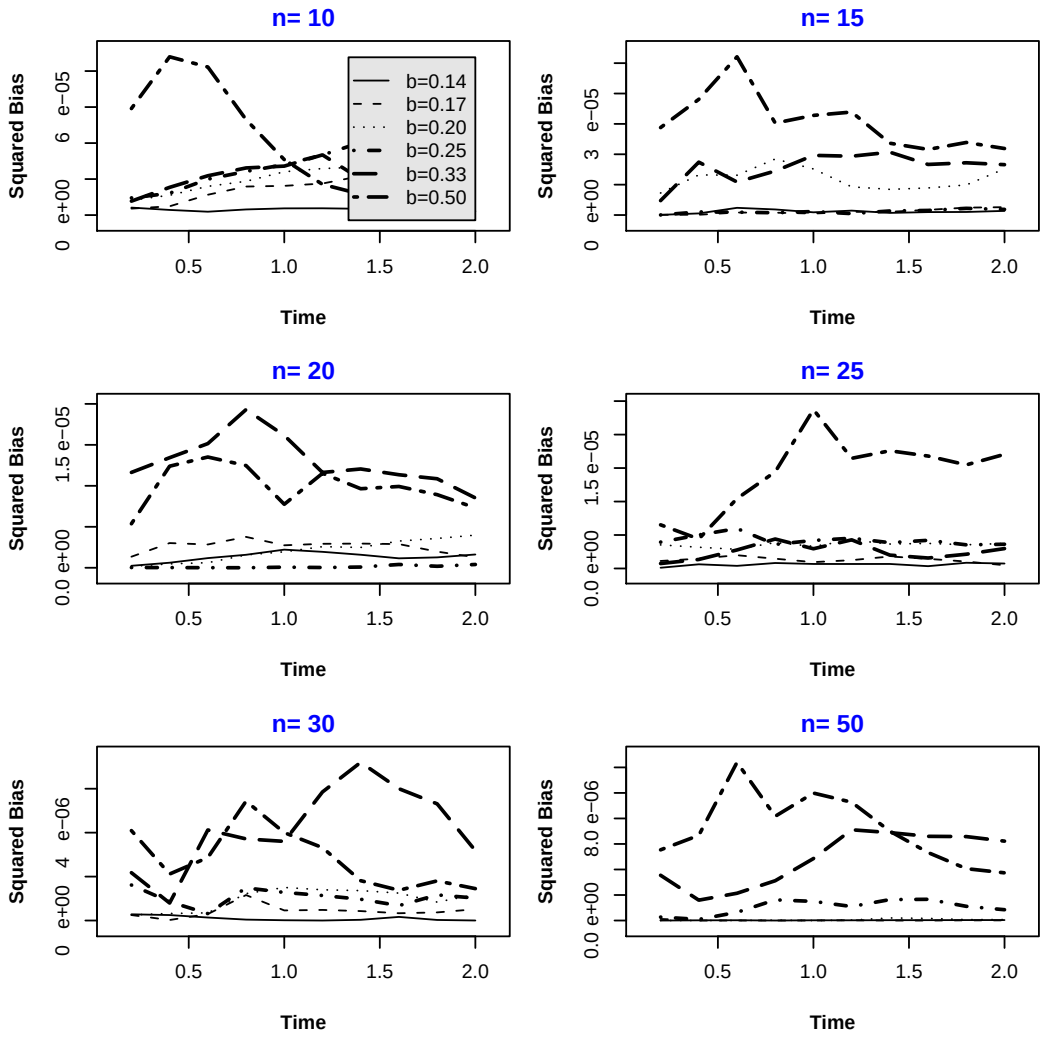

Figure 7.1: Graph of bias for different values of n and β under Weibull(1, .5) in the semiparametric case.

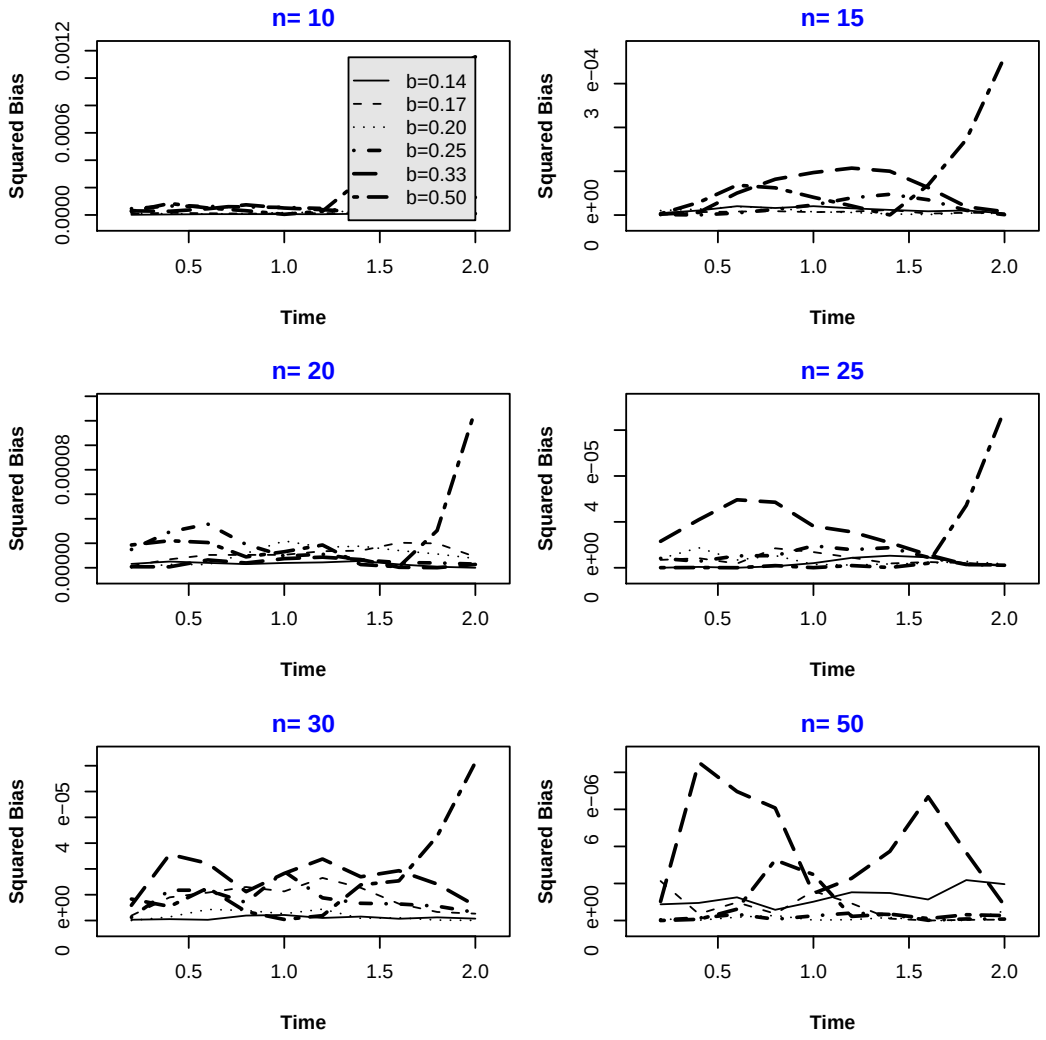

Figure 7.2: Biases for different values of n and β under Weibull(1, 1) in the semiparametric case.

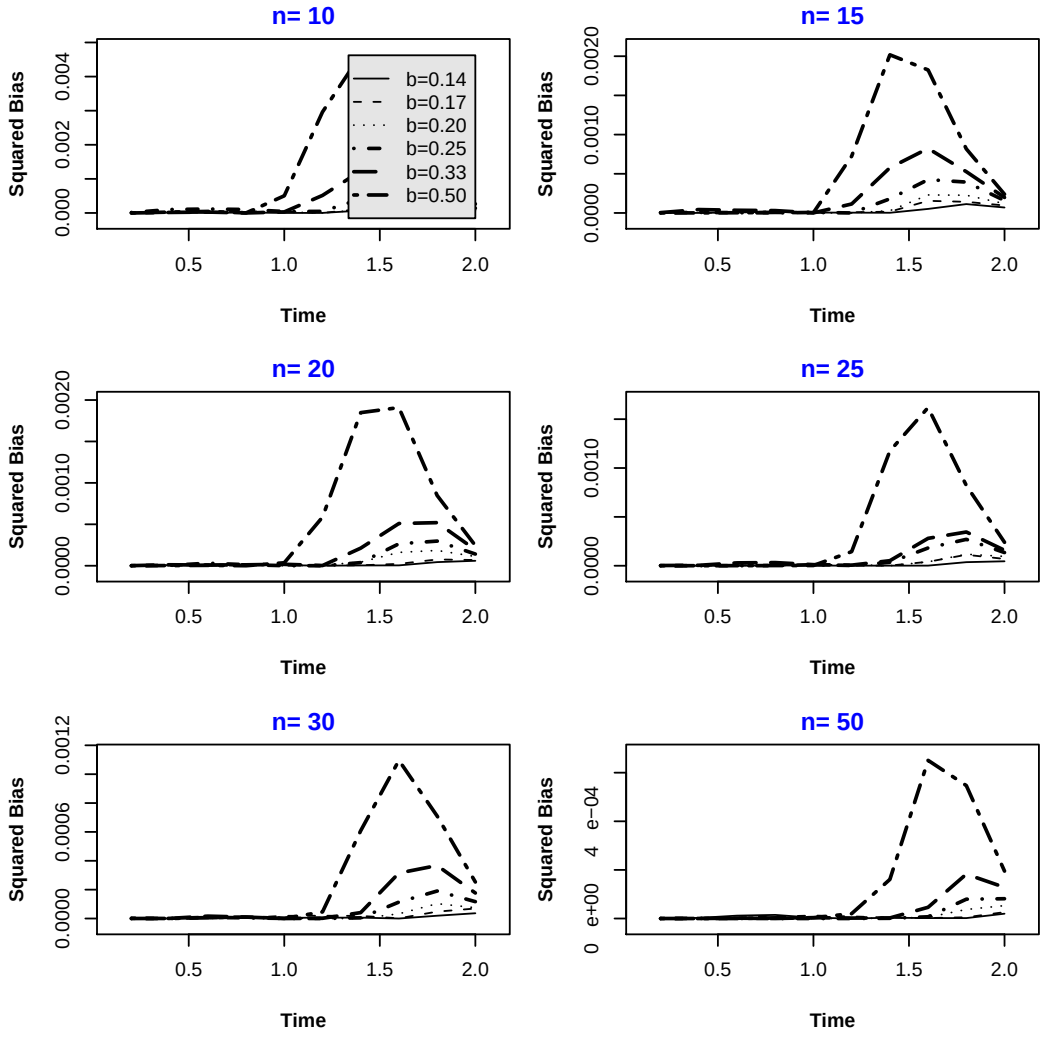

Figure 7.3: Biases for different values of n and β under Weibull(1, 2) in the semiparametric case.

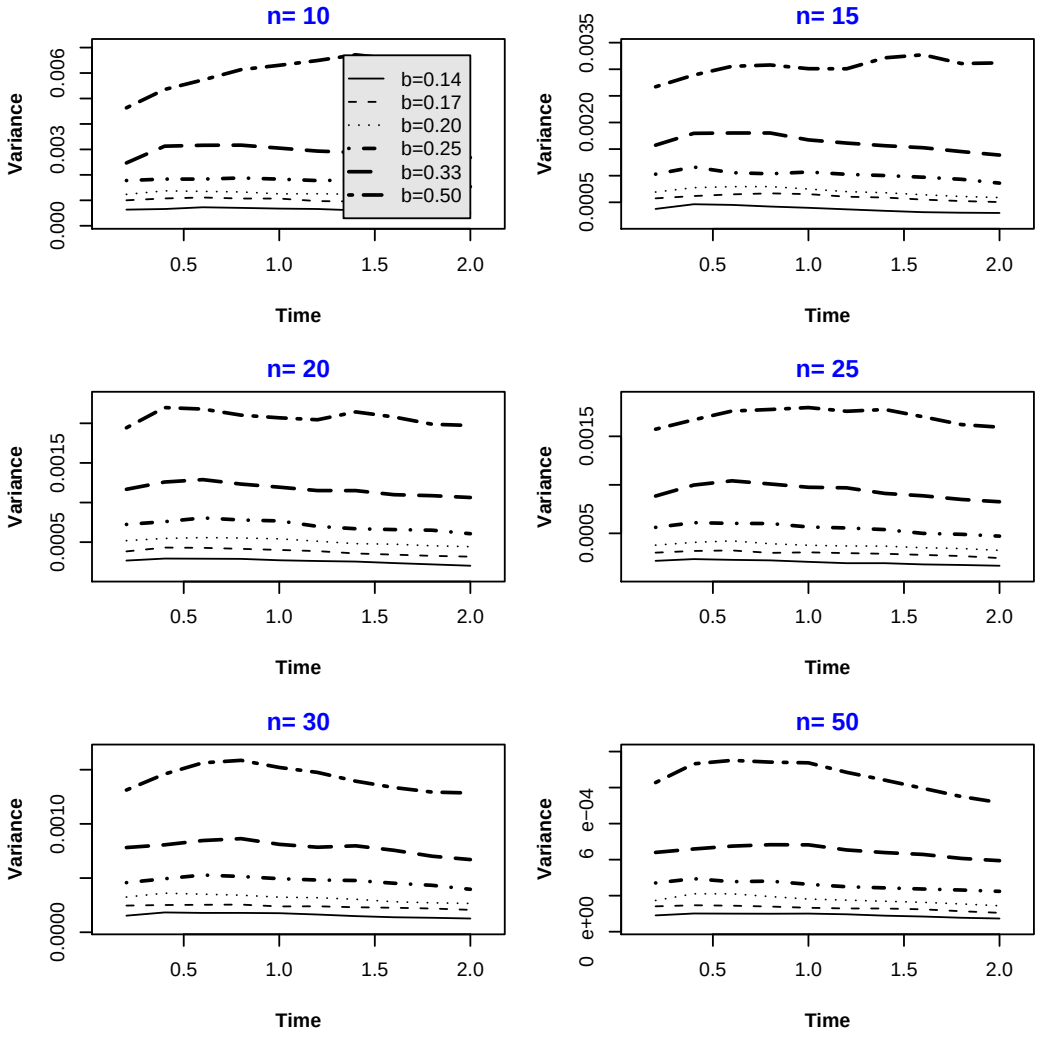

Figure 7.4: Variances for different values of n and β under Weibull(1, .5) in the semiparametric case.

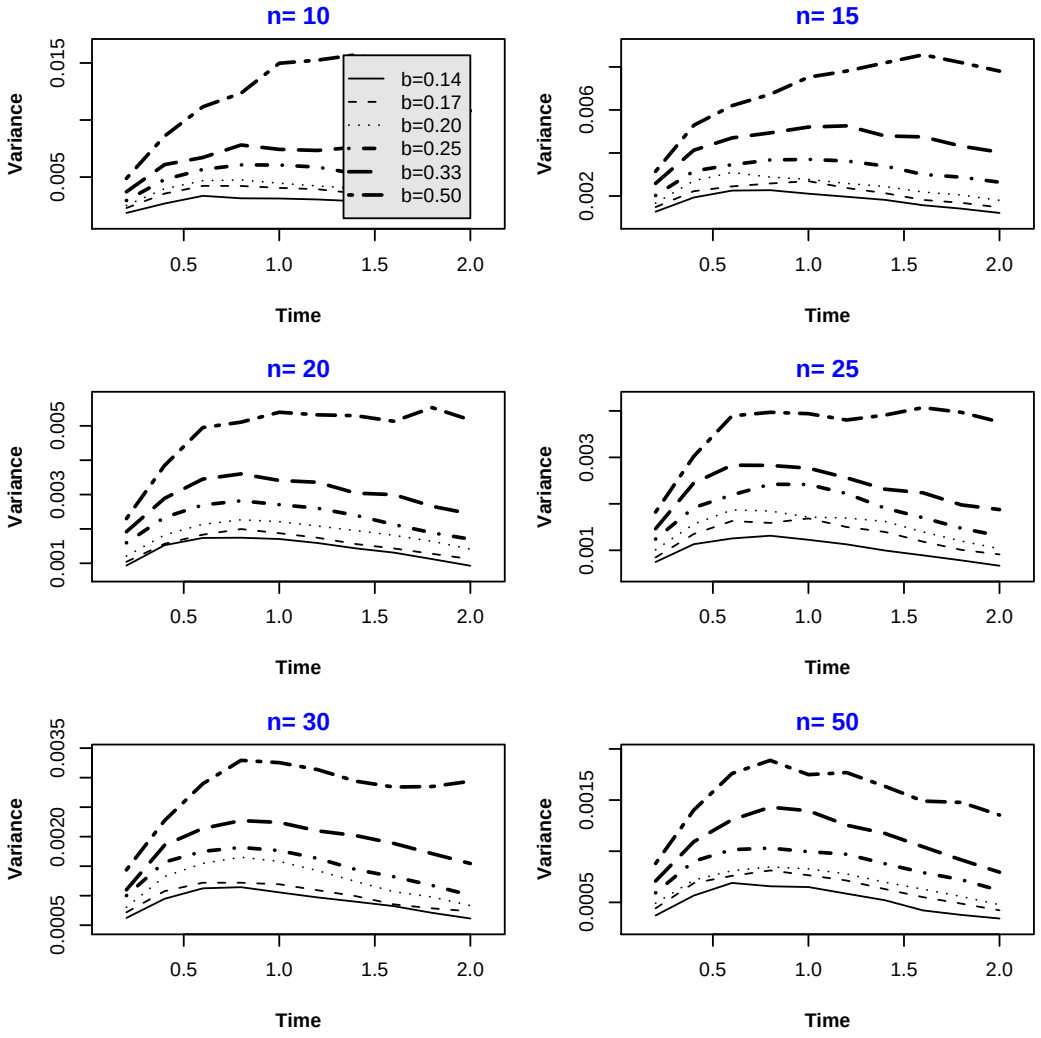

Figure 7.5: Variances for different values of n and β under Weibull(1, 1) in the semiparametric case.

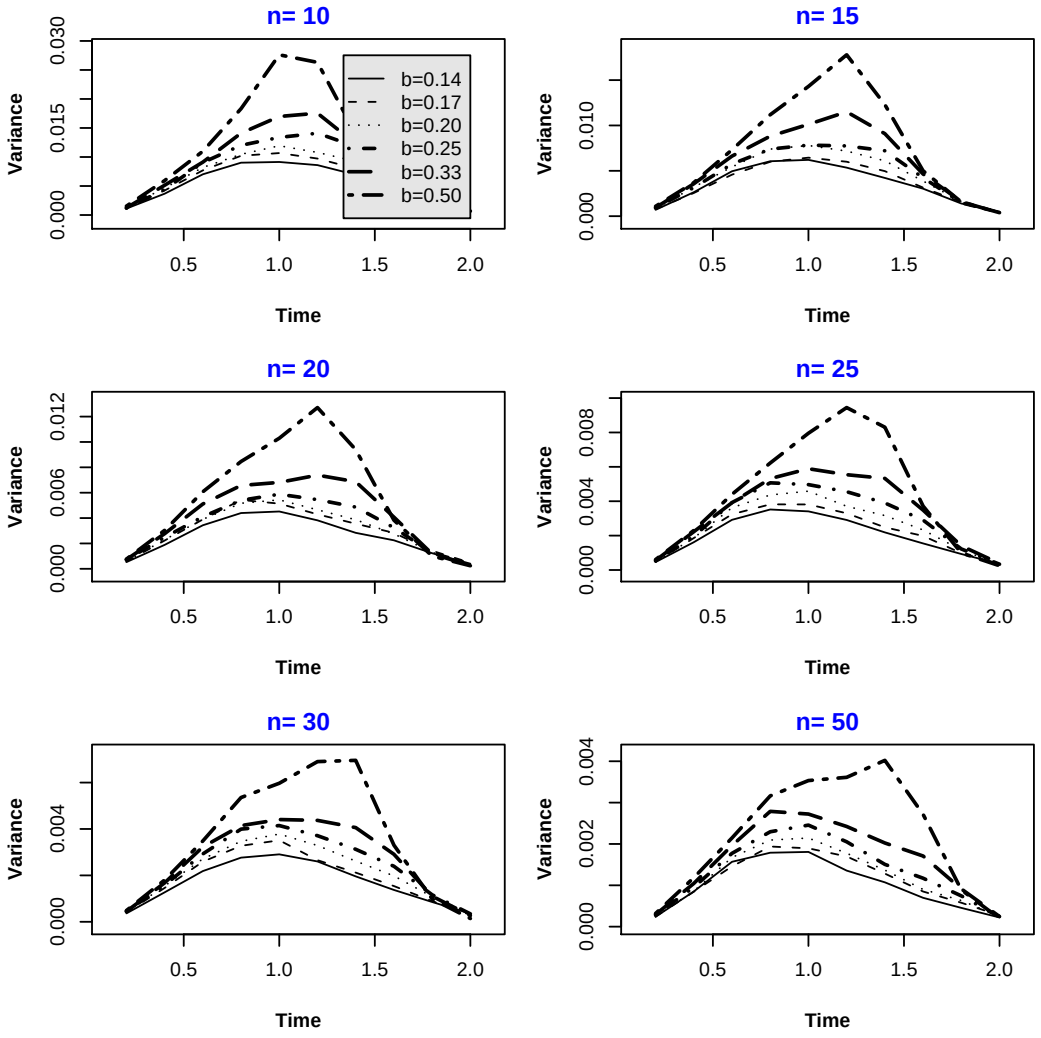

Figure 7.6: Variances for different values of n and β under Weibull(1, 2) in the semiparametric case.

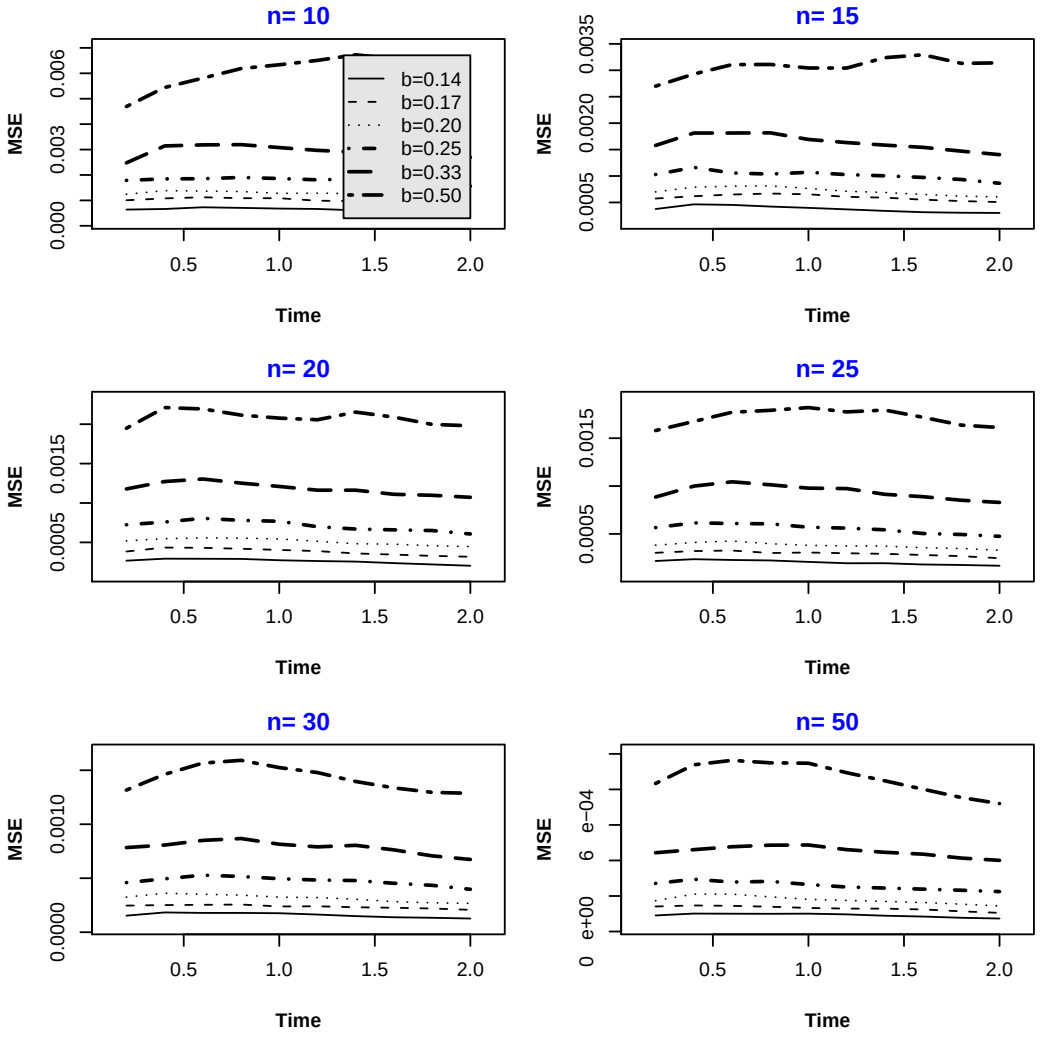

Figure 7.7: MSE for different values of n and β under Weibull(1, .5) in the semiparametric case.

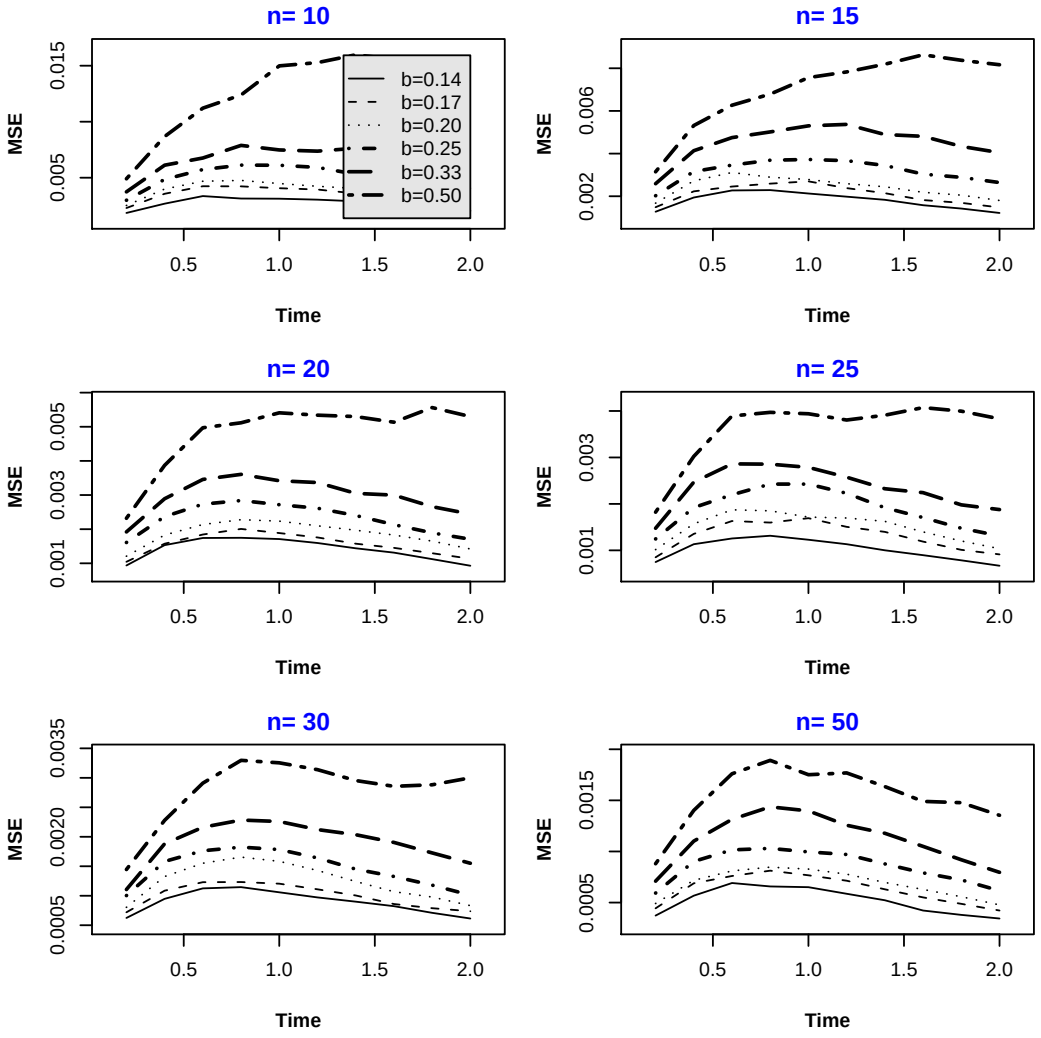

Figure 7.8: MSE for different values of n and β under Weibull(1, 1) in the semiparametric case.

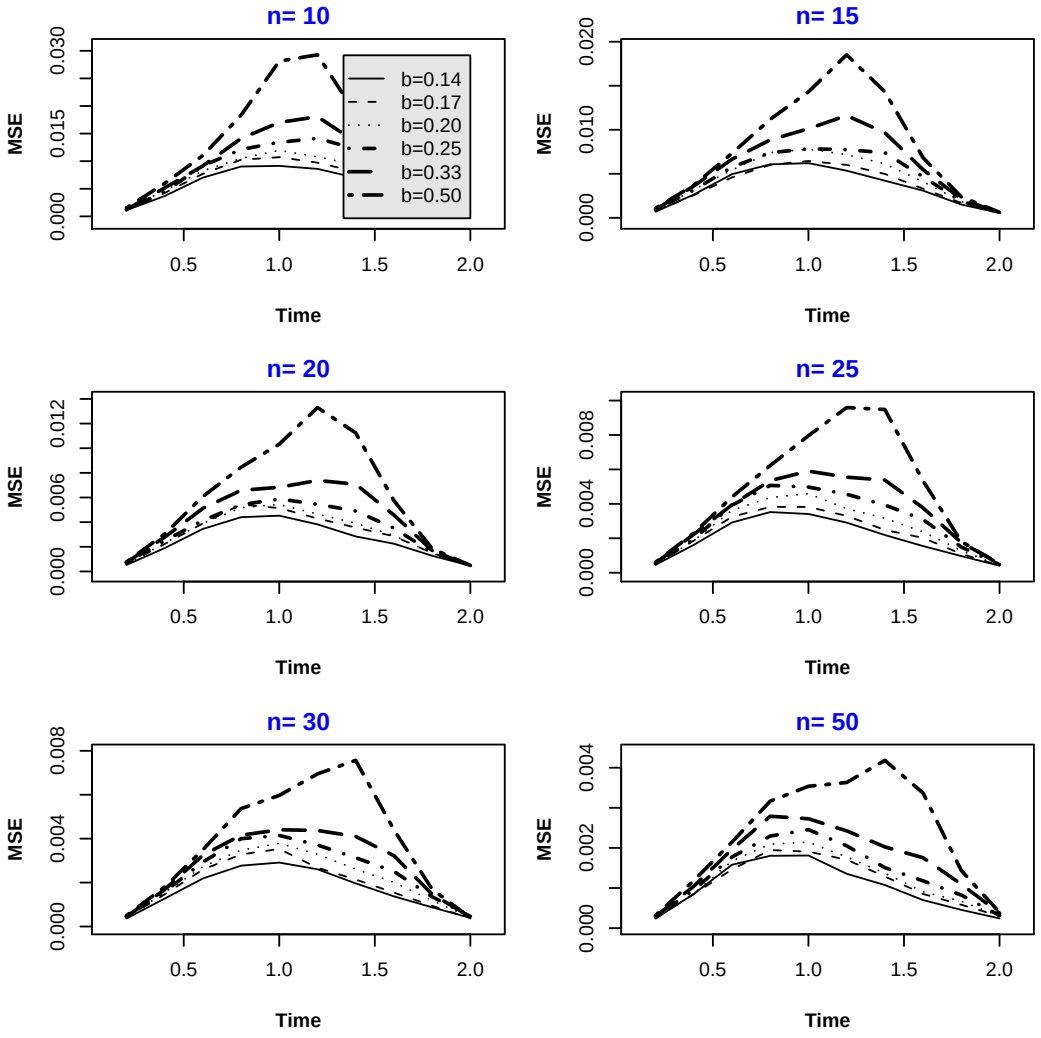

Figure 7.9: MSE for different values of n and β under Weibull(1, 2) in the semiparametric case.

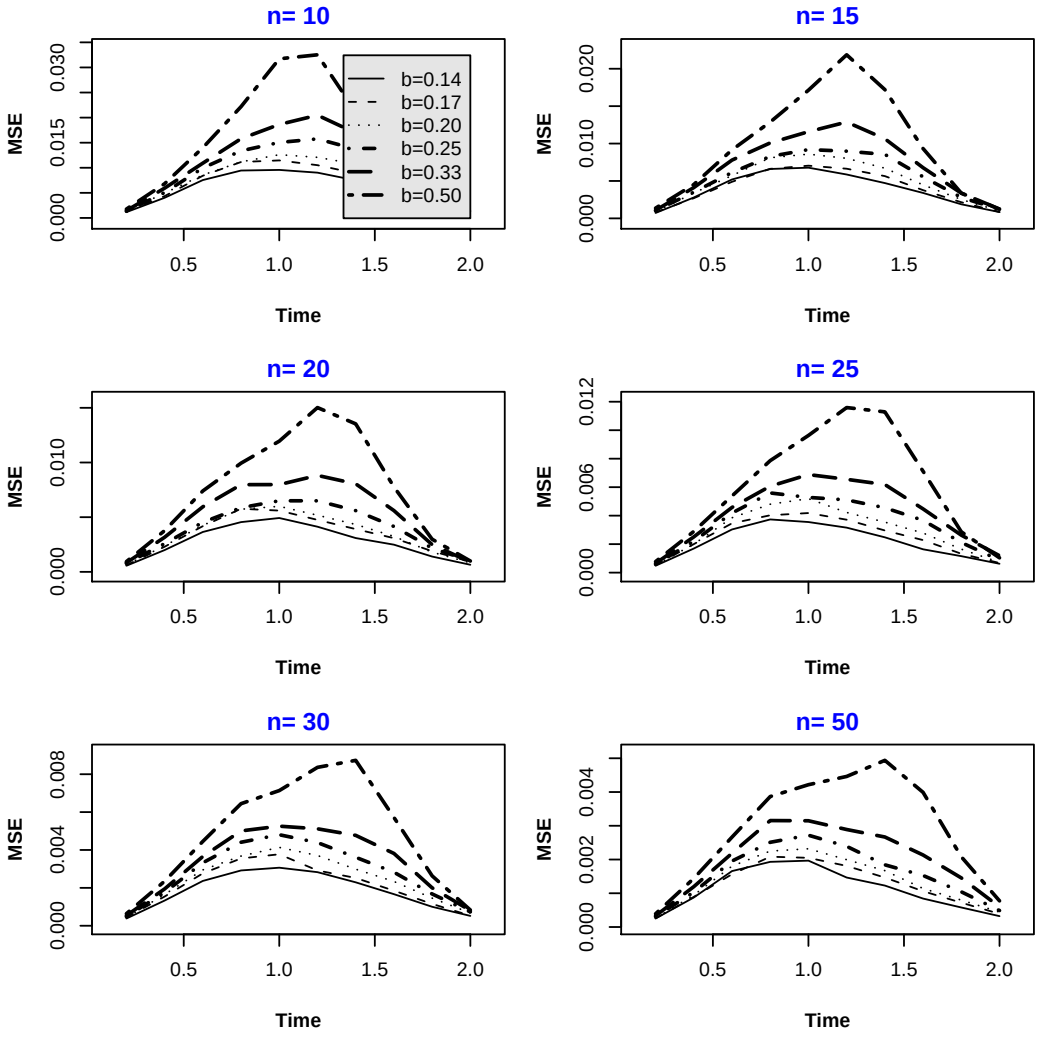

Figure 7.10: MSE for different values of n and β under Weibull(1, 2) in the semiparametric case.

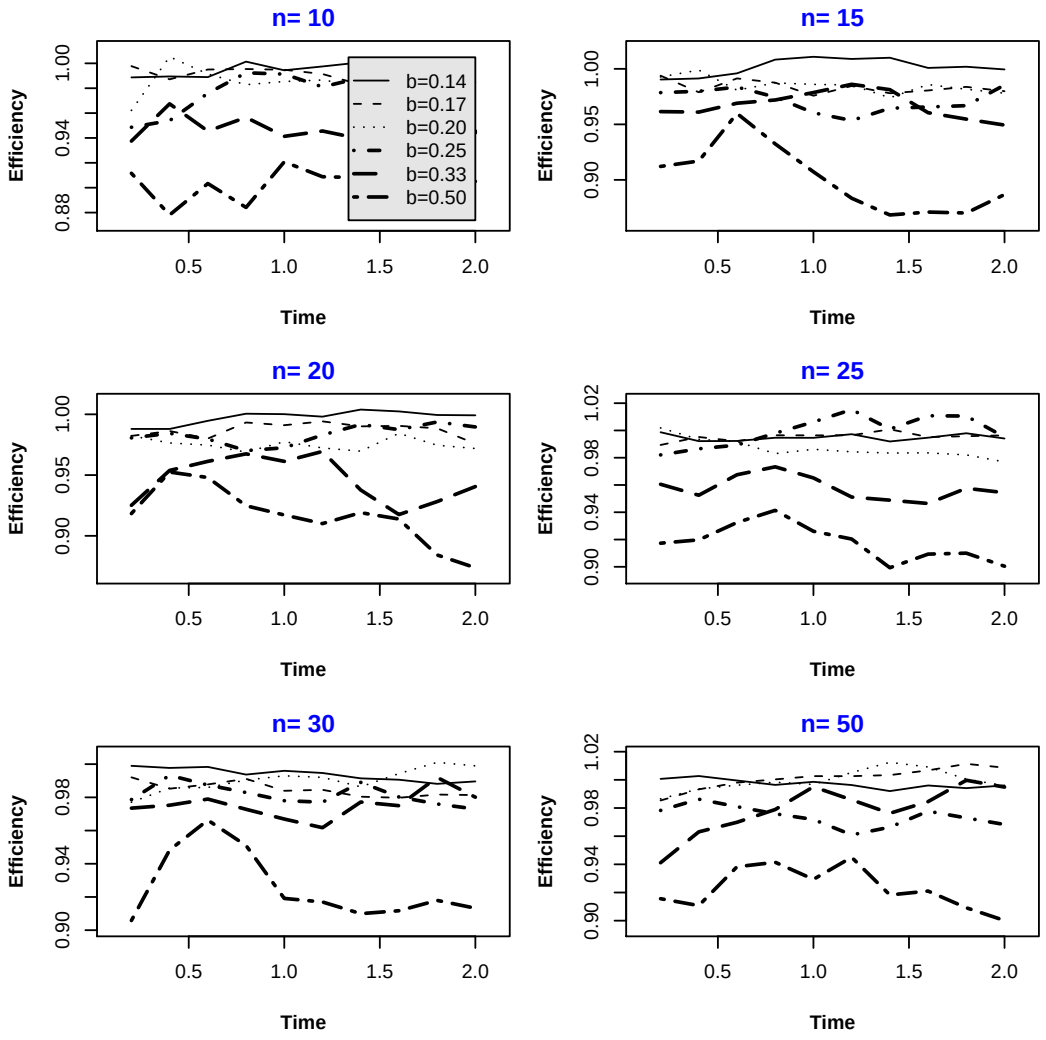

Figure 7.11: Efficiency for different values of n and β under Weibull(1, .5) in the semiparametric case.

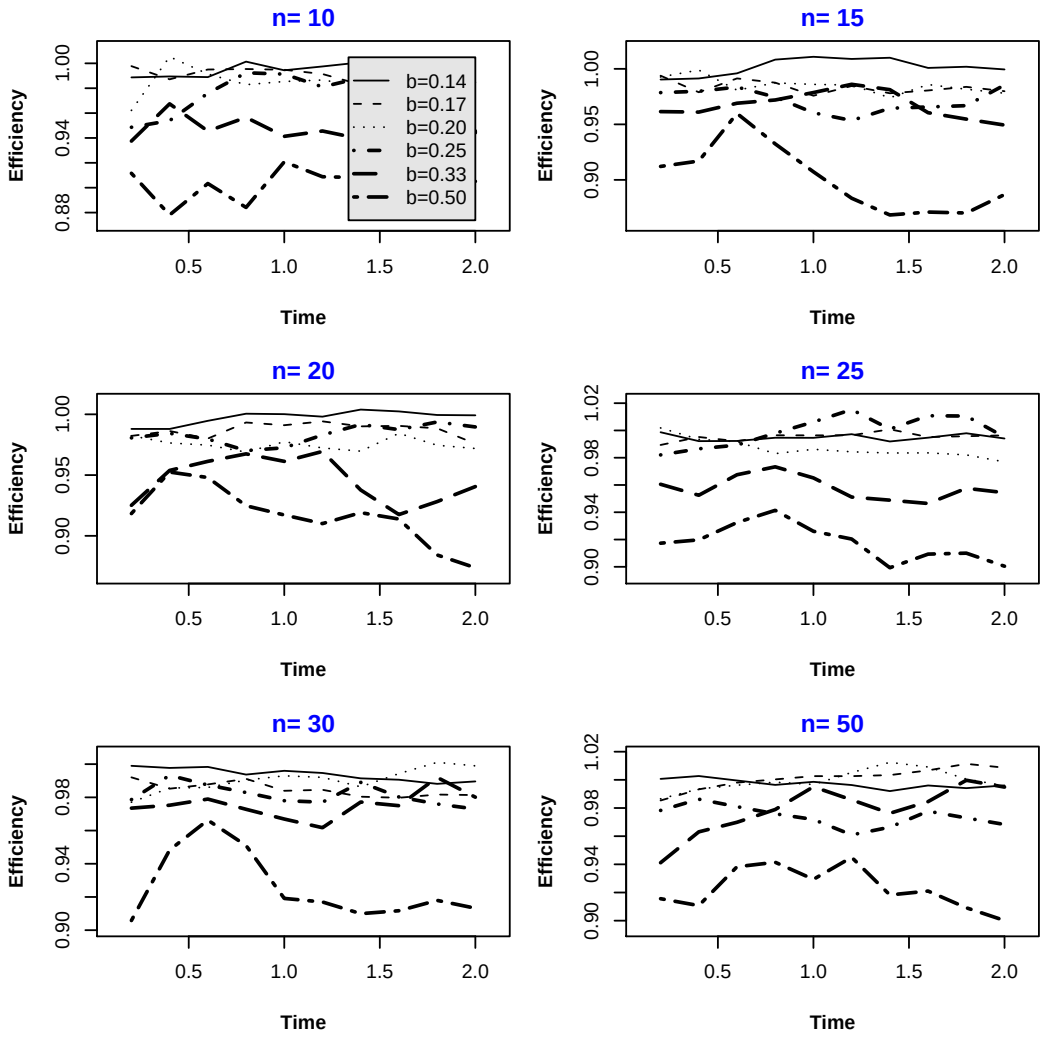

Figure 7.12: Efficiency for different values of n and β under Weibull(1, 1) in the semiparametric case.

Chapter 8

Application

In this chapter, we will apply the methodologies developed in previous chapters to real data sets that are in the literature. For each data set, we assume the inter-event time are i.i.d. Weibull distributed with shape parameter α and scale parameter μ. The estimate of the censoring parameter along with an estimate of α and μ will be given. The graph of the survivor function when the KG model holds and when it does not will also be presented.

8.1 Migrating Motor Complex Data (MMC data)

8.1.1 Description of the data

The data appears in Aalen and Husbeye [23]. The data reports Phase III [also called migrating motor complex] of the fasting state of 19 individuals [15 men and 4 women]. Phase III is also defined as fasting cycle and the time interval between two phase IIIs

is termed MMC period. Several MMC periods occurred in each individual [mean number is 4.2] with a censored MMC period terminating the records. The object was to estimate the average interevent time over the population of individuals. A pictorial representation of the data is given below.

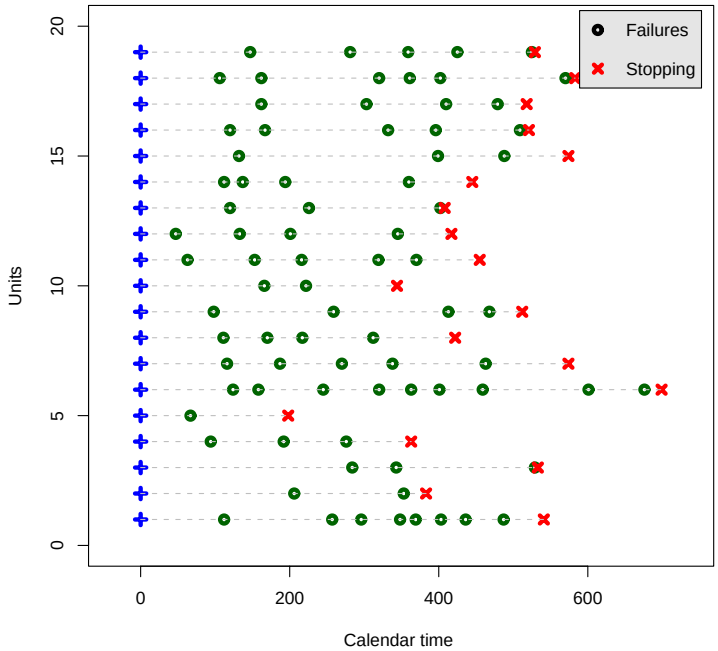

Figure 8.1: Migrating Motor Complex Data (MMC).

8.1.2 Parametric Estimation

For the parametric estimation, we assume that the interevent times are i.i.d. Weibull distributed with shape parameter μ and scale parameter α. Using the MMC data under the Weibull assumption, and ignoring informative censoring, the estimates of μ and α are respectively given by $\hat{\mu} = 2.09$ and $\hat{\alpha} = 117.99$ which corresponds to an estimated survivor function $\hat{\bar{F}}(t) = \exp\{-(t/117.99)^{2.09}\}$. On the other hand, if the

KG model is assumed, we obtain $\hat{\mu} = 2.28$ and $\hat{\alpha} = 119.75$ with an estimated survivor function $\hat{\bar{F}}(t) = \exp\{-(t/119.75)^{2.28}\}$. The estimated value of β is .04 which corresponds to a probability of an individual being censored at the first event occurrence of $.04/(1.04) = .038$. The graphs of the estimator of the gap-time in both cases is given in Figure 8.2. It can be seen that the two graphs are very close which suggests the KG assumption does not have much impact on the survivor function. This is due to the fact that the MMC data contains 19 subjects and the many complete events have been observed for each subject, and as a consequence the impact of the KG model will disappear.

8.1.3 Semiparametric Estimation

For the semiparametric estimation using MMC data, the Kaplan-Meier estimator was obtained ignoring the KG model and when it is taken into account. For the latter, the estimated value of β is .011. We observe that the differences among the values of the survivor function for different values of t are very small except for large values of t as shown on Figure 8.3. This again is due to reasons mentioned above.

8.2 Air Conditioning Data

Proschan [33] reports data on the times between failures of air conditioning units used on aircraft. The times were collected for 13 jet airlines and the purpose was to estimate the distribution of the time between two failures.

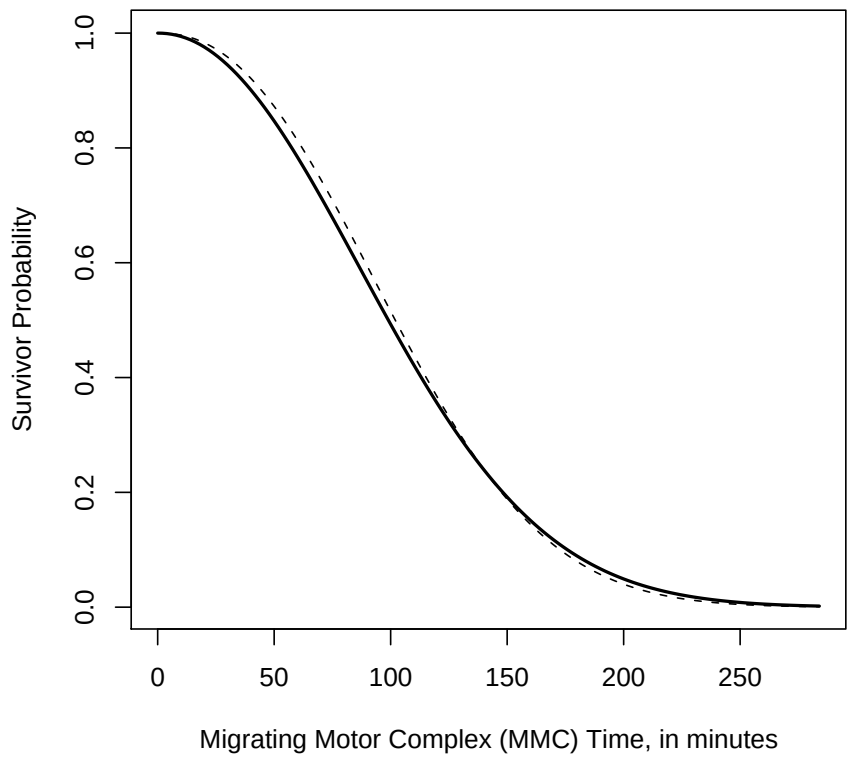

Figure 8.2: Survivor curves estimates for the MMC data. The thick curve is obtained when the KG model is ignored and the curve with break is obtained when the KG model is assumed.

8.2.1 Parametric Estimation

As in the MMC data, we assume that the interevent times are i.i.d. Weibull distributed with shape parameter μ and scale parameter α. Using the Air conditioning data under the Weibull assumption, and ignoring informative censoring, the estimates of μ and α are respectively given by $\hat{\alpha} = .924$ and $\hat{\mu} = 89.55$ which corresponds to an estimated survivor function $\hat{\bar{F}}(t) = \exp\{-(t/89.55)^{.924}\}$. If the KG model holds, we obtain $\hat{\alpha} = .95$ and $\hat{\mu} = 90.82$ with an estimated survivor function

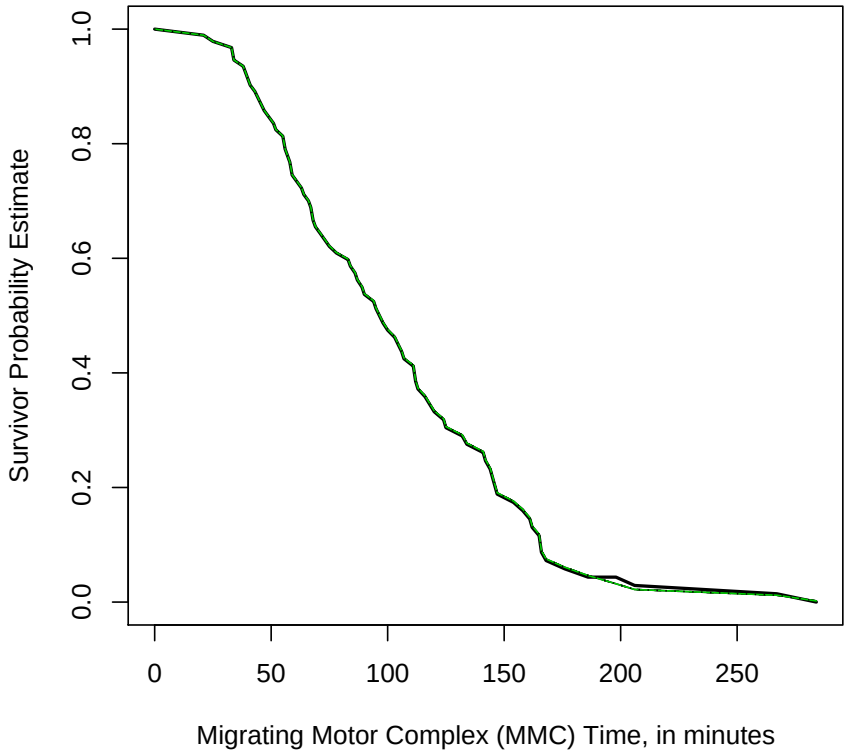

Figure 8.3: Survivor curves estimates for the MMC data. The thick curve is obtained when the KG model is ignored and the curve with break is obtained when the KG model is assumed.

$\hat{\bar{F}}(t) = \exp\{-(t/90.82)^{.924}\}$. The estimated value of the censoring parameter β is .06 which corresponds to a probability of a Jet being censored at the first event occurrence of $.06/(1.06) = .056$. The graphs of the estimator of the gap-time distribution in both cases are given in Figure 8.4. It can be seen that the two graphs are very close which suggest the KG does not have much impact on the survivor function. In the Air conditioning data, the number of subjects is 13 and the corresponding number of event occurrences is 213 and as a consequence the impact of the right-censoring random variable will disappear.

For the semiparametric estimation, the estimated value of β is .013. Again, there are no major differences between the two estimators.

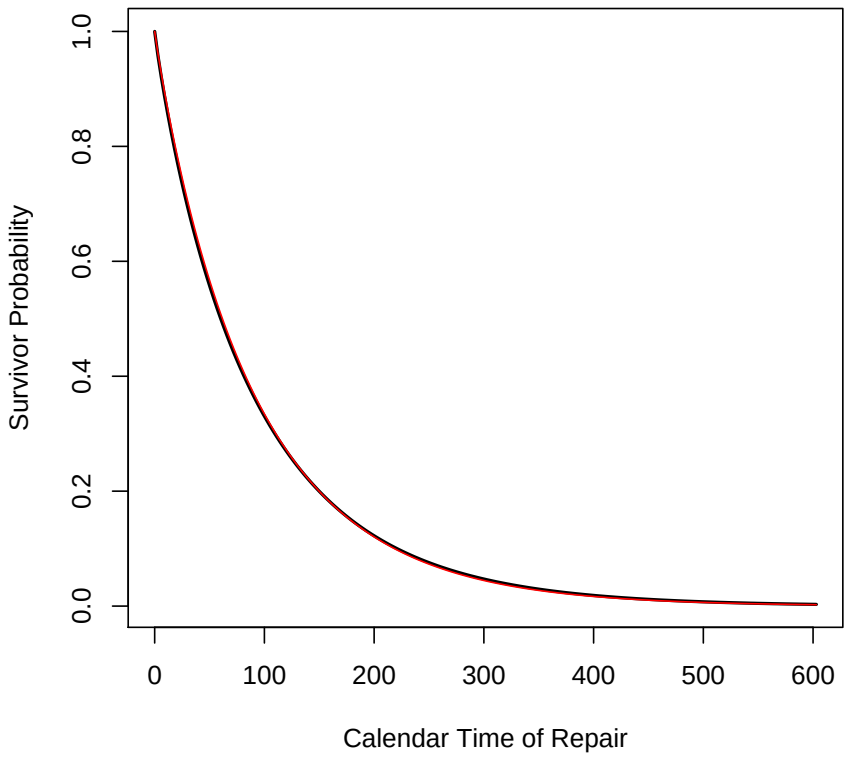

Figure 8.4: Survivor curves estimates for the Air conditioning data in the parametric case.

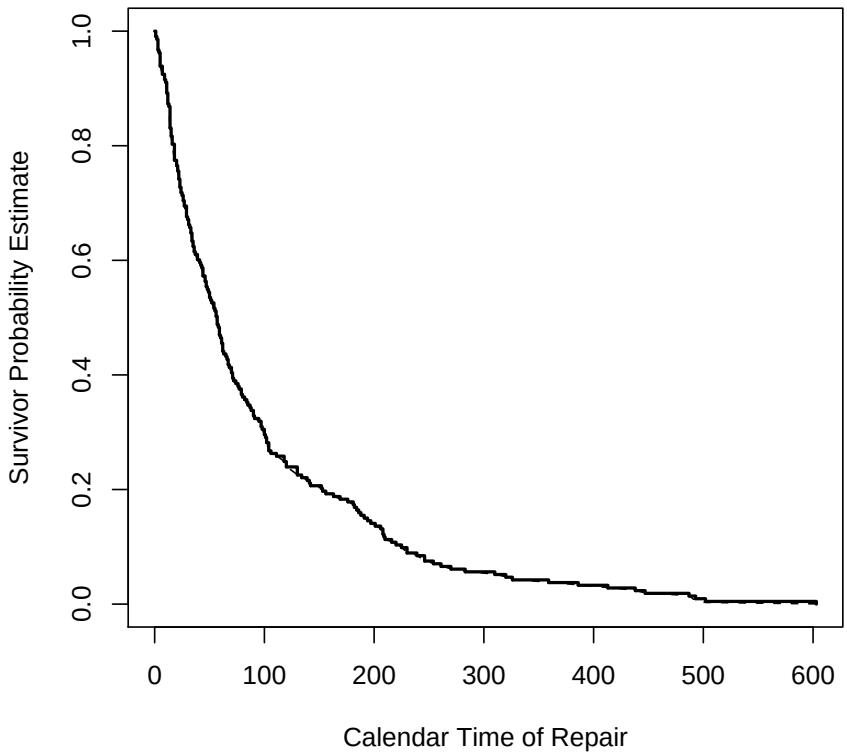

Figure 8.5: Survivor curves estimates for the Air conditioning data in the semiparametric case.

Chapter 9

Summary and Future Work

9.1 Summary

We have addressed the problem of estimating the gap-time distribution with recurrent event data under the assumption of informative censoring. Specifically, we have exploited a convenient and appealing model for informative censoring to derive parametric and semiparametric estimators of the survivor function using a counting process and stochastic integral approach. In the parametric case, the ML estimates of the parameters associated with F and G were obtained and were shown to be consistent and converge to a normal distribution when properly standardized using the results of Borgan [8]. A sufficient set of conditions for the asymptotic properties to hold were given in terms of gap-time using a doubly indexed time formulation of the original process as was done in Sellke [36]. These were easier to show because they did not contain the random quantity $R_i(w)$. Weak convergence to a Gaussian process of

the estimator of the survivor function was also obtained using the functional delta method, the continuous mapping theorem and the asymptotic properties of the MLEs. Similar results were also obtained by ignoring the KG structure. One of our main initial results was that our proposed estimator is more efficient than the one derived by ignoring informative censoring as the number of subjects in the study increases. The special case of the homogeneous Poisson process model was also investigated. It was straightforward in that case to obtain closed form expressions for the ML estimators as well as for the asymptotic covariance matrix. It was further shown that the ARE of the nonparametric estimator in Peña et al. [30] with respect to our proposed estimator is always bounded above by $\simeq 0.65$. That is, to obtain the same efficiency, 35 percent fewer observations would be needed for our proposed estimator as compared to the one proposed in their paper. This result parallels those obtained by Cheng and Lin [12] in the case of a single event. The finite sample properties of our proposed estimator were studied using computer simulation studies. The results were shown to be consistent with the asymptotic properties as the sample size increases. In the semiparametric case, an estimator of the cumulative hazard was first obtained by embedding the problem into a bivariate time scale counting process that utilizes both the calendar time and length of time since the last event occurrence. This reformulation allowed us to obtain expressions for the estimators of the parameters of interest as well as their asymptotic properties. An 'estimator' of the cumulative hazard which depends on the parameter β was first obtained. β was then estimated using likelihood profile and plugged in the former estimator to finally obtain an estimator of the cumulative hazard. Consequently, an estimator of the survivor function of the gap-time

was proposed using product integral representation technique. Asymptotic properties of the parameters were derived using Theorem 6.2. These estimators were compared to the nonparametric estimator proposed in Peña et al. [30] and were shown to be more efficient when the KG model holds. Using simulation studies, small sample results were also shown to be in concordance with the asymptotic results as sample sizes increase. However, there are still many interesting and important problems that need to be investigated.

9.2 Future Work

In terms of future work in this area, we have identified two categories. First, some technical issues need to be addressed regarding the parametric estimation, and secondly, the problem of correlated recurrence time data needs to be addressed.

9.2.1 Technical Issues

In the first part of this research, we have assumed that the survivor function has some known parametric form, and we have used the ML technique for estimation and inference. A fundamental assumption underlying classical results on the properties of ML estimators [cf. Wald [43] and LeCam [29]] is that the probability model is correctly specified. However, most of the time, this is not so since models are just an approximation to reality. If indeed the model is misspecified, many technical issues arise with the ML estimators.

1. The first question one could ask is: Does the ML estimators converge to some

limit? This issue was first considered by Berk [[5] and [6]]. He gave general conditions under which the ML estimator converges even if the model is not correctly specified. Later, White [46] gave simple conditions with broad applicability under which the ML estimators are strongly consistent. We will follow the approach of White [46] to address this technical issue in future research.

2. The second technical issue is that of convergence to a normal distribution in misspecified models. Again, we will use White's [46] approach to obtain general conditions under which the ML estimator converges to normal distribution.

9.2.2 Estimation with Frailties

The estimators derived in this dissertation are frailty-less estimators, that is, the interoccurrence times are assumed to be a random sample from some underlying distribution function F. However, there is a possibility of correlation between these gap-times specially in biomedical studies and which could be modeled using a frailty model. Frailties are unobservable random effect components that could lead to possible associations among the gap-times. The most common choice of the frailty distribution is a gamma distribution with shape and scale parameters equal to an unknown parameter α. For a given subject or unit, the parameter α has the effect of controlling the degree of association among the interoccurrence times. Many authors in survival analysis and reliability have used the gamma random variable to model frailty. For instance, Stocker and Peña [39] used a gamma frailty to derive estimators for the parameters of a general class of models for recurrent events when the model

incorporates a frailty component and derived their asymptotic properties. Peña et al. [30] proposed methods for estimating the parameter α and consequently a nonparametric estimator of the survivor function when the inter-event times are assumed to be correlated. In the future, we would like to derive parametric and nonparametric estimators of the gap-time distribution under the KG model by dropping the independence assumption, thereby ascertaining the impact of correlated inter-event times on the efficiency gained.

Bibliography

[1] Odd Aalen. Nonparametric inference for a family of counting processes. *Ann. Statist.*, 6(4):701–726, 1978.

[2] P. K. Andersen and R. D. Gill. Cox's regression model for counting processes: a large sample study. *Ann. Statist.*, 10(4):1100–1120, 1982.

[3] Per Kragh Andersen, Ørnulf Borgan, Richard D. Gill, and Niels Keiding. *Statistical models based on counting processes*. Springer Series in Statistics. Springer-Verlag, New York, 1993.

[4] Robert G. Bartle. *The elements of integration*. John Wiley & Sons Inc., New York, 1966.

[5] Robert H. Berk. Limiting behavior of posterior distributions when the model is incorrect. *Ann. Math. Statist. 37 (1966), 51–58; correction, ibid*, 37:745–746, 1966.

[6] Robert H. Berk. Consistency a posteriori. *Ann. Math. Statist.*, 41:894–906, 1970.

[7] Patrick Billingsley. *Convergence of probability measures*. Wiley Series in Probability and Statistics: Probability and Statistics. John Wiley & Sons Inc., New York, second edition, 1999. A Wiley-Interscience Publication.

[8] Ørnulf Borgan. Maximum likelihood estimation in parametric counting process models, with applications to censored failure time data. *Scand. J. Statist.*, 11(1):1–16, 1984.

[9] Pierre Brémaud. *Point processes and queues*. Springer-Verlag, New York, 1981. Martingale dynamics, Springer Series in Statistics.

[10] D Byar. Multiple tumor recurrence data for patients with bladder cancer. In D. Andrews and A. Herzberg, editors, *Data: A Collection of Problems for Many Files for Stdent and Research Worker*. New York: Springer-Verlag, 1980.

[11] Y. Y. Chen, M. Hollander, and N. A. Langberg. Small-sample results for the Kaplan-Meier estimator. *J. Amer. Statist. Assoc.*, 77(377):141–144, 1982.

[12] Philip E. Cheng and Gwo Dong Lin. Maximum likelihood estimation of a survival function under the Koziol-Green proportional hazards model. *Statist. Probab. Lett.*, 5(1):75–80, 1987.

[13] K. L. Chung and R. J. Williams. *Introduction to stochastic integration*. Probability and its Applications. Birkhäuser Boston Inc., Boston, MA, second edition, 1990.

[14] Sándor Csörgő. Estimation in the proportional hazards model of random censorship. *Statistics*, 19(3):437–463, 1988.

[15] Sándor Csörgő and Julian J. Faraway. The paradoxical nature of the proportional hazards model of random censorship. *Statistics*, 31(1):67–78, 1998.

[16] Thomas R. Fleming and David P. Harrington. *Counting processes and survival analysis*. Wiley Series in Probability and Mathematical Statistics: Applied Probability and Statistics. John Wiley & Sons Inc., New York, 1991.

[17] Santner T. Gail, M. and C. Brown. An analysis of comparative carcinogenesis experiements based on multiple times to tumor. *Biometrics*, 36:255–266, 1980.

[18] Ursula Gather and Jörg Pawlitschko. Estimating the survival function under a generalized Koziol-Green model with partially informative censoring. *Metrika*, 48(3):189–207 (1999), 1998.

[19] R. D. Gill. Nonparametric estimation based on censored observations of a Markov renewal process. *Z. Wahrsch. Verw. Gebiete*, 53(1):97–116, 1980.

[20] R. D. Gill. Testing with replacement and the product limit estimator. *Ann. Statist.*, 9(4):853–860, 1981.

[21] Kenneth Hoffman and Ray Kunze. *Linear algebra*. Second edition. Prentice-Hall Inc., Englewood Cliffs, N.J., 1971.

[22] Myles Hollander and Edsel Peña. Families of confidence bands for the survival function under the general random censorship model and the Koziol-Green model. *Canad. J. Statist.*, 17(1):59–74, 1989.

[23] E. Husbeye and O. Aalen. Statistical analysis of recurrent events forming renewal processes. *Statistics in Medicine*, 10:1227–1240, 1991.

[24] Jean Jacod. Multivariate point processes: predictable projection, Radon-Nikodým derivatives, representation of martingales. *Z. Wahrscheinlichkeitstheorie und Verw. Gebiete*, 31:235–253, 1974/75.

[25] E. L. Kaplan and Paul Meier. Nonparametric estimation from incomplete observations. *J. Amer. Statist. Assoc.*, 53:457–481, 1958.

[26] Klein J. Keiding, N. and C. Kamby. Semiparametric marshall-olkin models applied to the occurrence of metastases at multiple sites after breast cancer. *Biometrics*, 45:1073–1086, 1989.

[27] James A. Koziol and Sylvan B. Green. A Cramér-von Mises statistic for randomly censored data. *Biometrika*, 63(3):465–474, 1976.

[28] Paul H. Kvam and Edsel A. Peña. Estimating load-sharing properties in a dynamic reliability system. *J. Amer. Statist. Assoc.*, 100(469):262–272, 2005.

[29] L. Le Cam. On the assumptions used to prove asymptotic normality of maximum likelihood estimates. *Ann. Math. Statist.*, 41:802–828, 1970.

[30] E. A. Peña, R. L. Strawderman, and M. Hollander. Nonparametric estimation with recurrent event data. *J. Amer. Statist. Assoc.*, 96(456):1299–1315, 2001.

[31] Edsel A. Peña, Elizabeth H. Slate, and Juan R. Gonzalez. Semiparametric inference for a general class of models for recurrent events. *Journal of Statistical Planning and Inference*, 137:1727–1747, 2007.

[32] Edsel A. Peña, Robert L. Strawderman, and Myles Hollander. A weak convergence result relevant in recurrent and renewal models. In *Recent advances in reliability theory (Bordeaux, 2000)*, Stat. Ind. Technol., pages 493–514. Birkhäuser Boston, Boston, MA, 2000.

[33] Frank Proschan. Theoretical explanation of observing decreasing failure rate. *Technometrics*, 5:375–383, 1963.

[34] Rolando Rebolledo. Central limit theorems for local martingales. *Z. Wahrsch. Verw. Gebiete*, 51(3):269–286, 1980.

[35] Sidney I. Resnick. *A probability path*. Birkhäuser Boston Inc., Boston, MA, 1999.

[36] Thomas Sellke. Weak convergence of the Aalen estimator for a censored renewal process. In *Statistical decision theory and related topics, IV, Vol. 2 (West Lafayette, Ind., 1986)*, pages 183–194. Springer, New York, 1988.

[37] Galen R. Shorack and Jon A. Wellner. *Empirical processes with applications to statistics*. Wiley Series in Probability and Mathematical Statistics: Probability and Mathematical Statistics. John Wiley & Sons Inc., New York, 1986.

[38] Fotios Siannis. Applications of a parametric model for informative censoring. *Biometrics*, 60(3):704–714, 2004.

[39] R. Stocker and E. Peña. A general class of parametric models for recurrent event data. *Technometrics*, 49(2):210–220, 2007.

[40] van der. Vaart and R. Gill. Non and semiparametric maximum likelihood estimators and the von mises method-ii. *Scandinavian Journal of Statistics*, 20:271–288, 1993.

[41] Y. Vardi. Nonparametric estimation in renewal processes. *Ann. Statist.*, 10(3):772–785, 1982.

[42] Y. Vardi. Nonparametric estimation in the presence of length bias. *Ann. Statist.*, 10(2):616–620, 1982.

[43] Abraham Wald. Note on the consistency of the maximum likelihood estimate. *Ann. Math. Statistics*, 20:595–601, 1949.

[44] Mei-Cheng Wang and Shu-Hui Chang. Nonparametric estimation of a recurrent survival function. *J. Amer. Statist. Assoc.*, 94(445):146–153, 1999.

[45] Mei-Cheng Wang, Jing Qin, and Chin-Tsang Chiang. Analyzing recurrent event data with informative censoring. *J. Amer. Statist. Assoc.*, 96(455):1057–1065, 2001.

[46] Halbert White. "Maximum likelihood estimation of misspecified models" [Econometrica **50** (1982), no. 1, 1–25; MR 83b:62216]. *Econometrica*, 51(2):513, 1983.

[47] A. Link William. A model for informative censoring. *J. Amer. Statist. Assoc.*, 84(407):749–752, 1989.